Core Representation Theory

— ***EYNTKA*** **Series** —

Nathanael Chwojko-Srawley

August 22, 2026

Contents

Part I

Representation Theory

There are many domains of mathematics which have made huge amounts of progress, having very-well understood objects and morphisms, theorems, results, and properties. One of the most well-understood area's of mathematics has become linear algebra, where vector-spaces and linear transformations. One of the goals of this chapter will be do be able to study the following 3 objects where the operations becomes matrix addition and multiplication

1. (Finite) groups as a subgroup of $\mathrm{GL}(V)$, where the group operation becomes the matrix multiplication
2. Associative algebras
3. Lie Algebras

The further advantage of bringing these objects into linear algebra is that many field intersect with it: If the vector-spaces are infinite dimensional, we can make them Hilbert spaces and use tools from analysis. Vector-spaces are at the heart of differential geometry allowing us to use such theories to study groups!

For the reader who is really interested in number theory, we shall see in ref:HERE that matrix algebras have many parallels with some number theory results. The reader who goes onto learn class field theory[1] will to their delight find that these parallels continue. It was observed by Langland[2] that the study of simple k-algebras (algebras with trivial two-sided ideals) can be a higher-dimensional parallel to field extensions and the central results of class field mathematics known as the *Langland's Program*.

(The reader may want to go back to section **??** to review about direct sums of modules)

(Also, this entire link is really cool: from wiki. Also has some cool examples of duality!!!)

[1] See [7] for the continuation of this textbook into class field theory

[2] Or at least formalized by Langland, as these ideas were floating around during his time

1

Semisimple Rings and Modules

As mentioned in the introduction, representation theory is all about decomposing structures. In part **??**, we showed in chapter **??**, section **??**, that all finite groups can uniquely be represented as a series of extensions with simple factors (i.e. G_i/G_{i+1} is simple). The simplest type of composition series would be when G is a direct product of simple groups, which would be in the terminology of exact sequences a series of trivial extensions with simple factors: such a group is sometimes *semi-simple* group[1]. When it came to modules, we found that we may decompose it in different ways depending on what rings acts on it, in particular we saw in chapter **??** that if $R = k$ is a field k, then any k-module is free, and in chapter **??** we saw that if R is a Principal Ideal Domain, where we got that any R-module decompsoed into a free part R-isomorphic to R^k, and a torsion part $R/(a_1) \times \cdots \times R/(a_n)$. It is evident that the ring that acts on the abelian group can have a large impact on it's structure. In this section, we define a new type of ring, which will come to be called a *semi-simple* ring, that will force the module M to *always* be a direct product of *simple modules* (so that we are analysing the "building blocks" of modules).

To define this ring, we first shift focus and start by work with the module itself to find the "simplest" structure it may be[2], and explore in a manner which at the beginning parallels groups: we will look for *simple R-modules*, and direct product of simple modules (since they are the simplest extension of simple module).[3]. After defining such a strucutre, we will define which rings acting on moduels will *always* make the module semi-simple

[1]the term "semi-simple group" is not a completely standard terminology, as direct product of simple groups is not generally very heavily studied, see this

[2]"simple" here means no non-trivial sub-structures that can be used to build an extension, just like for simple groups or simple rings

[3]It might be tempting to say that this means we are exploring projective modules. However, we have yet to establish whether simple modules are free-modules. Are they?

1.1 Decomposing Modules

Definition 1.1.1: Decomposition Terminology

Let M be a nonzero R-module.

1. M is said to be *simple* (or *irreducible*) if the only submodules of M are 0 and M. Otherwise, M is *reducible*
2. M is said to be *indecomposable* if there do not exist submodules $M_1, M_2 \subseteq M$ such that $M = M_1 \oplus M_2$. Otherwise, M is said to be *decomposable.*
3. The module M is said to be *semisimple* (or *completely reducible*) if it is the direct summand of simple modules
4. The module M is said to be semi-indecomposable (or completely indecomposable) if it is a direct summand of indecomposable modules.
5. If M is semisimple, any direct summand of M is called a *constituent* of M (i.e. N is a constituent of M if there exists a submodulde N' such that $M = N \oplus N'$)

A moment should be taken to find the relations between these definitions:

1. Simple modules are semi-simple and indecomposable. The prototypical example of a simple module is a field k over itself.
2. Semi-Simple $\not\Rightarrow$ Simple: take k^2 over k (the 2 dimensional vector-space over k). The cannonical example of a semi-simple module would be vector-spaces, since they are all direct sums of the underlying field, which is a simple k-module.
3. Indecomposable $\not\Rightarrow$ Simple: this statement is equivalent to saying that not all short exact sequence spilt (i.e., not all modules are *projective*, see section **??**). For a concrete example, let $\mathbb{Z}^2$ be a module over $\mathbb{Z}$ with the action:

   $$r \cdot (a, b) = (a + rb, b)$$

   Then it should be checked that it is not simple since $\mathbb{Z} \times \{0\}$ is a submodule, but it is indecomposable[4].
4. Every semi-simple module is a semi-indecomposable module, however the converse is not true: $\mathbb{Z}$ is a semi-indecomposable $\mathbb{Z}$-module (more genreally any module module over a PID is semi-indecomposable due to the Decomposition Theorem from chapter **??**), however $\mathbb{Z}$ is not semi-simple a semi-simple $\mathbb{Z}$-module as $p\mathbb{Z}$ is a submodule (the general case will be immediate after lemma 1.1.7)[5]

As stated in the introduction, we want to determine a semi-simple module or semi-simple modules in terms of the ring that is acting on it. The following defintion was made for this purpose:

[4]Hint: Is it possible to write $\mathbb{Z} = n\mathbb{Z} \oplus m\mathbb{Z}$?

[5]This is a version of Krull-Schmidt's theorem for the decomposition of Noetherian or Artinian modules into the direct sum of indecomposable

Definition 1.1.2: Semi-Simple Ring

Let R be a ring. Then R is *semi-simple* if every R-module is semi-simple. Recall that R is *simple* if the only two-sided ideals are (0) and R.

Notice how a semi-simple ring was *not* defined as the direct product of simple rings. In the following proofs, we will show that the definition of semi-simple ring implies that R is the direct product of simple rings, however the converse is not true: there exists simple rings that are *not* semi-simple modules over themselves (and hence not simple R-modules). The problem is that of dimension; we will soon explore the relation between being semi-simple and Noetherian/Artinian. If we impose the condition that R be Noetherian and Artinian, then we are able to say that R is a semi-simple ring if and only if R is the direct product of simple rings.

We next establish many properties of simple and semi-simple modules, and show how closely related they are to the ring acting on them:

Lemma 1.1.3: Simple Module Characterization

Let M be an R-module. Then M is a simple R-module if and only if there exists a two-sided maximal ideal $\mathfrak{m} \subsetneq R$ such that

$$M \cong_R R/\mathfrak{m}$$

Proof:

($\Rightarrow$) Let M be a simple [non-trivial] R-module so that the only sub-modules of M are $\{0\}$ and M. Consider any $0 \neq x \in M$ and the R-module homomorphism:

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

Since $\varphi(R) \subseteq M$ is a sub-module and M is a non-trivial R-module, it must be that $\varphi(R) = M$. Hence:

$$R/\ker(\varphi) \cong M$$

But then, $R/\ker(\varphi)$ cannot have any submodules, since M doesn't, and so $\ker(\varphi)$ must be maximal.

($\Leftarrow$) Let $R/\mathfrak{m} \cong M$ as R-module where $\mathfrak{m}$ is maximal. If $0 \subsetneq N \subsetneq M$, then there would exist an ideal $\subsetneq J \subsetneq R/\mathfrak{m}$, but that would make $\mathfrak{m} \subseteq J$, contradicting maximality of $\mathfrak{m}$.

If R is commutative, then $R/\mathfrak{m}$ is a field. For the non-commutative case, $R/\mathfrak{m}$ need not be a field, not even a division ring, the canonical example being $M_n(k)$ for $n \geq 2$ (see the remark ref:HERE or go over section **??**). Do not confuse the isomorphism in the above statement as a ring isomorphism: the structure of M is isomorphic to the R-module structure of left-multiplication on $R/\mathfrak{m}$. Notice how this also shows that a simple R-module M is cyclic since $R \cdot \varphi(1) = \varphi(R \cdot 1) = \varphi(R/\mathfrak{m}) = M$[6]. If $R = k$ is a field, then M is isomorphic to a 1-dimensional vector-space over k. There do exist non-commutative rings R that are *not* of the form $M_n(k)$; we shall explore in ref:HERE that there is a family of quaternion algebras that are degree two extensions of a field k that is simple ring (more specifically, a simple k-algebra).

[6] In fact, any element must generate it, can you see why?

Using this result, we see that any semi-simple R-module is of the form:

$$M \cong_R \bigoplus_{i \in I} R/\mathfrak{m}_i$$

for maximal ideals $\mathfrak{m}_i \subseteq R$. In fact, we can further characterize $\mathfrak{m}_i$:

Corollary 1.1.4: Simple Module's And Annihilators

Let M be a simple left (or right) R-module. Then $M \cong R/\mathrm{Ann}(m)$ for any nonzero m, and $\mathrm{Ann}(m)$ is a maximal ideal.

***Proof*:**
Since M is a simple left R-module, for any $0 \neq m \in M$, $mR \subseteq M$ is a submodule (by identifying R in M by the map $r \mapsto r \cdot 1$). As it is nonzero and M is simple, $M \cong_R Rm$. Then we have a surjective R-module homomorphism

$$R \to Rm \qquad r \mapsto r \cdot m$$

Thus, $R/\mathfrak{m} \cong_R Rm$ for some ideal $\mathfrak{m}$ (note that the R-submodules of R are ideals). In particular, $\varphi(\mathfrak{m}) = \mathfrak{m}m = 0$, so $\mathfrak{m} = \mathrm{Ann}(m)$ by definition. Thus, we have the isomorphism chian:

$$M \cong_R Rm \cong_R R/\mathrm{Ann}(m)$$

Finally, since Rm is simple, $R/\mathrm{Ann}(m)$ must have no other two-sided ideal,s i.e. $\mathrm{Ann}(m)$ is a two-sided maximal ideal, completing the proof.

Since we've seen the limitations of the structure of simple modules, it is also natural to ask if there are any limitations on the structure of the hom-set given two simple modules:

Lemma 1.1.5: Schur's Lemma

Let M and N be simple R-modules, $\varphi : M \to N$ an R-module homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. As a consequence, $\mathrm{End}_R(M)$ is a division ring.

***Proof*:**
Since N is simple, either $\varphi(M) = N$ or $\varphi(M) = 0$. If $\varphi(M) = N$, it has to be that $\ker(\varphi) = 0$, since M is simple, and hence φ would be an isomorphism.

As a consequence, every element in $\mathrm{End}_R(M)$ is either 0 or is a unit, i.e. has a multiplicative inverse, which by definition makes $\mathrm{End}_R(M)$ a divising ring under $(+, \circ)$.

Since $M \cong_R R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subsetneq R$, we have that $\mathrm{End}_R(R/\mathfrak{m})$ is always a division ring, whether R is commutative or not. In a sense, this can be thought of as "generalization" to **??**, where it was shown that $R/\mathfrak{m}$ is a field if R is commutative.

Notice that if $M = V$ was an 1-dimensional vector-space over k, then $\mathrm{End}_R(V)$ is a known division ring, in particular it's ring-isomorphic to k. If V is a finite dimensional semi-simple module over k, i.e. it is n-dimensional, then $\mathrm{End}_R(V)$ is ring-isomorphic to $M_n(k)$. We might hope for something similar for the structure of $\mathrm{End}_R(M)$ is M is semi-simple. Take a moment to ponder this before reading the next lemma (hint: recall **??**)

Lemma 1.1.6: Hom's of Semisimple Modules

Let M be a semisimple R-module, so that:

$$M = S_1^{\oplus n_1} \oplus S_2^{\oplus n_2} \oplus \cdots \oplus S_k^{\oplus n_k}$$

where each S_i are simple R-modules, $S_i \ncong S_j$. Then as a ring:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \mathrm{End}_R(S_1^{\oplus n_1}) \oplus \cdots \oplus \mathrm{End}_R(S_k^{\oplus n_k})$$

Furthermore, for each $S_i^{\oplus n_i}$, we have that direct sums "pull out" to become matrices:

$$\mathrm{End}_R(S_i^{\oplus n_i}) \cong_{\mathbf{Ring}} M_{n_i}(\mathrm{End}_R(S)) = M_{n_i}(D_i)$$

where $D = \mathrm{End}_R(S_i)$ and $M_{n_i}(D_i)$ is the matrix ring over the division ring $\mathrm{End}_R(S_i)$.

***Proof*:**
We invoke **??** to get the result in terms of groups, and then use Schur's lemma and check that the ring operation is compatible:

$$\begin{aligned}
\mathrm{End}_R(M) &= \mathrm{Hom}_R\left(\bigoplus_i S_i^{\bigoplus n_i}, \bigoplus_j S_j^{\bigoplus n_j}\right)\\
&\cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R\left(\ S_i^{\bigoplus n_i}, S_j^{\bigoplus n_j}\right)\\
&\cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R(\ S_i, S_j)^{\bigoplus n_i n_j}
\end{aligned}$$

where when $S_i \neq S_j$, by Shur's lemma we must have:

$$\mathrm{Hom}_R(S_i, S_j)^{n_i n_j} = 0$$

Hence we get that $\mathrm{End}_R(M) \cong_{\mathbf{Grp}} \oplus_i \mathrm{Hom}_R(S_i^{n_i}, S_i^{n_i}) = \oplus_i \mathrm{End}_R(S_i^{n_i})$. Next, this map is compatible with the ring structure, since the map:

$$\varphi \to (\varphi|_{S^{n_i}} : S_i^{n_i} \to S_i^{n_i} \subseteq M)$$

is clearly clearly compatible with composition, and hence

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \bigoplus_i \mathrm{End}_R(S_i^{n_i})$$

For part 2, we follow the same steps as for part (1): re-indexing so that $S^n = S_1 \oplus S_2 \oplus \cdots \oplus S_n$ so that $S_i = S_j = S$, we get

$$\mathrm{Hom}_R\left(\bigoplus_i S_i, \bigoplus_j S_j\right) \cong_{\mathbf{Grp}} \bigoplus_{i,j} \mathrm{Hom}_R(S_i, S_j) \overset{!}{=} \bigoplus_{i,j} D \cong_{\mathbf{Grp}} M_n(D)$$

where $\stackrel{!}{=}$ comes from Shur's lemma with $D = \mathrm{End}_R(S)$ since $S_i = S_j = S$. The last equality comes from the fact that the matrix $M_n(D)$ when treated like a group is simply re-arranging where the addition is happening in $\bigoplus_{i,j} D$. To show that the ring structure is compatible, observe that

$$\varphi \mapsto \varphi_{ji}$$

is the desired map, where

$$\varphi \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \varphi_{12} & \cdots & \varphi_{1n} \\ \varphi_{21} & \varphi_{22} & \cdots & \varphi_{2n} \\ \vdots & \vdots & \ddots & \cdots \\ \varphi_{n1} & \varphi_{n2} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}$$

which is simply a formalization of what the group-isomorphism give us, showing us that composition is compatible, completing the proof. (see this for a more detailed explanation)

Hence, for any semi-simple R-module M, we get that:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(D_i) \cong_{\mathbf{Ring}} \prod_i \mathrm{End}_R(S_i^{\oplus n_i})$$

and since each S_i is a simple R-module, $S \cong_R R/\mathfrak{m}_i$ for appropriate maximal ideal $\mathfrak{m}_i \subseteq R$, $\mathrm{End}_R(S_i) \cong \mathrm{End}_R(R/\mathfrak{m}_i)$. Thus:

$$\mathrm{End}_R(M) \cong_{\mathbf{Ring}} \prod_i M_{n_i}(\mathrm{End}_R(R/\mathfrak{m}_i)) = \prod_i M_{n_i}(D_i)$$

If $R = k$ is a field making $M = V$ a vector-sapce over k (say n-dimensonal), then the only simple k-submodule up to isomorphism is k, and so this result reduces to:

$$\mathrm{End}_R(V) \cong_{\mathbf{Ring}} M_n(\mathrm{End}_k(k/(0))) = M_n(k)$$

where $\mathrm{End}_k(k) \cong_{\mathbf{Ring}} k$, aligning with the results we found in secion **??**. This shows that when generlizing from linear algebra, due to the fact that a ring is non-commutative, we may have multiple non-isomorphic maximal ideals giving us multiple products of matrix rings. With a better understanding of simple modules and their endomorphisms, we continue by characterizing semi-simple modules:

Lemma 1.1.7: Semi-Simple Module Equivalence

Let M be a semi-simple module. Then the following are equivalent:

1. M is an external direct sum of simple modules:
$$M \cong \bigoplus_{i \in I} S_i$$
2. M is is the internal direct sum of simple modules:
$$M = \bigoplus_{i \in \mathfrak{m}} S_i$$
3. If $N \subseteq M$, there exists a complimentary module, that is there exists another submodule $N' \subseteq M$ such that
$$M = N \oplus N'$$
i.e., every semi-simple R-module M is *projective* and *injective*.

***Proof*:**
(1) $\Rightarrow$ (2) is **??**.

For (2) $\Rightarrow$ (3), Let $N \subseteq M$. Then using Zorn's lemma, we take a maximal submodule $N' \subseteq M$ ordered by inclusion with the property that $N \cap N' = 0$ (upper-bounds naturally exist in this set, and 0 is in this set, and hence Zorn's lemma does apply). If $M = N + N'$, we're done, since we can use the isomorphism from **??**. For the sake of contradiction, let's say $N + N' \subsetneq M$. Then since $M = \sum_i S_i$, there must exist some $i \in I$ and $s_i \in S_i$ such that $s_i \notin N + N'$, since $0 \in S_j$ for all $j \in I$. Hence $S_i \subsetneq N + N'$, so $S_i \cap (N + N') \subsetneq S_i$, however S_i is simple, and hence:
$$S_i \cap (N + N') = 0$$
But then:
$$(S_i + N') \cap N \subseteq N' \cap N = 0$$
Thus, by maximality of N', $S_i + N' = N'$, so $S_i \subseteq N' \subseteq N + N'$, contradicting the assumption that $s_i \notin N + N'$, as we sought to show.

For (3) $\Rightarrow$ (1), Let $\mathcal{S}$ be the collection of simple submodule of M where their direct sum maps injectively into M. Since $0 \in \mathcal{S}$ and taking the sum's of all the modules is an upper bound, Let N be the maximal submodule with this property. By **??**,
$$N = \sum_{i \in I} S_i \cong \bigoplus_{i \in I} S_i$$
If the map is surjective, we are done. For the sake of contradiction, suppose it is not. Then by part (3), there exists an N' such that $M = N + N'$. If $N' = 0$, we found a contradiction. So, let's say $N' \neq 0$. Then it is enough to show that N' contains a simple module, say S, since then we can make the collection $\{(S_i)_{i \in I}, S\}$ which will map injectively into M, contradicting maximality.

(here is what the professor posted Suppose every submodule N of M has a complement (i.e. a submodule N' of M such that $M = N \oplus N'$ (internal direct sum)). Then every nonzero submodule N' of M contains a simple submodule.

Proof: first we show that every submodule K of M has the same property (every submodule has a complement): suppose L is a submodule of K. by assumption, $M = L \oplus L'$ for some submodule L' of M. then note that $K = L \oplus (L' \cap K)$.

Now, suppose N' is nonzero submodule of M. Take a nonzero $x \in N'$, so Rx is a nonzero submodule and by 1st isomorphism theorem we have $Rx \cong_R R/I$ for some proper left ideal I. By the previous paragraph, every submodule of Rx has a complement, i.e. every submodule of R/I has a complement. Take any maximal left ideal $\mathfrak{m}$ containing I. Then M/I is a submodule of $R/\mathfrak{m}$ with complement L' in R/I, i.e. $R/I = \mathfrak{m}/I \oplus L'$, so $L' \cong (R/I)/(\mathfrak{m}/I \cong R/\mathfrak{m}$ is simple, i.e. L' is a simple submodule of N' (via the isomorphism $Rx \cong R/I$))

This theorems shows that every vector-space over k is a semisimple k-module, however by remark internalSumVsSumRmk, $\mathbb{Z}$ is *not* a semi-simple $\mathbb{Z}$-module, since $2\mathbb{Z}$ is a submodule with no compliment (showing that not all free modules are semi-simple).We can characterize semi-simple rings by saying that R is semi-simple if and only if every R-module is projective. This also shows us that $\mathbb{Z}$ is not a semi-simple ring; though $\mathbb{Z}$ is projective over itself, any finite abliean group is a non-projective $\mathbb{Z}$-module. Thus, a module of a semi-simple ring is stronger then being a projective module. Hence being semi-simple is stronger (in a sense much stronger) than being projective, but may not be free: even though many example of semi-simple modules are free-modules, we would require the underlying ring to be semi-simple over itself. In fact, it is enough that the underlying ring be a semi-simple module over itself, giving us an easier condition for defining a semi-simple ring! We first intorduce an important result we will use:

Corollary 1.1.8: Extension Closure Of Semisimple Modules

Let M an R-module. Then submodule and quotient modules of M are semisimple if and only if M is semi-simple. Furthermore, if $(M_i)_{i\in I}$ is a collection of semi-simple modules, then $\bigoplus_{i\in I} M_i$ is a semi-simple module.

***Proof*:**
The direct sum of semi-simple modules is semi-simple by definition, and so we check the other two statements.

Let M be a semi-simple R-module so that $M = \bigoplus_{i\in I} S_i$ is by lemma 1.1.7. First, consider a surjection $M \twoheadrightarrow N$.

$$\varphi(M) = \varphi\left(\bigoplus_{i\in I} S_i\right) = \bigoplus_{i\in I} \varphi(S_i)$$

then by Schur's lemma, either $\varphi(S_i) = 0$ or $\varphi(S_i) \cong S_i$, and hence N is semisimple.

If $N \subseteq M$, then by lemma 1.1.7, we have that $M = N \oplus N'$, and so by **??**

$$M/N' \cong N$$

which we just showed makes N semi-simple.

Using these two facts, it is not difficult to show that if $N \subseteq M$ and $M/N \cong N'$ are semi-simple, then so is M, where if $x \in M$, $x = \sum_i n_i + \sum_j n'_j$.

These properties of semi-simple modules allow us to characterize semi-simple rings much more easily:

Corollary 1.1.9: Semi-Simple Rings Condition

Let R be a ring. Then R is a semi-simple ring if and only if R is a semi-simple R-module, that is we only need to check that R is a semi-simple R-module for one R-module structure (namely R itself)

***Proof*:**
By definition, R if R is a semi-simple ring then any R-module is a semi-simple R-module, including R over itself.

Now R is given some R-action becomes a semi-simple R-module, then by corollary 1.1.8 so is $\oplus_{i\in I}R$. We will use this to show that any R-module M is semi-simple, which will make R semi-simple by definition.

Let M is any other R-module, by by the universal property of the direct sum and free-modules:

$$M = \bigoplus_i Rm_i$$

for appropriate m_i (for ex. we can pick all nonzero elemetnts of M). Then we get a surjective map:

$$\bigoplus_{i\in I} R \to M \qquad (r_i)_{i\in I} \mapsto \sum r_i m_i$$

hence, M is the quotient of a semi-simple module, and hence semi-simple. Thus, R is semi-simple, as we sought to show.

Due to this, we get a nice description of semi-simple rings which is even stronger than semi-simple modules:

Lemma 1.1.10: Decomposing Semi-Simple Rings

Let R be a semi-simple ring. Then:

$$R = I_1 \oplus \cdots \oplus I_n$$

for a finite collection of I_k, where I_k are simple left-submodules of R, i.e. *minimal* left ideals[a] of R

[a] These are *left ideals* as we are working with *left* submodules

***Proof*:**
Since R is a semi-simple ring, it is a semi-simple R-module and so by definition:

$$R = \bigoplus_{i\in\mathcal{C}} I_i$$

for simple submodules I_i (i.e. minimal left ideals I_i). To make this a finite sum, since $1 \in R$, we get for an appropraite finite subset $A \subseteq \mathcal{C}$

$$1 \in I_{\alpha_1} + \cdots + I_{\alpha_n}$$

But then, since these are all ideals,

$$R = R \cdot 1 \subseteq I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n} \subseteq R$$

and hence $R = I_{\alpha_1} \oplus \cdots \oplus I_{\alpha_n}$, completing the proof.

1.1.11 Remark As a consequence of this, any semi-simple ring R is a left-Noetherian/Artinian ring. Interestingly, we will soon show that this also implies that R is a *right*-Noetherian/Artinian ring (see rightNoethArtRemark)

Furthermore, this shows how it is necessary for the condition that R be left-Noetherian/Artinian so that the notion of being semi-simple coincide between the two definitions.

Since we characterized simple R-modules for any ring R to be R-isomorphic to $R/\mathfrak{m}$, and we've given a structure of R if R is a semisimple ring, we can make lemma 1.1.3 more precise:

Corollary 1.1.12: Structure Of Simple Modules Of Semi-Simple Ring

Let R be a semi-simple ring so that $R = I_1 \oplus I_2 \oplus \cdots \oplus I_n$ for minimal left ideal I_k (equivalently, simple left R-modules). Then if M is a simple left R-module,

$$M \cong_{R\textbf{-Mod}} I_i$$

for some $1 \leq i \leq n$.

***Proof*:**
Let M be a simple nonzero left R-module, and consider the R-module homomorphism:

$$\varphi : R \to M \qquad r \mapsto r \cdot x$$

where $x \neq 0$. Since M is simple, it is nonzero, so φ is surjective. Since $R = \bigoplus_i^n I_i$, for some i's

$$\varphi|_{I_i} \neq 0$$

and since $\varphi|_{I_i} = \pi_i \circ \varphi$ we have the R-module homomorphism:

$$\varphi|_{I_i} I_i \to M$$

and since both modules are simple and this is a nonzero homomorphism, by Schur's lemma this map must be an isomorphism, as we sought to show.

Note that there can be multiple I_i's that are isomrophic to M, which we will show in a following theorem.

This shows that any simple R-module over a semi-simple ring R is in fact isomorphic to a simple

R-module over R (i.e. R-isomorphic to a minimal ideal of R). Thus, studying simple R-modules over semi-simple rings comes down to studying the ring R itself, and that there are only *finitely many* simple R-modules. Furthermore, we have that $R/(\mathfrak{m}_1 \oplus \cdots \mathfrak{m}_{n-1}) \cong_R \mathfrak{m}_n$. Here is a good example of where the ring structure of the left hand side cannot exist in the right hand side. For example, if R is commutative, then $R/\mathfrak{m}$ is a field that is R-module isomorphic to a minimal ideal, but ideals cannot be fields (over the same identity[7]). Thus, in the case where R is semi-simple, for a maximal ideal $\mathfrak{m}$, $R/\mathfrak{m}$ is R-isomorphic to a minimal ideal of R.

Exercise 1.1.1

1. Let S be a simple module. Show that any nonzero submodule of the injective enveloppe $E(S)$ is indecomposable, but not simple.
2. Show that if M is a simple R-module, then it necessarily has to be cyclic (the converse, however, is not true).
3. Show that $A = \mathbb{Q}[x, y]/(xy - yx - 1)$ is a simple ring, but no a semi-simple ring (this is a special example of an algebra called a *Weyl* Algebra)
4. Let R be a ring, not necessarily commutative, and let $[R, R] = \{rs - sr \mid r, s \in R\}$. Show that the trace map induces an isomorphism
$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$
From this perspective, matrix rings are the natural "nonabelian" extension of a ring!

1.2 Artin-Wederburn

We just showed how studying simple and semi-simple R-modules comes down to studying the ring R, especially if the ring R is semi-simple. Thus, it is fruitful to classify semi-simple rings, which leads us to the main classification result for semi-simple rings, namely we build on lemma 1.1.10 and show that we can find more precisely how to represent each minimal ideal $\mathfrak{m}_i$, and furthermore show that this classification is exhaustive, and that any semi-simple ring uniquely breaks down into the result we'll show. We will then finish by showing that if we upgrade our ring to k-algebras (that is, we can put a vector-space structure on our ring), we get some stronger results due to the fact we can count dimensions.

First, we try to relate what we've shown for the hom-sets of simple modules with the strucure of simple modules. You may perhaps recall that we commented in exercise ref:HERE (in 9.1.1) and example **??** that R is note necessarily isomorphic to R^{op}. There is in fact a very nice way of linking R^{op} and R, which will be demonstrated after formally define R^{op} for reference

[7] $k \times \{0\} \subseteq k \times k$ is an ideal and a field with identity $(1, 0)$

Definition 1.2.1: Opposite Ring

Let R be a ring. Then we define a new ring labelled R^{op} that has the same underlying set and addition operation, whith the multiplication operation $*$ being:

$$x * y = yx$$

where the multiplication yx happens in R

If R is commutative, then $R = R^{\text{op}}$, and so we only get non-trivial results when R is not commutative. An example of $R \ncong R^{\text{op}}$ would be a free-algebra over two variables (say $k\langle x, y\rangle$).

Proposition 1.2.2: Properties Of Opposite Ring

Let R be a ring and R^{op} its oppposite ring. Then:

1. $(R^{\text{op}})^{\text{op}} = R$
2. $(R \times S)^{\text{op}} = R^{\text{op}} \times S^{\text{op}}$
3. $M_n(R)^{\text{op}} = M_n(R^{\text{op}})$ where the isomorphism is $A \mapsto A^t$ (essentially since $(AB)^t = B^t A^t$)

***Proof*:**
exercise

With this, we can make a link between R and R^{op}:

Lemma 1.2.3: Yonneda's Lemma for opposite Rings

Let R be a ring and consider the ring $\text{End}_R(R)$ to be the set of all R-module homomorphisms from R (as an R-module) to itself. Then $\varphi : R^{\text{op}} \to \text{End}_R(R)$, $r \mapsto (x \mapsto xr)$ is a ring isomorphism:

$$R^{\text{op}} \cong_{\mathbf{Ring}} \text{End}_R(R)$$

***Proof*:**
Let φ_r be the given map. Let's first check that $\varphi_r \in \text{End}_R(R)$. Then we see that:

$$\begin{aligned}
\varphi_r(\lambda x + \lambda y) &= (\lambda x + \lambda y)r \\
&= \lambda(xr) + \lambda(yr) \\
&= \lambda\varphi_r(x) + \lambda\varphi_r(y)
\end{aligned}$$

showing that $\varphi_r \in \text{End}_R(R)$. Next, we check that φ is ineed a ring-homomorphism. By the properties of modules, it is easy to see that:

1. $\varphi_1 = \text{id}$, and hence φ sends the identity to the identity.
2. $\varphi_{r+r'} = \varphi_r + \varphi_{r'}$
3. $\varphi_{rr'} = \varphi_{r'} \circ \varphi_r$ (don't forget we are working with R^{op})

Finally, showing bijectivity, for injectivity we see that if $\varphi_r = 0$, then $0 = \varphi_r(1) = r$, and hence $\varphi_r = \varphi_0$. For surjectivity, let $\varphi \in \mathrm{End}_R(R)$. Let's say $\varphi(1) = r$. Then $\varphi(x) = x\varphi(1) = xr$, so $\varphi = \varphi_r$, completing the proof.

In the special case where R is commutative, then $R \cong R^{\mathrm{op}}$, showing that in this case $R \cong \mathrm{End}_R(R)$. Using this, we will use lemma 1.2.3 along with lemma 1.1.6 to upgrade lemma 1.1.10:

Theorem 1.2.4: Artin-Wedderburn, one Direction

Let R be a semisimple ring. Then:

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_i is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i

The key realization about this theorem is that while the multiplication on the left hand side can "send you" all over the place, the multiplication on the right hand side only sends you back into your particular component. To see this more clearly, let's say R was a k-algebra for a field k so that it has a basis. If R was a semi-simple ring, then that means that when multiplying elements of R (so figuring out where the basis elements go while multiplying), we can isolate the problem down to the components on the right hand side, and figure out where they go through matrix multiplication!

***Proof*:**
First, by lemma 1.1.10, for we get simple R-submodule of R, i.e. minimal ideals of R,

$$R = I_1 \oplus \cdots \oplus I_n$$

Now, we can always re-order the sums (i.e. $I_1 \oplus I_2 \cong I_2 \oplus I_1$, since $R \cong I_1 \oplus \cdots \oplus I_n$ by **??**). Hence, we will group all the I_i's that belong to the same isomrophism class (as R-modules), letting us re-write R as:

$$R \cong_{\mathbf{Ring}} I_1^{n_1} \oplus \cdots \oplus I_k^{n_k}$$

where $I_i \ncong I_j$ for $i \neq j$. Hence by lemma 1.2.3 and lemma 1.1.6:

$$\begin{aligned} R^{\mathrm{op}} &\cong_{\mathbf{Ring}} \mathrm{End}_R(R) \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k) \end{aligned}$$

where $D_i = \mathrm{End}_R(I_i)$ is a divison ring. Finally, Taking the opposite ring on both sides, we get:

$$\begin{aligned} R &\cong_{\mathbf{Ring}} (R^{\mathrm{op}})^{\mathrm{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1)^{\mathrm{op}} \times \cdots \times M_{n_k}(D_k)^{\mathrm{op}} \\ &\cong_{\mathbf{Ring}} M_{n_1}(D_1^{\mathrm{op}}) \times \cdots \times M_{n_k}(D_k^{\mathrm{op}}) \end{aligned}$$

and it is clear the D_i^{op} is still a division ring, hence completing the proof.

Thus, if we have a simple-ring then the ring breaks down into the product of matrix rings over division rings. In the special case where $R = k$ is a field, then k has only one minimal ideal, namely itself, and $\mathrm{End}_k(k) \cong M_1(k) \cong k$, and hence we reduce to $k \cong k$. If $R = k^n$ is a product of the same field, then the minimal ideals of R are $R = k \oplus \cdots \oplus k$, so that we get back $k^n \cong_{\mathbf{Ring}} k^n$.

We will now show that the converse of this theorem is true, that the product of matrix rings over division rings is semi simple. We start by showing that matrix ring are simple modules[8].

Proposition 1.2.5: Matrix Ring over Division Rings are semi-simple

Let $R = M_n(D)$ be a matrix ring over a divison ring. Let $C = D^{\oplus n}$ be the left $M_n(D)$-module where the action is left matrix multiplication (imagine C as a column). Then C is a simple $M_n(D)$-module, and

$$M_n(D) \cong_{M_n(D)\textbf{-Mod}} C^{\oplus n} = (D^{\oplus n})^{\oplus n}$$

showing that $M_n(D)$ is semisimple ring and $M_n(D) \cong_{\textbf{Ring}} C^{\oplus n}$

***Proof*:**
Let $R = M_n(D)$ and $C = D^{\oplus n}$ (C for *column*). For notational simplicity, let $E_{ij} \in R$ be the elementary matrix which is 1 in position ij and zero everywehre else. Let $e_i \in C$ be the column where it is 1 at position i and zero everywehre else:

$$E_{ij} = \begin{pmatrix} 0 & & & & 0 \\ & \ddots & & \iddots & \\ & & 1 & & \\ & \iddots & & \ddots & \\ 0 & & & & 0 \end{pmatrix} \qquad e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

Then $E_{ij}e_k = \delta_{jk}e_i$ where δ_{jk} is the Kronecker Delta:

$$\delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

Now, we will start by showing that C is a simple $M_n(D)$-module. Since $C = D^{\oplus n}$,

$$C = \bigoplus_i De_i$$

for some basis $e_1, e_2, ..., e_n$. To show that C is simlpe, it is enough to show that for any $0 \neq x \in C$, $C = Rx$ since then any nonzero submodule would in fact be C. Given our above notation, we have that:

$$x = \sum_k d_k e_k$$

for appropriate $d_k \in D$, where at least one of these is nonzero, let's say $d_j \neq 0$. Then consider $E_{ij}x \in Rx$:

$$E_{ij}x = \sum_k \delta_{jk} d_k e_k = d_j e_i$$

since $d_j \neq 0$. Since D is a division ring, we get that $d_j^{-1}e_i \in Rx$, for all $1 \leq i \leq n$. But then, we get:

$$C = \sum De_i \subseteq Rx \subseteq C$$

[8]This will be an example of R being both a simple R-module and a simple ring

showing that $C = Rx$ for any $0 \neq x \in C$, showing that C is indeed simple.

Now, we can re-write this ring as the internal direct sum of matrices I_i which are zero outisde the ith column:

$$R = M_n(D) = I_1 \oplus \cdots \oplus I_n$$

$$I_k = \begin{pmatrix} 0 & \cdots & a_{1,k} & \cdots & 0 \\ 0 & & a_{2,k} & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & & a_{2,n-1} & & 0 \\ 0 & \cdots & a_{n,k} & \cdots & 0 \end{pmatrix}$$

This decomposition works since each I_i is stable under *left* multiplication by elements of $M_n(D)$ (since left multiplication is a *column* transformation, right multiplication would be *row* transformation). Finally, $C \cong_{R\textbf{-mod}} I_i$ by simply moving the column to the right place, and so each I_i is simple. Hence, we have decomposed $M_n(D)$ as a dierct product of simple left $M_n(D)$-modules, and hence $M_n(D)$ is a semisimple $M_n(D)$-module, which by corollary 1.1.9 makes $M_n(D)$ a semi-simple ring. Finally, since $C^{\oplus n}$ represents the n rows of n-length columns, we see that the action of $M_n(D)$ on $C^{\oplus n}$ is identical to that of matrix multiplying elements of $C^{\oplus n}$, which gives us that $M_n(D) \cong_{\textbf{Ring}} C^{\oplus n}$, as we sought to show.

1.2.6 Remark In fact, $M_n(D)$ is a simple ring, meaning the only *two-sided* ideals of R are 0 and R; the above proof can be modified to show this. This gives us a nice class of non-commutative simple rings (since all simple commutative rings are fields).

Notice that it is quite sepcial that the ring opration and the module operation coincided, this because of the special way in which we decompoed the $M_n(D)$-module $M_n(D)$ and by the fact that it is a module over itself. In the previous proposition, we showed that $D^{\oplus n}$ is a simple $M_n(D)$-module. In fact, all simple modules of $M_n(D)$ are isomorphic to $D^{\oplus n}$:

Corollary 1.2.7: Structure of Simple $M_n(D)$-Module

Let $R = M_n(D)$ be a matrix ring over a division ring. Then any simple $M_n(D)$-module is isomorphic to $D^{\oplus n}$.

***Proof*:**
Let M be a simple $M_n(D)$-module. By proposition 1.2.5

$$M_n(D) \cong_{\textbf{Ring}} D^{\oplus n} \oplus \cdots_{\textbf{Ring}} \oplus D^{\oplus n}$$

And by corollary 1.1.12,

$$M \cong D^{\oplus n}$$

Corollary 1.2.8: Endomorphism Of Simple $M_n(D)$-Module

Let S be a simple $M_n(D)$-module. Then

$$\operatorname{End}_{M_n(D)}(S) \cong_{\mathbf{Ring}} D^{\mathrm{op}}$$

***Proof*:**
Since S is simple, $S \cong_R D^{\oplus n}$ for some n. Let $\varphi_d : S \to S$, $\varphi_d(x) = x \cdot d$ where $x \cdot d$ is x times dI. Then:

$$\varphi_d(Ax) = (Ax) \cdot d = A(x \cdot d) = A\varphi_d(x)$$

and

$$\varphi_d(x+y) = (x+y) \cdot d = (x+y)dI = xdI + ydI = x \cdot d + y \cdot d = \varphi_d(x) + \varphi_d(y)$$

hence $\varphi_d \in \operatorname{End}_R(S)$. Conversely, if $\varphi \in \operatorname{End}_R(S)$, Since $S \cong D^{\oplus n}$, and

$$D^{\oplus n} \cong \begin{pmatrix} D^{\oplus n} & 0^{\oplus n} & \cdots & 0^{\oplus n} \end{pmatrix}$$

we will treat S as a $M_n(D)$ submodule of $M_n(D)$. Now, since φ is a group homomorphism, we know that:

$$\varphi \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11} & \cdots & \varphi_{n1} \\ \vdots & \ddots & \vdots \\ \varphi_{n1} & \cdots & \varphi_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \varphi_{11}(1) + \cdots + \varphi_{n1}(1) \\ \vdots \\ 0 \end{pmatrix}$$

hence let $d = \sum_i^n \varphi_{i1}(1)$, and let the right-action on $\operatorname{End}_{M_n(D)}(S)$ be right multiplication by the identity matrix dI. Then: Then for all $x \in S$, we have:

$$\begin{aligned}
\varphi \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} &= \varphi \left(\begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \right) \\
&\overset{!}{=} \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \varphi \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \\
&= \begin{pmatrix} x_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ x_n & \cdots & 0 \end{pmatrix} \cdot d
\end{aligned}$$

where $\overset{!}{=}$ comes from the fact that we can treat that matrix as an element of $M_n(D)$, and since φ is an $M_n(D)$-module homomorphism, we can pull it out. But then, the action is completely dependent on on d, an so we have a bijection between right multiplication by $d \in D$ an $M_n(D)$-endomrophisms. It is an easy taks to check that:

$$\varphi_{dd'} = \varphi_{d'} \circ \varphi_d$$

showing that

$$D^{\mathrm{op}} \cong_{\mathbf{Ring}} \operatorname{End}_{M_n(D)}(S)$$

as we sought to show.

The next thing we will need is to show that if R, S are semisimple rings, then so is $R \times S$. Since R and S are semi-simple R-modules and S-modules, if we can show that $R \times S$ is a semi-simple $(R \times S)$-module, we are done. For this to be the case, we must be able to decompose $R \times S$ into simple $(R \times S)$-submodules. Forunately, many universal propeties play in our favour to be able to easily identify simple $(R \times S)$-modules of $R \times S$ given we know the simple R-modules and S-modules.

To see the key idea of the following proof, we define some terms

Definition 1.2.9: Idempotent And Orthogonal Elements

Let R be a ring. Then $e \in R$ is said to be *idempotent* if

$$e^2 = e$$

Furthermore, if e_1, e_2 are idempotent elements in R, if

$$e_1 e_2 = e_2 e_1 = 0$$

then they are said to be *orthogonal*. If e is idempotent and cannot be written as the sum of two (commuting) idempotent elements, then e is said to be a *primitive* idempotent element.

Example 1.2.10: Idempotent Elements

Naturally, for any ring R, $1 \in R$ is an idempotent element.

For a more interesting example the product ring $R \times S$, and take $e = (1, 0) \in R \times S$, $f = (0, 1) \in R \times S$. Then:

1. $e^2 = e$ and $f^2 = f$ (i.e. they are both *idempotent*)
2. $ef = fe = 0$ (implying also that $ef \in Z(R \times S)$)
3. $e + f = 1$ (i.e. e and f are *not* primitive)

What happens if the number of terms in the product increases?

Next, if M is an R-module an N is an S-module, we can M and N to be $(R \times S)$-module by ignoring what happens to S (resp. R), i.e. think of the action being defined by the following function

$$R \times S \twoheadrightarrow R \to \mathrm{End}_{\mathbf{ab}}(M)$$

and similarly for a S-module N. Hence, we can define $M \oplus N$ as an $(R \times S)$-module. Conversely, if M is an $(R \times S)$-module, then using what we've shown in example 1.2.10, given $e = (1, 0)$ and $f = (0, 1)$:

$$M = eM + fM$$

Notice that eM is killed by Rf, so it's a $A \cong A \times B/(\{0\} \times B)$-module, and similarly for fM. In other words, idempoent elements characterize the splitting of modules! This gives us the following:

Proposition 1.2.11: Direct Product Decomposition

If A-**Mod** and B-**Mod** are the categoies of A-modules and B-modules, and A-**Mod** $\times$ B-**Mod** is the category where every object and morphism is a product of objects or morphisms. Then there is an equivalence:

$$A\text{-}\mathbf{Mod} \times B\text{-}\mathbf{Mod} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} (A \times B)\text{-}\mathbf{Mod}$$

sending:

$$\begin{aligned}(M, N) &\mapsto M \oplus N \\ (eM, fM) &\leftarrow\!\shortmid M\end{aligned}$$

where we further get: $\varphi(\psi(M \oplus N)) \cong M \oplus N$ and $\psi(\varphi(M)) \cong M$

***Proof*:**
easy to check

Proposition 1.2.12: Classifying Simple $(A \times B)$-Modules

Let P be a simple $(A \times B)$-module. Then

$$P \cong_{A \times B\text{-Mod}} M \oplus (0) \qquad \text{or} \qquad P \cong_{A \times B\text{-Mod}} (0) \oplus N$$

where M is a simple A-module and N is a simple B-module.

***Proof*:**
By proposition 1.2.11, $P \cong_{A \times B\text{-Mod}} M \oplus N$ where M is an A-module and N is a B-module. BUt then:

$$P \cong_{A \times B\text{-Mod}} (M \oplus (0)) \oplus ((0) \oplus N)$$

since P is simple, one of these has to be zero, and the other one has to be simple.

Corollary 1.2.13: Product Of Semi-Simple Rings

Let R and S be semi-simple rings. Then $R \times S$ is a semi-simple ring. Furtheromre, if $R_1, R_2, ..., R_n$ are the simple R-module up to isomorphism and $S_1, S_2, ..., S_m$ are the simple S-modules, then the simple $R \times S$-modules are of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

***Proof*:**
By proposition 1.2.11, since R is a semi-simple R-module and S is a semi-simple S-module, $R \times S$ is a semi-simple $(R \times S)$-module, and so $R \times S$ is a semi-simple ring. Then by proposition 1.2.12,

if $P \subseteq R \times S$ is a simple $(R \times S)$-module, then they it must be of the form:

$$R_i \oplus (0) \qquad (0) \oplus (S_j)$$

Corollary 1.2.14: Structure Of $\prod_i M_{N_i}(D_i)$: Artin Wedderburn part II

Let $R = M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. Then

1. R is semi-simple
2. There exist exactly k simple R-modules up to isomorphism, each of the form:

$$0 \oplus \cdots \oplus D_i^{\oplus n_i} \oplus \cdots \oplus 0$$

***Proof*:**
The ring R is semi-simpleby corollary 1.2.13 using induction. Furthermore, each $M_{n_i}(D_i)$ has $D_i^{\oplus n_i}$ is it's unique simple $M_{n_i}(D_i)$-module, since there are k matrix rings in the productt, there must be exactly k simple R-modules, as we sought to show.

1.2.15 Remark We can as before get that

$$R^{\mathrm{op}} \cong M_{n_1}(D_1^{\mathrm{op}}) \times \cdots \times M_{n_k}(D_k^{\mathrm{op}})$$

and hence R^{op} is left Noetherian/Artinian, which we know now is true if and only if R is right Noetherian/Artinian (Since a left R^{op})-module is a right R-module). By leftNoethArtRemark, we have that any semisimple ring is left-Noetherian/Artinian, and hence any semisimple ring is both left and right Noetherian/Artinian, even though R is not necesarily commutative!

Thus, we have the following statemnt, were the last thing we need to prove is uniquness

Theorem 1.2.16: Artin-Wedderburn

Let R be a ring. Then R is a semi-simple ring if and only if

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

where each D_k is a division ring and $M_{n_i}(D_i)$ is the matrix ring over D_i, and each D_i n_i are unique up to isomorphism

***Proof*:**
All that is left to prove is the uniqueness. Let:

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_s}(D_s) \cong_{\mathbf{Ring}} R \cong_{\mathbf{Ring}} M_{n'_1}(D'_1) \oplus \cdots \oplus M_{n_s}(D'_s)$$

By corollary 1.2.14, R has exactly s and t simple module, and so $s = t$. Hence, for any $D_i^{\oplus n_i}$, there must exist a $(D'_i)^{\oplus n'_i}$ such that

$$D_i^{\oplus n_i} \cong_R (D'_i)^{\oplus n'_i}$$

Since $D_i^{\oplus n_i} \cong_R (D_i')^{\oplus n_i'}$, $\mathrm{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i'}(D_i')}((D_i')^{\oplus n_i'})$. By corollary 1.2.8,

$$D_i^{\mathrm{op}} \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i}(D_i)}(D_i^{\oplus n_i}) \cong_{\mathbf{Ring}} \mathrm{End}_{M_{n_i'}(D_i')}((D_i')^{\oplus n_i'}) \cong_{\mathbf{Ring}} (D_i')^{\mathrm{op}}$$

which, taking op on both sides, we see that:

$$D_i \cong_{\mathbf{Ring}} D_i'$$

Finally, we need to show that $D^{\oplus n} \cong_R D^{\oplus m}$ implies $n = m$. Treating $D^{\oplus n}$ and $D^{\oplus m}$ as D-modules, then we get that each have a basis of dimension n and m. Importantly, the results of the Replacement Theorem (**??**) applies to divions rings (notice that commutativity was not necessary), and so **??** and **??** give us that $n = m$, completing the proof.

This theorem also shows us that if R is a [finitely-generated] semi-simple ring, then R is indeed the direct product of simple rings (since $M_{n_i}(D_i)$ is a simple ring). Notice too that this theorem just gave use the answer to the classification of all *finite simple rings*: namely they would have ot be of the form $M_n(\mathbb{F}_q)$[9] for some finite field of order q (which is a power of some prime p). Contrast this to how hard it is to find all finite simple groups! Furthermore, the finite dimensional assumption was key: the *Weyl Algebra*:

$$W = k\langle x, y\rangle/(xy - yx - 1)$$

is a simple algebra that is not the matrix ring of a division algebra.

Example 1.2.17: Artin Wedderburn

$\mathrm{End}_{M_2(k)}(M_2(k)) = \mathrm{End}_{M_2(k)}((k^2)^{\oplus 2}) = M_2(\mathrm{End}_{M_2(k)}(k^2)) = M_2(k)$

Corollary 1.2.18: Artin-Wedderburn For k-Algebras

Let R be a semi-simple k-algebra for some field k. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

where each D_i is a divsion k-algebra, finite dimensional as a k-vector space:

$$\dim_k R = \sum n_i^2 \dim_k(D_i)$$

and has exactly l simple R-modules as we've classified before.

***Proof*:**
By the Artin-Wedderburn theorem, we have that

$$R \cong_{\mathbf{Ring}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Let's now further assume R is a k-algebra for some field k, so that there is a map $k \hookrightarrow Z(R)$.

[9]Why is the division ring a field? The answer lies in **??**

Then by the above isomorphism:

$$\begin{aligned} k \mapsto Z(R) &\cong Z(M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)) \\ &\cong Z(M_{n_1}(D_1)) \times \cdots \times Z(M_{n_l}(D_l)) \\ &\cong Z(D_1) \times \cdots \times Z(D_l) \end{aligned}$$

where the last one comes form the fact that $Z(M_{n_i}(D_i))$ are the diagonal elements, and $z \mapsto zI$. This shows that D_i each have a natural k-algebra strucure induced by R, and so the above isomrophsim was already a k-algebra isomorphism:

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(D_1) \times \cdots \times M_{n_l}(D_l)$$

Finally, treating $M_{n_i}(D_i)$ as k-vector spaces we have:

$$M_{n_i}(D_i) \cong_{k\text{-}\mathbf{Vect}} D^{\oplus n_i^2}$$

which gives the desired result, completing the proof.

Corollary 1.2.19: Special Case: k Algebraically Closed

Let $k = \overline{k}$ be an algebraically clsoed field and D a k-algebra that is finite-dimensional as a k-vector-space. Then $D = k$

***Proof*:**
Let's say $k \subsetneq D$. Then pick any $\delta \in D \setminus k$. Then

$$k \subseteq k[\delta] \subseteq D$$

the key thing to observe is that $k[\delta]$ is still commutative!! This is essentially the same reasoning for why any cyclic group is commutative! Since D is a finite-dimensional vector-sapce over k, some power of δ^n must be a k-linear combination of every smaller term. Since D is a division ring, every δ has an inverse, and so every element of $k[\delta]$ is in fact a unit. Hence $k[\delta]/k$ is an algebraic field extension. However, k is an *algebraically cloased field*, and hence $k[\delta] = k$. Since δ was arbitary, we must have that $D = k$, as we sought to show.

Corollary 1.2.20: Artin-Wedderburn For $\overline{k}$-Algebras

Let R be a semi-simple k-algebra for some algebraically closed field $k = \overline{k}$. Then

$$R \cong_{k\text{-}\mathbf{Alg}} M_{n_1}(k) \times \cdots \times M_{n_l}(k)$$

showing that R is a finite dimensional vector-space over R with dimension:

$$\dim_k(R) = \sum_i^l n_i^2$$

and has exactly l simple R-modules as we've classified before.

As a consequence of this, if R is a semi-simple $\mathbb{C}$-algebra, the simple R-modules are all R-isomorphic to $\mathbb{C}^{n_i}$ for appropriate power n_i. As a final word of warning, it is natural to wonder if all simple rings are of the form $M_n(D_i)$

Exercise 1.2.1

1. Let k be a field, k^n be a direct product of fields a k^n-module over itslef. Let $0 \oplus \cdots \oplus k \oplus \cdots \oplus 0 = k_i \subseteq k^n$. Then $\mathrm{End}_{k^n}(k_i) \cong_{\mathbf{Ring}} k$, $\mathrm{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} k^n$, and $\mathrm{End}_{k^n}(k^n) \cong_{\mathbf{Ring}} M_n(\mathrm{End}_{k^n}(k_i)) \cong_{\mathbf{Ring}} M_n(k)$. But that would imply $k^n \cong_{\mathbf{Ring}} M_n(k)$ – what's the contradiction?

2. Let k be an algebraically closed field, and let R be a finite-dimensional k-algebra. Let M be a simple R-module. Prove that if r is an element of the centre $Z(R)$ of R, there exists $c \in k$ such that $rm = cm$ for all $m \in M$

1.3 Central Simple Algebras

Definition 1.3.1: Algebra Representation

Let A be a finite dimensional (associative) algebra over k. Then a *representation* of A is a k-homomorphism:

$$\rho : A \to \mathrm{End}_k(V)$$

for some vector space V.

Recall that one definition of an R-module is given by a ring homomorphism $\rho : R \to \mathrm{End}_{\mathbf{Ab}}(M)$ for an abelian group M. If we restrict to k-homomorphisms and k-endomorphisms, we can define the A-module $\rho : A \to \mathrm{End}_k(V)$, hence we can think of representation of and algebra A as a left A-modules. This makes sense, as representation theory can be thought of the study of associative algebras (the study of group representation is the study of group algebras, the study of the representation of lie algebras is universal enveloping algebras, the study of the representation of quiver algebras is the study of path algebras, and so forth). Because of this, we shall see that many of the results for the representation of associative algebras outlined bellow to be familiar to anyone who has taken a basic course in representation theory.

Next, we define the object that shall represent the generalization of number rings:

Definition 1.3.2: Central Simple Algebra

Let A be an algebra over a field k. Then A is called *simple* if it is a simple ring. It is called a *central simple algebra* (CSA) if it is simple, is finite dimensional over k, and its center is k.

The idea behind this definition is that it generalizes field extensions F/k, for we have a distinguished field k, and we are interested in the "algebraic case", hence the extension ought to be finite. Contrasting to fields, a CSA need not be commutative and need be a division ring

Example 1.3.3: Simple and Central Simple Algebras

1. Naturally, $M_n(k)$ is a central simple algebra of dimension n^2
2. It is possible to be simple and have center k but not be finite: the Weyl algebra $k[x, \delta_x]$ is infinite dimensional over k.
3. $\mathbb{C}$ is a central simple algebra over $\mathbb{C}$, but not over $\mathbb{R}$ (as it's center contains more than $\mathbb{R}$
4. Let $A = \mathbb{H}$ be the quaternions (or Hamiltonians). Then this is a CSA over $\mathbb{R}$ with dimension 4. We shall soon define a quaternion algebra, and see that all 4-dimensional central simples algebra's over a field k is a quaternion algebra, and it shall be limited to 2-by-2 matrices or division algebras.

Let us now generalize some of basic representation theory results. Let E be an irreducible representation of A (take for example a minimal left ideal of A).

Lemma 1.3.4: Schur's Lemma - updated

Let E be an irreducible representation of a k-algebra A, and let $K \subseteq \mathrm{End}_k(E)$ be the endomorphisms that commute with the operations of A: it is the centralizer of A in $\mathrm{End}_k(E)$, $K = C_{\mathrm{End}_k(E)}(A)$. Then K is a division ring.

Note that this makes K an A-algebra

***Proof*:**
Take $u \in K$. We want to show that $u^{-1} \in K$. Consider $u^{-1}(0) = V \subseteq E$. For $x \in V$, we have:

$$0 = a \cdot u(x) = u(a \cdot x)$$

hence, V is stable under A, and as $u \neq 0$ we must have $V = 0$ as u is invertible in $\mathrm{End}_k(E)$. It follows that $u^{-1} \in K$.

Notice from this that we may say that the representation E must be faithful (the map must be injective). Indeed, the kernel of this representation would be an ideal of A but

Lemma 1.3.5: Minimal Centralizer of Representations

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Then

$$A = B$$

***Proof*:**
Let $x \in E$ and $b \in B$. We shall show that there exists an $a \in A$ such that $a(x) = b(x)$.

Let $Ax = V \subseteq E$ be a stable subspace. Then V admits a (stable) compliment: $E = V \oplus V^{\perp}$. Let $\pi : E \to V$ be the natural projection map. Then certainly π is A-linear and commutes with K, namely $\pi \in K$. Then:

$$\pi(b(x)) = b(\pi(x)) = b(x)$$

namely, $b(x) \in V = aX$, implying $a(x) = b(x)$, as we sought to show.

Lemma 1.3.6: Generator lemma

Let E be a semi-simple an faithful representation of A, $K = C_{\mathrm{End}_k(E)}(A)$, $B = C_{\mathrm{End}_k(E)}(K)$ (i.e. the centralizer of A and K in $\mathrm{End}_k(E)$). Let $x_1, ..., x_n \in E$ and $b \in B$. Then there exits values $a_i \in A$ such that:

$$ax_1 = bx_1, \; ax_2 = bx_2 \qquad \cdots \qquad ax_n = bx_n$$

***Proof*:**
Let F be the direct sum space of n copies of representation E. Then F is formed of elements $(y_1, ..., y_n)$, $y_i \in E$, on which A acts by

$$a \cdot (y_1, ..., y_n) = (ay_1, ..., ay_n)$$

The representation F is still semi-simple. The centralizer of A in $\mathrm{End}_k(F)$ is formed of matrices (u_{ij}), $u_{ij} \in K$, and it follows that, if we let B operate on F by the formula:

$$b \cdot (y_1, ..., y_n) = (by_1, ..., by_n)$$

the elements of $\mathrm{End}_k(F)$ thus defined will still be in the centralizer of the centralizer of A (in the theory of semi-groups, this is sometimes called the bicommutant). By lemma 1.3.5 applying to the point $x = (x_1, ..., x_n) \in F$, and to $b \in B$ giving us the desired result.

Serre write K_n for $M_n(K)$ for a division ring K, but we shall stick to $M_n(D)$ where our division ring is D instead of K. If D contains k in its center, we have:

$$M_n(D) = M_n(k) \otimes D$$

as can be easily seen.

We present one more important result that has two useful corollaries. Let A be a k-algebra. Then A acting on itself by left multiplication gives something called the *regular left representation.* If

$A = M_n(D)$, it has basis e_{ij}, $1 \leq i, j \leq n$. The elements e_{i,j_0} for a fixed j_0 gives a subspace of $M_n(D)$, say $N_{j_0} \subseteq M_n(D)$. It is easy to see that N_{j_0} is a minimal left ideal (namely, an irreducible representation). Then we see that for any other choice of j_0, say j_1, we get an isomorphic irreducible representation N_{j_1}. Then this gives the following classical result:

Theorem 1.3.7: Representation of Simple Algebra

Let A be a simple k-algebra. Then all representation of A is isomorphic to the direct sum of irreducible representation

***Proof*:**
use corollary 1.2.20.

Corollary 1.3.8: Representation of Simple Algebra

Let A be a simple k-algebra. Then:

1. All representations of A are semi-simple (completely reducible in Serre's words)
2. Two representation of A which have the same dimension are isomorphic

***Proof*:**
immediate from the above

Let us now turn towards the tensor product of simple algebras.

Lemma 1.3.9: Centralizer of Tensor Product of Algebras

Let A, A' be two k-algebras, B, B' be sub k-algebras of A, A' (respectively) with C, C' their centralizers on A, A' (respectively). Then the centralizer of $B \otimes B'$ in $A \otimes A'$ is $C \otimes C'$.

***Proof*:**
Let us first find the centralizer of $B \otimes 1$. For this, let a'_i be a basis of A'. Any $x \in A \otimes A'$ can be written uniquely in the form:

$$x = \sum_i a_i \otimes a'_i \qquad \text{for appropriate } a_i \in A$$

If x commutes with $b \otimes 1$, then $a_i b = b a_i$ that is $a_i \in C$ for all i. We thus see that the centralizer of $B \otimes 1$ is $C \otimes A'$. Similarly, that of $1 \otimes B'$ is $A \otimes C'$. It follows that the centralizer of $B \otimes B'$ is equal to the intersection of the centralizers of $B \otimes 1$ and $1 \otimes B'$, namely

$$(C \otimes A') \cap (A \otimes C') = C \otimes C'$$

as we sought to show.

Thus, the centre of the tensor product of two algebras is the tensor product of the centre of the two algebras. Applying this to the special case of $A = M_n(k) \otimes D$ for some division ring D, then:

$$Z(A) = k \otimes Z(D)$$

Corollary 1.3.10: Centre of simple algebra is a Field

Let A be a simple k-algebra, finitely generated over k. Then $Z(A)$ is a field.

***Proof*:**
Consider $Z(A)$. As A is a finitely generated k-algebra, $Z(A)$ is a finitely generated k-algebra. For any $0 \neq z \in Z(A)$, consider zA. As z is central,we have that zA is two-sided ideal. As A is simple, it must be that $zA = A$, that is z is there exists a $w \in A$ such that $za = 1$. Therefore, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field. Since z was an arbitrary nonzero element, $Z(A)$ is a division ring. It is by definition commutative, and hence it is a field.

We shall next require to see the interaction between the tensor product of central simple algebras. We require the following lemma:

Theorem 1.3.11: Ideal Generation Tensor Product

Let A be a k-algebra and K' a left division ring with center k. Then every ideal $N \subseteq A \otimes K'$ is generated (as a left module over K') by its intersection with $A \otimes 1$.

Note that neither A nor K' are assumed finite over k

***Proof*:**
We will first note that we can make K' operate on the left on $A \otimes K'$ by:

$$k'(a \otimes k'') = a \otimes k'k''$$

If N is an ideal of the algebra $A \otimes K'$, it is in particular a K'-module subspace of $A \otimes K'$.

Let a_i be a basis of A over k; the elements $a_i \otimes 1$ equally form a basis of the K'-module $A \otimes K'$. Let x be a primitive element of N with respect to this basis (Bourbaki, Alg.II, paragraph 5):

$$x = \sum_i a_i \otimes k'_i$$

Consider the element $x \cdot m$ ($m \in K'$); we have $x \cdot m \in N$ (N is an ideal), and on the other hand:

$$x \cdot m = \sum_i a_i \otimes k'_i m$$

It then follows from the properties of primitive elements that we have:

$$k'_i m = nk'_i \text{ for all } i \quad (n \in K')$$

Since one of the k'_i is equal to 1, we have $m = n$, and the preceding equality simply means that k'_i commutes with all $m \in K'$, i.e., that $k'_i \in k$. We can then write: $x = a \otimes 1$, by setting: $a = \sum_i k'_i a_i$. We have thus shown that every primitive element x is in $A \otimes 1$; the theorem follows immediately since every vector space is generated by its primitive elements.

Theorem 1.3.12: Tensor Product CSA

Let A, A' be simple k-algebras finitely generated over k, where one of them is a CSA. Then $A \otimes A'$ is a simple k-algebra

***Proof*:**
Suppose that $A' = M_n(k) \otimes K'$ is central, in other words that the center of the left field K' is reduced to k. It will suffice to show that $A \otimes K'$ is a simple algebra. Indeed, if this is demonstrated, $A \otimes K' = M_m(k) \otimes K''$, and $A \otimes A' = M_m(k) \otimes M_n(k) \otimes K'' = M_{mn}(k) \otimes K'' = M_{nm}(K'')$. The algebra $A \otimes A'$ will thus be isomorphic to a matrix algebra for which simplicity is immediately verified. Then the general case of $A \otimes K'$ being simple follow from theorem 1.3.11, completing the proof.

Corollary 1.3.13: Tensor Product of Two CSA's

The tensor product of two CSA is a CSA

***Proof*:**
In the previous proof, it was enough for one of them to be a CSA.

Corollary 1.3.14: Opposite Algebra Tensoring

Let A be a CSA over k and A^{op} the opposite algebar. Then:

$$A \otimes A^{\mathrm{op}} \cong M_{n^2}(k)$$

***Proof*:**
Let us denote by E the algebra A considered as a vector space over k. We can make A act naturally on E by the left or right by multiplication to create a left or right A-module. This amounts to identifying A and A^{op} as sub-algebras of the algebra $\mathrm{End}_k(E)$ of k-endomorphisms of E. Since the algebras A and A^{op} commute in $\mathrm{End}_k(E)$, we can define a canonical homomorphism of $A \otimes A^{\mathrm{op}}$ into $\mathrm{End}_k(E)$ which extends the isomorphisms: $A \to \mathrm{End}_k(E)$ and $A^{\mathrm{op}} \to \mathrm{End}_k(E)$. The algebra $A \otimes A^{\mathrm{op}}$ being simple, this homomorphism is an isomorphism; moreover, as;

$$[A \otimes A^{\mathrm{op}} : k] = [A : k]^2 = n^2 = [\mathrm{End}_k(E) : k]$$

this isomorphism maps $A \otimes A^{\mathrm{op}}$ onto $\mathrm{End}_k(E) = M_{n^2}(k)$, as we sought to show.

1.3.1 Brauer Group

Let A be a CSA over k. As A is simple, we get by Artin-Wedderburn that it is isomorphic to $M_n(D) \cong M_n(k) \otimes D$. For example, if $n = 1$ and $k = \mathbb{R}$ we get (by a classical theorem from algebraic topology) that $A \cong \mathbb{R}$ or $A \cong \mathbb{H}$ (the quaternions). If $n > 1$, we get $M_n(\mathbb{R})$ and $M_n(\mathbb{H})$ (which are no longer division algebras). We see that the division ring is one of the "characterizing" feature of the CSA (the other being its matrix ring), especially that up to Morita-equivalence we get the same category. We thus have the following natural equivalence we may define:

Definition 1.3.15: Brauer Equivalence

Let k-**CSA** be a collection of central simple algebras over k. Then define the equivalence relation $A \sim A'$ where the two are equivalent if and only if their associated division rings are isomorphic.

Proposition 1.3.16: Brauer Group

The Brauer equivalence on a collection of CSA's forms a commutative group under the tensor product. This group will be denoted $\mathrm{Br}(k)$.

In Serre, this group is denoted G_k.

***Proof*:**
By corollary 1.3.13, it is closed under the tensor product, the equivalence class containing k is certainly the identity, and corollary 1.3.14 shows that every equivalence class has an inverse. Associativity and commutativity is a classical result for tensor products.

Example 1.3.17: Brauer Groups

1. Let $k = \overline{k}$. Then $\mathrm{Br}(\overline{k}) = 0$ by corollary 1.2.20.
2. If $k = \mathbb{R}$, then we shall show in the next section that $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$, namely we shall have $\mathbb{R}$ and $\mathbb{H}$, the quaternions.
3. Let k be a finite field. Then $\mathrm{Br}(k) = 0$, as all finite division rings are fields.
4. If $k = \mathbb{Q}_p$, then $\mathrm{Br}(k) \cong \mathbb{Q}/\mathbb{Z}$ (Serre references Algebren - Chap. VII - 2nd paragarph).
5. If $k = \mathbb{Q}$, $\mathrm{Br}(\mathbb{Q})$ will be a subgroup of
$$\mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Q}/\mathbb{Z})^{\oplus \mathbb{N}}$$
which can be thought of as a result similar to that in number theory where we study the "global field" $\mathbb{Q}$ by looking at all its localizations (note that $\mathbb{Z}/2\mathbb{Z}$ comes from $\mathrm{Br}(\mathbb{R})$!) Serre references Deuring for the proof (see Sere p.8)

Theorem 1.3.18: Field Extensions and CSA

Let A be a CSA over k, L/k a field extension (either finite or infinite). Then $A \otimes L$ is simple

***Proof*:**
If A is a division ring, then all ideals of $A \otimes L$ will be given with in part with the intersection of the ideals of L, which is simply (0), making $A \otimes L$ simple.

If A is any CSA, write $A \cong M_n(k) \otimes D$. Then:

$$A \otimes L \cong M_n(k) \otimes (D \otimes L) \stackrel{!}{=} M_n(k) \otimes M_m(D')$$

where for $\stackrel{!}{=}$, D' is the division ring naturally given by $D \otimes_k L$! But this gives the result.

Corollary 1.3.19: Index of CSA

Let A be a simple, central and finite algebra over k. Then $[A : k]$ is a perfect square.

***Proof*:**
Let L be an algebraically closed extension of k. The algebra $A \otimes L$ can be endowed with an algebra structure over L, and we have:

$$[A \otimes L : L] = [A : k]$$

But $A \otimes L$ is a simple algebra, central over L, and finite over L. It then follows from example 1.3.17[1] that it is a matrix algebra over L, and $[A \otimes L : L]$ is indeed a perfect square.

Looking at the field extension of a k-algebra, it is natural to consider that $-\otimes_k L$ gives a homomorphism of Brauer groups $\varphi_{k,L} : \mathrm{Br}(k) \to \mathrm{Br}(L)$, namely due to the following result from linear algebra:

$$(A \otimes_k L) \otimes_L (A' \otimes_k L) \cong (A \otimes_k A') \otimes_k L$$

The kernel $\ker \varphi_{k,L}$ consists of all k-algebra's that become trivial (matrix algebras) when extending to L. Serre calls these algebras the algebras that *admit* L *as a field of decomposition.* (or admit a decomposition by L). More recently, the term *splitting field* is adopted due to the parallels to number theory. Naturally, if $L = \overline{k}$, then every algebra admits a decomposition as $\mathrm{Br}(\overline{k}) = 0$. We only require a finite extension:

Theorem 1.3.20: Splitting Field is Finite

The Brauer group $\mathrm{Br}(k)$ is the union of $\ker \varphi_{k,L}$ going over each L/k where L is a finite field extension. In particular, there exists a finite extension L/k such that for a CSA A over k, $[A \otimes_k L] = [1]$ in $\mathrm{Br}(L)$.

***Proof*:**
By example 1.3.17[1] we have that for a CSA A over $\overline{k}$, $A \otimes_k \overline{k} \cong M_n(D)$. Then the right hand side is generated by finitely many elements of the left hand side. We need only take the elements that map to the generators of the right hand side and to choose the appropriate field elements to get find L/k, as we sought to show.

1.4 Build up to Quaternion Algebras

Theorem 1.4.1: Skolem-Noether Theorem

Let A, B be CSA's over k, and let $f, g : A \to B$ be k-algebra isomorphism. Then there exists a unit $b \in B$ such that:
$$g(a) = bf(a)b^{-1}$$
In particular, every automorphism is an inner automorphism

***Proof*:**
Let's first prove the special case of $B = M_n(k)$. Then the isomorphisms $f, g : A \to M_n(k)$ define actions on k^n, or said differently these define two A-modules (with two different actions): V_f, V_g. Then using f, g we have an A-module isomorphism $g \circ f^{-1} : V_f \to V_g$. But any isomorphism between vector-spaces is representable as (unit) matrix in $M_n(k)$, and so we we get our value by linear algebra.

For the general case, we know that $B \otimes_k B^{\mathrm{op}} \cong M_n(k)$ and $A \otimes_k B^{\mathrm{op}}$ is simple as B^{op} is still simple (use theorem 1.3.12). As $B \otimes_k^B$ op is a (finite dimensional) matrix algebra, we may apply our special case to:
$$f \otimes 1,\ g \otimes 1 : A \otimes_k B^{\mathrm{op}} \to B \otimes_k B^{\mathrm{op}} \cong M_n(k)$$
which shows that:
$$(f \otimes 1)(a \otimes c) = b(g \otimes 1(a \otimes c)b^{-1}$$
which is true for all $a \in A$ and $c \in B^{\mathrm{op}}$. Thus, it is true for $a = 1$, giving us:
$$1 \otimes c = b(1 \otimes c)b^{-1}$$
for all $c \in B^{\mathrm{op}}$. Thus,
$$b \in Z_{B \otimes_k B^{\mathrm{op}}}(k \otimes B^{\mathrm{op}}) = B \otimes k$$
Hence, as constants move through the tensor we get that $b = b \otimes_k 1$. And so, by setting $z = 1$, we find:
$$f(a) = b' g(a) b'^{-1}$$
as we sought to show.

Theorem 1.4.2: Centralizer of Subalgebras

Let A be a CSA over k, $B \subseteq A$ a simple subalgebra, $C = C_A(B)$ the centralizer of B in A. Then C is simple, $B = C_A(C)$, and:
$$[B : k][C : k] = [A : k]$$

***Proof*:**
Let E be B with only the structure of a k-vector space. Then B acts on E via left multiplication; this action isomorphically embeds B into $\mathrm{End}_k(E)$. Then the centralizer of B in $\mathrm{End}_k(E)$ is nothing more than the algebra of right translations by E, that is B^{op}.

Given this, we can consider the simple algebra $A \otimes_k \mathrm{End}_k(E)$. We can now embed B into this tensored algebra in two ways: via $B \otimes 1$ or $1 \otimes B$. Their centralizers are $C \otimes \mathrm{End}_k(E)$ and $A \otimes B^{\mathrm{op}}$ respectively.

By theorem 1.4.1, there exists an inner automorphism of $A \otimes_k \mathrm{End}_k(E)$ mapping $B \otimes 1$ to $1 \otimes B$; it is then a simple algebra exercise to see that this inner automorphism induces a map on the centralizers. As a consequence, as $A \otimes B^{\mathrm{op}}$ is simple, so is $C \otimes \mathrm{End}_k(E)$, which must make C simple.

Furthermore:

$$[C \otimes \mathrm{End}_k(E) : k] = [C : k][B : k]^2 \qquad \text{and} \qquad [A \otimes B^{\mathrm{op}} : k][A : k][B : k]$$

From which we get $[A : k] = [B : k][C : k]$. What's left to show is the centralizer of C in A is B. Say that B' is this centralizer. Then $B \subseteq B'$. Then as $[A : k] = [B : k][C : k]$, we get:

$$[B' : k] = [B : k]$$

But then as they contain we must get $B = B'$, completing the proof.

2

The Jacobson Radical and Non-Semisimple Algebras

The Artin-Wedderburn theorem (theorem 1.2.16) is a complete answer to a narrow question: it classifies the semisimple rings and, in the process, shows that a semisimple k-algebra is nothing more than a finite product of matrix rings over division algebras. When R is semisimple, every module splits into simples, every short exact sequence splits, and the entire representation theory of R collapses onto the finite list of its simple modules. This is the best possible situation, and it is also the exception. The algebras whose representation theory occupies most of this book are precisely the ones that fail to be semisimple: the group algebra $k[G]$ when $\mathrm{char}(k)$ divides $|G|$ (where Maschke's theorem breaks down), the truncated polynomial algebra $k[x]/(x^n)$, universal enveloping algebras of Lie algebras, and the path algebra of a quiver with an oriented cycle. For all of these, some module fails to be a direct sum of simples, some sequence refuses to split, and Wedderburn has nothing to say.

This chapter builds the tools that take over where Wedderburn stops. The first of these, and the subject of the present section, is the *Jacobson radical* $\mathrm{Jac}(A)$: the two-sided ideal that measures, in a single invariant, exactly how far A is from being semisimple. It is the obstruction that vanishes precisely when A is semisimple, and dividing it out always produces a semisimple ring $A/\mathrm{Jac}(A)$ whose simple modules are the same as those of A. In the sections that follow, the radical becomes the engine behind the theory of indecomposable modules and the Krull-Schmidt theorem (which replaces "decomposition into simples" with "decomposition into indecomposables"), and behind the construction of projective covers, which repair the failure of projectivity that semisimplicity had guaranteed for free. The radical is where the non-semisimple story begins.

2.1 The Jacobson Radical

Throughout chapter 1 the simple modules did all the work: a semisimple ring was assembled out of its minimal left ideals, and every module was a direct sum of simples. When a ring is *not* semisimple, the simple modules are still present, but they no longer see everything. Some elements of the ring act as zero on every simple module without being zero themselves, and these elements form an ideal that records the discrepancy. To isolate them, we begin one level down, with modules, and ask which part of a module is invisible to all of its simple quotients.

The maximal submodules of a module M are exactly the kernels of the surjections $M \twoheadrightarrow S$ onto simple modules: a submodule $N \subsetneq M$ is maximal precisely when M/N is simple, and every simple quotient of M is of the form M/N for such an N (the classification of simples as quotients by maximal ideals is lemma 1.1.3). An element lying in every maximal submodule is one that every simple quotient sends to zero. Collecting these gives the radical of a module.

Definition 2.1.1: Radical of a Module

Let M be a left A-module. The *radical* of M, written $\operatorname{rad}(M)$, is the intersection of all maximal submodules of M:

$$\operatorname{rad}(M) = \bigcap_{\substack{N \subsetneq M \\ N \text{ maximal}}} N$$

If M has no maximal submodules, the intersection is empty and $\operatorname{rad}(M) := M$ by convention.

Since each maximal submodule is the kernel of a map onto a simple module, an equivalent description is

$$\operatorname{rad}(M) = \bigcap_{\substack{\varphi\colon M \to S \\ S \text{ simple}}} \ker(\varphi)$$

so that $\operatorname{rad}(M)$ is the set of elements killed by *every* homomorphism from M into a simple module. The quotient $M/\operatorname{rad}(M)$ is the largest quotient of M that embeds into a product of simple modules; it is the *semisimple quotient* of M, the part of M that its simple quotients can detect. Everything inside $\operatorname{rad}(M)$ is invisible to them.

Example 2.1.2: The Radical of a Cyclic $\mathbb{Z}$-Module

Take $A = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The maximal submodules of M correspond to the maximal ideals of $\mathbb{Z}$ containing $12\mathbb{Z}$, that is to the primes dividing 12, namely 2 and 3. The corresponding maximal submodules are $2\mathbb{Z}/12\mathbb{Z}$ and $3\mathbb{Z}/12\mathbb{Z}$, with simple quotients $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$. Their intersection is

$$\operatorname{rad}(\mathbb{Z}/12\mathbb{Z}) = \frac{2\mathbb{Z}}{12\mathbb{Z}} \cap \frac{3\mathbb{Z}}{12\mathbb{Z}} = \frac{6\mathbb{Z}}{12\mathbb{Z}}$$

so $\operatorname{rad}(M) = 6\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$, and the semisimple quotient is $M/\operatorname{rad}(M) \cong \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. The radical has stripped $\mathbb{Z}/12\mathbb{Z}$ down to its squarefree quotient: the extra factors of 2 and 3 that no simple quotient can see are exactly what lies in $6\mathbb{Z}/12\mathbb{Z}$.

Applying this construction to the ring itself, viewed as a module over itself, produces the central object of the chapter. The radical of A as a left A-module is the intersection of its maximal *left* ideals.

Definition 2.1.3: Jacobson Radical

Let A be a ring. The *Jacobson radical* of A, written $\mathrm{Jac}(A)$, is the radical of A regarded as a left module over itself, that is the intersection of all maximal left ideals of A:

$$\mathrm{Jac}(A) = \mathrm{rad}({}_A A) = \bigcap_{\substack{\mathfrak{m} \subseteq A \\ \mathfrak{m} \text{ maximal left ideal}}} \mathfrak{m}$$

The definition is one-sided: it uses left ideals, and a priori $\mathrm{Jac}(A)$ need only be a left ideal. One of the first surprises is that the resulting set is symmetric, a two-sided ideal that could equally have been defined with maximal *right* ideals. This symmetry, and several other faces of the same object, are the content of the next theorem. Before stating it, recall that an element $u \in A$ is *left-invertible* if there exists $v \in A$ with $vu = 1$.

Theorem 2.1.4: Characterizations of the Jacobson Radical

Let A be a ring and let $a \in A$. The following subsets of A coincide, and their common value is $\mathrm{Jac}(A)$:

1. the intersection of all maximal left ideals of A;
2. the intersection of the annihilators $\mathrm{Ann}_A(S)$ of all simple left A-modules S;
3. the set $\{a \in A \big| 1 - xa \text{ is left-invertible for all } x \in A\}$.

Consequently, $\mathrm{Jac}(A)$ is a two-sided ideal of A, and it is the largest two-sided ideal of A that annihilates every simple left A-module.

The three descriptions look at the same ideal from three angles. The first is the definition. The second recasts it representation-theoretically: $\mathrm{Jac}(A)$ is exactly the set of elements that act as zero on every simple module, which is why it is invisible to the simple representation theory. The third is an intrinsic ring-theoretic criterion that never mentions modules at all, and it is the one that makes the two-sidedness fall out for free, since it is symmetric in a way the definition is not.

***Proof*:**
We prove (1) = (2), then (2) = (3), and finally read off the two-sidedness and maximality statements.

Step 1: (1) = (2). Let J_1 denote the intersection of all maximal left ideals and J_2 the intersection of the annihilators of all simple left modules. If S is a simple left A-module, then for any $0 \neq s \in S$ we have $S = As$ (as As is a nonzero submodule of the simple module S), and by corollary 1.1.4 the map $a \mapsto as$ gives $S \cong A/\mathrm{Ann}_A(s)$ with $\mathrm{Ann}_A(s)$ a maximal left ideal. The annihilator of the whole module is

$$\mathrm{Ann}_A(S) = \bigcap_{0 \neq s \in S} \mathrm{Ann}_A(s)$$

an intersection of maximal left ideals, so $J_1 \subseteq \mathrm{Ann}_A(S)$ for every simple S, giving $J_1 \subseteq J_2$. Conversely, if $\mathfrak{m}$ is any maximal left ideal, then $S = A/\mathfrak{m}$ is a simple left module and $\mathrm{Ann}_A(S) \subseteq \mathfrak{m}$ (since $\mathrm{Ann}_A(S)$ kills the coset $1 + \mathfrak{m}$, so $\mathrm{Ann}_A(S) \cdot 1 \subseteq \mathfrak{m}$). Hence $J_2 \subseteq \mathfrak{m}$ for every maximal left ideal $\mathfrak{m}$, giving $J_2 \subseteq J_1$. Thus $J_1 = J_2$.

Step 2: (2) $\subseteq$ (3). Let $a \in J_2$, so a annihilates every simple left module, and fix $x \in A$. We must show $1 - xa$ is left-invertible. If not, then $A(1 - xa)$ is a proper left ideal, hence contained in some maximal left ideal $\mathfrak{m}$ (using that $1 \notin A(1-xa)$ and Zorn's lemma). The quotient $S = A/\mathfrak{m}$ is simple, so a annihilates it: $a \cdot (1 + \mathfrak{m}) = 0$, meaning $a \in \mathfrak{m}$. But $xa \in \mathfrak{m}$ as well, since $\mathfrak{m}$ is a left ideal, and $1 - xa \in \mathfrak{m}$ since $1 - xa \in A(1 - xa) \subseteq \mathfrak{m}$. Adding, $1 = (1 - xa) + xa \in \mathfrak{m}$, contradicting that $\mathfrak{m}$ is proper. Hence $1 - xa$ is left-invertible for every x, so $a \in J_3$, the set in (3).

Step 3: (3) $\subseteq$ (2). Let $a \in J_3$, and suppose toward a contradiction that a does *not* annihilate some simple left module S. Then $as \neq 0$ for some $s \in S$, and since As is a nonzero submodule of the simple module S we have $As = S$. In particular $s \in As$, so $s = xas$ for some $x \in A$, giving $(1 - xa)s = 0$. By hypothesis $1 - xa$ is left-invertible, say $v(1 - xa) = 1$; applying v gives $s = v(1 - xa)s = 0$, contradicting the choice of s. Hence a annihilates every simple module, so $a \in J_2$.

Combining the three steps, $J_1 = J_2 = J_3 = \mathrm{Jac}(A)$.

Step 4: two-sidedness and maximality. As the intersection of the left ideals $\mathrm{Ann}_A(S)$, description (2) shows $\mathrm{Jac}(A)$ is a left ideal. But each $\mathrm{Ann}_A(S)$ is in fact a two-sided ideal: if $a \in \mathrm{Ann}_A(S)$ and $b \in A$, then for all $s \in S$ we have $(ab)s = a(bs) = 0$, so $ab \in \mathrm{Ann}_A(S)$. An intersection of two-sided ideals is two-sided, so $\mathrm{Jac}(A)$ is a two-sided ideal. Finally, if I is any two-sided ideal annihilating every simple left module, then $I \subseteq \mathrm{Ann}_A(S)$ for every simple S, hence $I \subseteq \bigcap_S \mathrm{Ann}_A(S) = \mathrm{Jac}(A)$; thus $\mathrm{Jac}(A)$ is the largest such ideal, completing the proof.

The left-invertibility criterion (3) deserves a second look, because it is what makes the radical robust. It says a lies in the radical exactly when $1 - xa$ can never be turned into a non-unit by any left multiplier x. A standard manipulation upgrades "left-invertible" to "two-sided invertible" for these particular elements: if $a \in \mathrm{Jac}(A)$ and $v(1 - xa) = 1$, then $v = 1 + vxa = 1 - (-vx)a$, and since $a \in \mathrm{Jac}(A)$ the element v is itself left-invertible by the same criterion; a left-invertible element with a left-invertible left inverse is a unit. So for $a \in \mathrm{Jac}(A)$ the element $1 - xa$ is a two-sided unit for every x, and by the right-handed version of the whole theorem $1 - ax$ is a unit as well. This is the working form of the radical: $\mathrm{Jac}(A)$ is the set of elements a such that $1 - xay$ is a unit for all $x, y \in A$, and it is exactly this that will drive Nakayama's lemma below.

Description (2) also settles the two-sided nature of the radical without any of the asymmetry of the original definition, and it makes precise the sense in which the radical is invisible to representation theory. If A is semisimple, every module is a sum of simples, so an element killing all simples kills everything: $\mathrm{Jac}(A) = 0$. The content of the non-semisimple theory is that $\mathrm{Jac}(A)$ can be nonzero, and it is this ideal that the simple modules fail to detect.

Before turning to that theory, the radical of a ring is anchored by two computations. The first is the archetype of a local non-semisimple algebra.

Example 2.1.5: The Radical of a Truncated Polynomial Algebra

Let k be a field and let $A = k[x]/(x^n)$ with $n \geq 1$, writing $\bar{x}$ for the image of x. This is a commutative local ring: an element $f(\bar{x}) = c_0 + c_1\bar{x} + \cdots + c_{n-1}\bar{x}^{n-1}$ is a unit if and only if $c_0 \neq 0$, since when $c_0 = 0$ the element is nilpotent ($\bar{x}^n = 0$ forces $f(\bar{x})^n = 0$), and when $c_0 \neq 0$ one inverts using the geometric series, which terminates because $\bar{x}^n = 0$:

$$(c_0 + \bar{x}g)^{-1} = c_0^{-1}\big(1 + c_0^{-1}\bar{x}g\big)^{-1} = c_0^{-1}\sum_{j=0}^{n-1}(-c_0^{-1}\bar{x}g)^j$$

The non-units are therefore exactly the elements with $c_0 = 0$, which is the ideal $(\bar{x})$. In a commutative ring the maximal ideals are the maximal left ideals, and here there is a single maximal ideal, namely $(\bar{x})$, because $A/(\bar{x}) \cong k$ is a field while any element outside $(\bar{x})$ is a unit. Hence

$$\mathrm{Jac}(A) = (\bar{x}) = \{c_1\bar{x} + c_2\bar{x}^2 + \cdots + c_{n-1}\bar{x}^{n-1} \big| c_i \in k\}$$

This ideal is nilpotent: $(\bar{x})^n = (\bar{x}^n) = 0$, so already $\mathrm{Jac}(A)^n = 0$. The quotient $A/\mathrm{Jac}(A) = A/(\bar{x}) \cong k$ is a field, hence semisimple, which is the general phenomenon established in theorem 2.1.9. The single simple A-module is $k = A/(\bar{x})$, and $\bar{x}$ acts on it as zero, confirming that $\mathrm{Jac}(A) = (\bar{x})$ annihilates every simple module.

The second computation is the model of a non-commutative, non-local radical, where the one-sided origin of the definition is easiest to see in matrices.

Example 2.1.6: The Radical of Upper-Triangular Matrices

Let k be a field and let $A = T_n(k)$ be the ring of $n \times n$ upper-triangular matrices over k. Write $N \subseteq A$ for the strictly upper-triangular matrices (zero on and below the diagonal). The claim is $\mathrm{Jac}(A) = N$.

N annihilates every simple module. For each $1 \le i \le n$ there is a one-dimensional A-module $S_i = k$ on which an upper-triangular matrix $a = (a_{pq})$ acts by its ith diagonal entry, $a \cdot v = a_{ii}v$; this is a ring homomorphism $A \to k$ because the (i,i) entry of a product of upper-triangular matrices is the product of the (i,i) entries. The S_i are pairwise non-isomorphic simple A-modules. Any strictly upper-triangular matrix has all diagonal entries zero, so it acts as 0 on each S_i; thus $N \subseteq \mathrm{Ann}_A(S_i)$ for all i, and $N \subseteq \mathrm{Jac}(A)$ by theorem 2.1.4.

Nothing outside N lies in the radical. If $a \in A \setminus N$, some diagonal entry a_{ii} is nonzero, so a acts invertibly on S_i, and a does not annihilate the simple module S_i. By part (2) of theorem 2.1.4, $a \notin \mathrm{Jac}(A)$. Combining, $\mathrm{Jac}(A) = N$.

As a check on the intrinsic criterion, N is nilpotent with $N^n = 0$ (each multiplication by a strictly upper-triangular matrix pushes nonzero entries one further diagonal above the main diagonal), so for $a \in N$ the element $1 - xa$ has the form $1 - (\text{nilpotent})$ and is a unit via the terminating geometric series $\sum_{j\ge 0}(xa)^j$. The quotient is $A/N \cong k^n$, the diagonal matrices, which is semisimple, and there are exactly n simple modules $S_1, \ldots, S_n$, matching the n factors of k^n.

The two examples share a feature: in each, $\mathrm{Jac}(A)$ turned out to be nilpotent, and the geometric series that inverts $1 - xa$ terminated precisely because a power of the radical vanished. This is not special to small examples. For the finite-dimensional and, more generally, left Artinian algebras that concern us, the radical is always nilpotent. The route to that fact runs through Nakayama's lemma, which converts the invertibility criterion into a statement about modules being forced to zero.

The lemma answers a question that recurs throughout module theory: if multiplying a module by the radical does not shrink it, must the module already be zero? The commutative version, over a local ring or with a general ideal contained in the radical, is proved in the companion volume 'NateCommAlg' [2]and will not be re-derived here; the module form over an arbitrary ring is the form needed here, and its proof uses only the invertibility criterion from theorem 2.1.4.

Lemma 2.1.7: Nakayama's Lemma

Let A be a ring and M a finitely generated left A-module.

1. If $\mathrm{Jac}(A)\,M = M$, then $M = 0$.
2. If $N \subseteq M$ is a submodule with $M = N + \mathrm{Jac}(A)\,M$, then $N = M$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

1. Suppose $M \neq 0$ and $JM = M$. Since M is finitely generated, choose a generating set $m_1, \ldots, m_r$ with r minimal, so no fewer than r elements generate M; as $M \neq 0$ we have $r \geq 1$. From $M = JM$ we may write the last generator as
$$m_r = \sum_{i=1}^{r} a_i m_i \qquad \text{with } a_i \in J$$
Isolating m_r gives $(1-a_r)m_r = \sum_{i=1}^{r-1} a_i m_i$. Because $a_r \in J = \mathrm{Jac}(A)$, the element $1-a_r$ is a unit (the working form of theorem 2.1.4: $1 - a_r$ is left-invertible, and for radical elements this upgrades to a two-sided inverse). Multiplying on the left by $(1 - a_r)^{-1}$ expresses m_r as an A-combination of $m_1, \ldots, m_{r-1}$, so these $r - 1$ elements already generate M, contradicting the minimality of r. Hence $M = 0$.
2. Apply part (1) to the finitely generated module M/N. Its finite generation is inherited from M (the images of generators of M generate the quotient). The hypothesis $M = N + JM$ gives, in the quotient, $M/N = J\,(M/N)$, since the image of JM is $J(M/N)$ and the image of N is 0. By part (1), $M/N = 0$, that is $N = M$, completing the proof.

The finite-generation hypothesis is not optional. Over the local ring $A = \mathbb{Z}_{(p)}$ (integers localized at p, whose radical is $p\mathbb{Z}_{(p)}$) the module $M = \mathbb{Q}$ satisfies $pM = M$ yet $M \neq 0$; it fails to be finitely generated, and the minimal-generator argument has nothing to grip. Part (2) is the form used in practice: it says that generators of $M/\mathrm{Jac}(A)M$ lift to generators of M. Once $\mathrm{Jac}(A)M$ is known (below) to be a "small" submodule, this becomes the statement that a minimal generating set of M is detected by the semisimple quotient $M/\mathrm{Jac}(A)M$, which is the seed of the theory of projective covers in a later section.

We now specialize to the setting where the radical is best behaved. A ring A is *left Artinian* if it satisfies the descending chain condition on left ideals; every finite-dimensional algebra over a field is left Artinian, since a descending chain of left ideals is a descending chain of subspaces and dimension cannot drop forever. In this setting the radical is not only an intersection of maximal ideals but a nilpotent ideal, and it is the largest one.

Theorem 2.1.8: The Radical Is Nilpotent for Artinian Rings

Let A be a left Artinian ring. Then $\mathrm{Jac}(A)$ is nilpotent: there exists $m \geq 1$ with $\mathrm{Jac}(A)^m = 0$. Moreover, $\mathrm{Jac}(A)$ contains every nilpotent left ideal and every nilpotent two-sided ideal of A, so it is the largest nilpotent one-sided or two-sided ideal.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: J is nilpotent. The descending chain of two-sided ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

is a descending chain of left ideals, so by the descending chain condition it stabilizes: there is m with $I := J^m = J^{m+1} = J\,I$. Suppose for contradiction that $I \neq 0$. Consider the set Σ of left ideals $L \subseteq A$ with $IL \neq 0$; it is nonempty because $I \cdot I = J^{2m} = I \neq 0$ shows $I \in \Sigma$ (using $J^{2m} = (J^m)^2 = I \cdot I$ and $J^{2m} = J^m = I$ since the chain stabilized). By the descending chain condition, Σ has a minimal element L_0. Choose $x \in L_0$ with $Ix \neq 0$; then $Ix \subseteq L_0$ is a left ideal with $I(Ix) = I^2x = Ix \neq 0$ (as $I^2 = I$), so $Ix \in \Sigma$, and minimality forces $Ix = L_0$. In particular $x \in L_0 = Ix$, so $x = ax$ for some $a \in I \subseteq J$. Then $(1-a)x = 0$, and since $a \in J$ the element $1-a$ is a unit, giving $x = 0$ and hence $Ix = 0$, a contradiction. Therefore $I = J^m = 0$.

Step 2: J contains every nilpotent one-sided or two-sided ideal. Let I be a nilpotent left ideal, say $I^t = 0$. For any $a \in I$ and any $x \in A$, the element xa lies in I (a left ideal absorbs left multiplication), so $(xa)^t \in I^t = 0$; thus xa is nilpotent. A nilpotent element y with $y^t = 0$ makes $1-y$ a unit, with inverse the terminating sum $1 + y + y^2 + \cdots + y^{t-1}$. Hence $1 - xa$ is a unit, in particular left-invertible, for every $x \in A$, so $a \in \mathrm{Jac}(A)$ by part (3) of theorem 2.1.4. This holds for every $a \in I$, so $I \subseteq J$. A nilpotent two-sided ideal is in particular a nilpotent left ideal, so the same conclusion applies, completing the proof.

For a left Artinian ring, then, "the Jacobson radical" and "the largest nilpotent ideal" name the same object; the two definitions of the radical current in the literature (intersection of maximal left ideals versus largest nilpotent ideal) agree exactly when A is Artinian. The nilpotence is what powers the terminating geometric series seen in the truncated-polynomial and upper-triangular examples, and it is the property that fails for general rings. It also gives a concrete way to detect non-semisimplicity: a single nonzero nilpotent element of a two-sided ideal already forces the radical to be nonzero. This is exactly the obstruction that appears for modular group algebras.

Having pinned down $\mathrm{Jac}(A)$ as a nilpotent ideal, the pieces are in place for the structural payoff. Dividing A by its radical removes precisely the elements that the simple modules could not see, and what remains is semisimple.

Theorem 2.1.9: Semisimplicity of the Quotient by the Radical

Let A be a left Artinian ring. Then $A/\mathrm{Jac}(A)$ is a semisimple ring, and hence Artin-Wedderburn (theorem 1.2.16) applies to it. Moreover, the simple left A-modules are exactly the simple left $A/\mathrm{Jac}(A)$-modules: every simple A-module is annihilated by $\mathrm{Jac}(A)$ and so factors through $A/\mathrm{Jac}(A)$, and every simple $A/\mathrm{Jac}(A)$-module is a simple A-module via the quotient map $A \twoheadrightarrow A/\mathrm{Jac}(A)$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$.

Step 1: $\bar{A}$ is semisimple. By corollary 1.1.9 it suffices to show that $\bar{A}$ is semisimple as a left module over itself, that is that $\mathrm{rad}(\bar{A}) = 0$. The maximal left ideals of $\bar{A}$ are exactly the images $\mathfrak{m}/J$ of the maximal left ideals $\mathfrak{m}$ of A (which all contain $J = \mathrm{Jac}(A)$), by the correspondence theorem. Their intersection is

$$\operatorname{Jac}(\bar{A}) = \bigcap_{\mathfrak{m}\supseteq J} \mathfrak{m}/J = \Big(\bigcap_{\mathfrak{m}} \mathfrak{m}\Big)\Big/ J = J/J = 0$$

So $\operatorname{Jac}(\bar{A}) = 0$. It remains to convert this into semisimplicity, which is where the Artinian hypothesis is spent: $\bar{A}$ is Artinian (a quotient of the Artinian ring A), and for an Artinian ring $\operatorname{Jac}(\bar{A}) = 0$ forces $\bar{A}$ to be semisimple. Concretely, since $\bar{A}$ is Artinian we may pick finitely many maximal left ideals $\mathfrak{m}_1/J, \dots, \mathfrak{m}_r/J$ whose intersection is already 0 (take a minimal finite subintersection, which exists by the descending chain condition and equals the full intersection $\operatorname{Jac}(\bar{A}) = 0$). The map

$$\bar{A} \longrightarrow \bigoplus_{i=1}^{r} \bar{A}/(\mathfrak{m}_i/J)$$

is injective, since its kernel is $\bigcap_i (\mathfrak{m}_i/J) = 0$. Each summand $\bar{A}/(\mathfrak{m}_i/J)$ is a simple $\bar{A}$-module, so $\bar{A}$ embeds as a submodule of a semisimple module and is therefore semisimple by corollary 1.1.8. Thus $\bar{A}$ is a semisimple $\bar{A}$-module, and by corollary 1.1.9 a semisimple ring. Artin-Wedderburn now applies to $\bar{A}$.

Step 2: the simple modules coincide. Let S be a simple left A-module. By part (2) of theorem 2.1.4, $J = \operatorname{Jac}(A)$ annihilates every simple A-module, so $JS = 0$. Hence the A-action on S factors through the quotient $\bar{A} = A/J$: setting $(\bar{a})s := as$ is well-defined because J acts as zero. The A-submodules and $\bar{A}$-submodules of S are the same subsets (a subgroup of S is A-stable if and only if it is $\bar{A}$-stable, since the actions agree), so S is simple as an $\bar{A}$-module precisely because it is simple as an A-module.

Conversely, let T be a simple left $\bar{A}$-module. Pulling back the action along the surjection $\pi\colon A \twoheadrightarrow \bar{A}$, that is defining $a \cdot t := \pi(a)\, t$, makes T an A-module. Its A-submodules are exactly its $\bar{A}$-submodules (again the two actions have the same invariant subgroups, because π is surjective), so T is simple as an A-module. These two passages are mutually inverse, so the simple A-modules and the simple $\bar{A}$-modules are identified, as we sought to show.

This is the structural role of the radical: it is the exact obstruction one divides out to recover the Wedderburn world. The simple modules of A are unchanged by the passage to $A/\operatorname{Jac}(A)$, so the finite list of simples that a semisimple ring enjoys is available for *any* Artinian ring, read off from its semisimple quotient. What the radical takes away is not the simples but the way they fit together: over A the simples can be glued into indecomposable modules that do not split, and the gluing data lives in $\operatorname{Jac}(A)$. The theorem says that for an Artinian A there are finitely many simple A-modules, one for each simple factor of the Wedderburn decomposition of $A/\operatorname{Jac}(A)$, and this is the starting point for the character theory and the block theory developed later.

The special case $\operatorname{Jac}(A) = 0$ returns to chapter 1.

Corollary 2.1.10: Semisimplicity via the Radical

Let A be a left Artinian ring. Then A is semisimple if and only if $\operatorname{Jac}(A) = 0$.

***Proof*:**
If A is semisimple, then by corollary 1.1.9 it is a semisimple left module over itself, so its radical (the intersection of its maximal left ideals) is 0; that is, $\mathrm{Jac}(A) = \mathrm{rad}({}_A A) = 0$. Indeed, a semisimple module is a direct sum of simples, and its radical, the intersection of maximal submodules, is 0: writing $A = \bigoplus_i I_i$ as a sum of minimal left ideals (lemma 1.1.10), each $\bigoplus_{j \neq i} I_j$ is a maximal submodule, and their intersection is 0.

Conversely, if $\mathrm{Jac}(A) = 0$, then $A = A/\mathrm{Jac}(A)$, and by theorem 2.1.9 the quotient $A/\mathrm{Jac}(A)$ is semisimple; hence A itself is semisimple, completing the proof.

The corollary makes the radical the precise measure of the failure of semisimplicity: A is semisimple exactly when the measurement returns zero, and Artin-Wedderburn (theorem 1.2.16) describes what A looks like in that case, namely a product $M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$. For a general Artinian A, the same product describes $A/\mathrm{Jac}(A)$ rather than A, and the difference between A and this product is stored in the nilpotent ideal $\mathrm{Jac}(A)$. The truncated-polynomial algebra $k[x]/(x^n)$ with $n \geq 2$ is the simplest case where the measurement is nonzero: its radical $(\bar{x})$ is a nonzero nilpotent ideal, so it is not semisimple, even though its semisimple quotient is the field k.

The most consequential source of non-semisimple algebras in representation theory is the group algebra $k[G]$ when the characteristic of k divides the order of the group. Maschke's theorem guarantees semisimplicity precisely when $\mathrm{char}(k) \nmid |G|$; the radical detects, and measures, the failure in the opposite case.

Example 2.1.11: The Radical of a Modular Group Algebra
Let $G = \mathbb{Z}/p\mathbb{Z}$ be cyclic of order p and let k be a field. Writing g for a generator, the group algebra is

$$k[G] \cong k[t]/(t^p - 1)$$

via $g \mapsto t$, since g has order p. The structure of $\mathrm{Jac}(k[G])$ depends entirely on whether $\mathrm{char}(k)$ divides $|G| = p$.

The modular case $\mathrm{char}(k) = p$. In characteristic p the Frobenius identity gives $t^p - 1 = (t-1)^p$, so

$$k[G] \cong k[t]/\big((t-1)^p\big) \cong k[s]/(s^p) \qquad (s := t - 1)$$

the last isomorphism being the change of variable $s = t - 1$. This is precisely the truncated polynomial algebra of example 2.1.5 with $n = p$, so its radical is

$$\mathrm{Jac}(k[G]) = (s) = (g - 1)$$

the ideal generated by $g - 1$, which is the augmentation ideal (the kernel of the augmentation map $k[G] \to k$ sending $g \mapsto 1$). It is nilpotent, $(g-1)^p = g^p - 1 = 0$, and nonzero, so $k[G]$ is not semisimple: Maschke's theorem fails exactly because $p = \mathrm{char}(k)$ divides $|G|$. The unique simple $k[G]$-module is the trivial module k, on which g acts as the identity, so $g - 1$ acts as zero, confirming $\mathrm{Jac}(k[G]) = (g-1)$ annihilates the only simple module.

The coprime case $\mathrm{char}(k) \nmid p$. If the characteristic does not divide p, then the polynomial $t^p - 1$ is separable (its derivative pt^{p-1} is nonzero and shares no root with it), so it has no repeated factors. Over a field containing a primitive pth root of unity ζ it splits as $t^p - 1 = \prod_{i=0}^{p-1}(t - \zeta^i)$, and the Chinese remainder theorem gives

$$k[G] \cong \prod_{i=0}^{p-1} k[t]/(t - \zeta^i) \cong k^p$$

a product of p copies of k, which is semisimple with radical $\mathrm{Jac}(k[G]) = 0$. (Over a field without the roots of unity, $t^p - 1$ still factors into distinct irreducible separable factors and $k[G]$ is a product of finite separable field extensions of k, again semisimple with zero radical.) This is Maschke's theorem in the coprime case, seen through the radical: separability of $t^p - 1$ is exactly the condition $\mathrm{char}(k) \nmid |G|$, and it forces $\mathrm{Jac}(k[G]) = 0$.

The contrast in the two cases is the whole point of the chapter in miniature. When $\mathrm{char}(k) \nmid |G|$ the group algebra is semisimple, its radical is zero, and the representation theory of G over k is governed by Wedderburn and character theory. When $\mathrm{char}(k)\big||G|$ the radical becomes a nonzero nilpotent ideal, the group algebra acquires indecomposable modules that are not simple, and one enters modular representation theory, where the radical, the indecomposables, and the projective covers built in the following sections are the working tools.

2.2 Indecomposable Modules and Krull-Schmidt

The previous chapter rested on a single organizing principle: over a semisimple ring, every module splits into simple pieces, and Artin-Wedderburn turns that splitting into a matrix description of the ring itself. The trouble is that semisimplicity is a strong hypothesis. The ring $k[x]/(x^2)$ is a two-dimensional k-algebra, and the module $k[x]/(x^2)$ over itself has a proper nonzero submodule $(x)/(x^2)$, yet that submodule has no complement: there is no way to write $k[x]/(x^2) = (x)/(x^2) \oplus N$. So this module does not break into simples at all. Once the semisimple world is left behind, the simple modules stop being the building blocks, and the decomposition theory of lemma 1.1.7 collapses.

What survives is a weaker notion of "building block". A module that cannot be split into two nonzero summands is *indecomposable* (definition 1.1.1), and every module of finite length is at least a finite direct sum of indecomposables, for the trivial reason that one keeps splitting until nothing splits further. The content of this section is that this crude decomposition is in fact canonical: the list of indecomposable summands, taken up to isomorphism, is uniquely determined by the module. This is the Krull-Schmidt theorem, and it is the correct replacement for the semisimple decomposition when simples are no longer available. The engine driving uniqueness is a structural fact about the endomorphism ring of an indecomposable module of finite length, that it is a *local ring*, which is itself a consequence of Fitting's lemma. So the section runs Fitting's lemma to the local-ring theorem to Krull-Schmidt, each result feeding the next.

Throughout, A is an algebra over a field k (the arguments need only that modules have finite length, so the results apply verbatim to modules of finite length over any ring). Modules are left modules. The starting point is worth isolating, since it is the gap between this chapter and the last. A simple module is indecomposable, because a simple module has no proper nonzero submodule at all, let alone a complemented one. The converse fails, and it fails already in dimension two: the $k[x]$-module $k[x]/(x^2)$ has the unique proper nonzero submodule $(x)/(x^2)$, so it is not simple, but that submodule is not a summand (any complement would have to be one-dimensional and x-stable, and $(x)/(x^2)$ is the only such subspace), so the module is indecomposable. An indecomposable module of finite length can therefore sit strictly between "simple" and "decomposable", and it is exactly this middle ground that Krull-Schmidt organizes.

The first tool is a splitting result for a single endomorphism. Given $\varphi \in \mathrm{End}_A(M)$, the iterated kernels $\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots$ increase and the iterated images $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$ decrease. When M has finite length, both chains must stabilize, and Fitting's lemma says that at a common stabilization point the module splits into the eventual kernel and the eventual image.

Lemma 2.2.1: Fitting's Lemma

Let M be an A-module of finite length and let $\varphi \in \mathrm{End}_A(M)$. Then there exists a positive integer n such that

$$M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$$

Consequently, if M is indecomposable, then every $\varphi \in \mathrm{End}_A(M)$ is either nilpotent or an automorphism.

***Proof*:**
Write $\ell = \ell(M)$ for the length of M, which is finite by hypothesis.

Step 1: The kernel and image chains stabilize. The submodules $\ker(\varphi^i)$ form an increasing chain and the submodules $\mathrm{im}(\varphi^i)$ form a decreasing chain:

$$\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \cdots \qquad \mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^2) \supseteq \cdots$$

A module of finite length is both Noetherian and Artinian, so an increasing chain of submodules cannot strictly ascend more than ℓ times and a decreasing chain cannot strictly descend more than ℓ times. Setting $n = \ell$ makes both chains stationary from index n onward: for all $m \geq n$,

$$\ker(\varphi^m) = \ker(\varphi^n) \qquad \mathrm{im}(\varphi^m) = \mathrm{im}(\varphi^n)$$

Indeed, once the kernel chain repeats, it is stationary forever after: if $\ker(\varphi^i) = \ker(\varphi^{i+1})$ and $x \in \ker(\varphi^{i+2})$, then $\varphi(x) \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, so $x \in \ker(\varphi^{i+1}) = \ker(\varphi^i)$, giving $\ker(\varphi^{i+2}) = \ker(\varphi^{i+1})$; iterating shows $\ker(\varphi^j) = \ker(\varphi^i)$ for all $j \geq i$. A chain that strictly increased past step ℓ would produce more than ℓ strict inclusions, violating finite length, so the first repeat occurs by step ℓ. The same bound applies to the image chain, and $n = \ell$ dominates both stabilization thresholds.

Step 2: The two submodules intersect trivially. Fix n as above and take $y \in \ker(\varphi^n) \cap \mathrm{im}(\varphi^n)$. Write $y = \varphi^n(x)$ for some $x \in M$. Then $\varphi^n(y) = 0$ gives $\varphi^{2n}(x) = 0$, so $x \in \ker(\varphi^{2n}) = \ker(\varphi^n)$ by the stabilization of the kernel chain. Hence $y = \varphi^n(x) = 0$, and $\ker(\varphi^n) \cap \mathrm{im}(\varphi^n) = 0$.

Step 3: The two submodules span M. Take any $z \in M$. Since $\mathrm{im}(\varphi^n) = \mathrm{im}(\varphi^{2n})$, the element $\varphi^n(z) \in \mathrm{im}(\varphi^n)$ also lies in $\mathrm{im}(\varphi^{2n})$, so $\varphi^n(z) = \varphi^{2n}(w)$ for some $w \in M$. Then

$$z = \varphi^n(w) + \big(z - \varphi^n(w)\big)$$

where $\varphi^n(w) \in \mathrm{im}(\varphi^n)$, and the second term lies in $\ker(\varphi^n)$ because $\varphi^n(z - \varphi^n(w)) = \varphi^n(z) - \varphi^{2n}(w) = 0$. Thus $M = \ker(\varphi^n) + \mathrm{im}(\varphi^n)$, and together with Step 2 this gives the internal direct sum $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$.

Step 4: The dichotomy for indecomposable M. Suppose now that M is indecomposable. The decomposition $M = \ker(\varphi^n) \oplus \mathrm{im}(\varphi^n)$ forces one of the two summands to be M and the other to be 0. If $\ker(\varphi^n) = M$, then $\varphi^n = 0$, so φ is nilpotent. If instead $\ker(\varphi^n) = 0$ and $\mathrm{im}(\varphi^n) = M$, then φ is injective, since $\ker(\varphi) \subseteq \ker(\varphi^n) = 0$, and φ is surjective, since $\mathrm{im}(\varphi) \supseteq \mathrm{im}(\varphi^n) = M$. An injective and surjective endomorphism is an automorphism. In either case φ is nilpotent or an automorphism, completing the proof.

The decomposition in Fitting's lemma is the "eventual kernel" and "eventual image" of φ, and it works for any single endomorphism of any finite-length module, indecomposable or not. The sharp consequence appears only in the indecomposable case: an indecomposable module of finite length admits no interesting endomorphisms in the sense that the only two behaviors available to a $\varphi \in \mathrm{End}_A(M)$ are "eventually kills everything" (nilpotent) and "moves everything bijectively" (automorphism). There is no room in between, precisely because any intermediate behavior would produce a nontrivial Fitting decomposition and split the module. This is the mechanism that will make the endomorphism ring so rigid.

To see the dichotomy in a case where it can be checked by hand, take the endomorphism given by multiplication by x on $M = k[x]/(x^3)$. Multiplication by x has kernel $(x^2)/(x^3)$ and image $(x)/(x^3)$, which are not complementary. But the lemma is about a *power*: φ^3 is multiplication by $x^3 = 0$, so $\ker(\varphi^3) = M$ and $\mathrm{im}(\varphi^3) = 0$, giving the trivial decomposition $M = M \oplus 0$. This is the nilpotent branch of the dichotomy, and indeed multiplication by x is nilpotent. Multiplication by $1 + x$, on the other hand, is invertible (with inverse $1 - x + x^2$), landing in the automorphism branch. Every element of $\mathrm{End}_{k[x]}(M)$ is one or the other.

The endomorphism ring $\mathrm{End}_A(M)$ of an indecomposable finite-length module is close to being a division ring but not quite: Schur's lemma (lemma 1.1.5) gave a division ring only for *simple* M, and indecomposable is weaker. The right substitute is that $\mathrm{End}_A(M)$ is a *local ring*. A ring is local when its non-units form a two-sided ideal, equivalently when it has a unique maximal left ideal; the intuition is that "being a non-unit" is closed under addition, so failures of invertibility do not conspire to produce an invertible element. Fitting's lemma is exactly what closes the non-units under addition here, because a non-unit of $\mathrm{End}_A(M)$ is precisely a nilpotent endomorphism, and controlling sums of nilpotents is where the argument lives.

Theorem 2.2.2: Endomorphism Rings of Indecomposables are Local

Let M be a nonzero indecomposable A-module of finite length. Then $\mathrm{End}_A(M)$ is a local ring: the set of non-units of $\mathrm{End}_A(M)$ is a two-sided ideal, and it consists exactly of the nilpotent endomorphisms of M.

***Proof*:**
Write $E = \mathrm{End}_A(M)$ and let $\mathfrak{m} \subseteq E$ be the set of endomorphisms that are not units, that is, not automorphisms of M.

Step 1: The non-units are the nilpotents. By lemma 2.2.1, every $\varphi \in E$ is either an automorphism or nilpotent, and these are mutually exclusive (a nilpotent $\varphi \neq 0$ kills some nonzero element, and $\varphi = 0$ is not invertible since $M \neq 0$). Hence $\mathfrak{m}$ is exactly the set of nilpotent endomorphisms.

Step 2: $\mathfrak{m}$ is closed under left and right multiplication by E. Let $\varphi \in \mathfrak{m}$, so φ is not an automorphism, and let $\psi \in E$ be arbitrary. If $\psi\varphi$ were an automorphism, then φ would be injective (as $\psi\varphi$ is) and ψ surjective; an injective endomorphism of a finite-length module is bijective (its image has the same length as M, hence equals M), so φ would be an automorphism, a contradiction. Thus $\psi\varphi \in \mathfrak{m}$. Symmetrically, if $\varphi\psi$ were an automorphism, then φ would be surjective, hence bijective by the length argument, again contradicting $\varphi \in \mathfrak{m}$. So $\varphi\psi \in \mathfrak{m}$, and $\mathfrak{m}$ absorbs multiplication on both sides.

Step 3: $\mathfrak{m}$ is closed under addition. This is the step that uses indecomposability. Let $\varphi, \psi \in \mathfrak{m}$ and suppose, for contradiction, that $\varphi + \psi$ is an automorphism. Set

$$\alpha = (\varphi + \psi)^{-1}\varphi \qquad \beta = (\varphi + \psi)^{-1}\psi$$

so that $\alpha+\beta = (\varphi+\psi)^{-1}(\varphi+\psi) = \mathrm{id}_M$. By Step 2, $\mathfrak{m}$ absorbs left multiplication, so $\alpha, \beta \in \mathfrak{m}$; in particular α is nilpotent, say $\alpha^r = 0$. From $\beta = \mathrm{id}_M - \alpha$ the standard geometric-series identity gives an inverse for β:

$$(\mathrm{id}_M - \alpha)\big(\mathrm{id}_M + \alpha + \alpha^2 + \cdots + \alpha^{r-1}\big) = \mathrm{id}_M - \alpha^r = \mathrm{id}_M$$

so $\beta = \mathrm{id}_M - \alpha$ is an automorphism of M. But $\beta \in \mathfrak{m}$ means β is not an automorphism, a contradiction. Hence $\varphi + \psi \in \mathfrak{m}$.

Step 4: Conclusion. Steps 2 and 3 show $\mathfrak{m}$ is a two-sided ideal of E, and Step 1 identifies it as the set of non-units. A ring whose non-units form a two-sided ideal is local: any element outside $\mathfrak{m}$ is a unit, so $\mathfrak{m}$ is the unique maximal left ideal (a proper left ideal contains no units, hence lies in $\mathfrak{m}$). Therefore $E = \mathrm{End}_A(M)$ is local, as we sought to show.

The theorem sharpens Schur's lemma in exactly the direction needed. For a simple module, lemma 1.1.5 gave a division ring, where every nonzero element is a unit. For an indecomposable module of finite length, one gets the next best thing: a local ring, where the non-units are corralled into a single ideal and every element outside it is a unit. The residue $\mathrm{End}_A(M)/\mathfrak{m}$ is a division ring, and one can think of the local ring as a division ring with a nilpotent "fuzz" $\mathfrak{m}$ attached. The content used below is the geometric series in Step 3, which turns "id_M minus a nilpotent" into a unit; this is what will let one element of a decomposition be swapped for another in the exchange argument below.

The running example makes the local structure explicit.

Example 2.2.3: The Endomorphism Ring of a Truncated Polynomial Module

Fix $n \geq 1$ and let $M = k[x]/(x^n)$, viewed as a module over $A = k[x]$ (equivalently over $k[x]/(x^n)$, which acts through the same operators). This module is indecomposable of length n: its submodules are exactly the ideals $(x^j)/(x^n)$ for $0 \leq j \leq n$, forming a single chain

$$0 = (x^n)/(x^n) \subseteq (x^{n-1})/(x^n) \subseteq \cdots \subseteq (x)/(x^n) \subseteq M$$

and a chain of submodules has no way to split off a complementary summand (a decomposition $M = M_1 \oplus M_2$ with both $M_i \neq 0$ would give two submodules neither containing the other).

Every A-endomorphism of M is multiplication by some polynomial: since M is generated by the image of 1, a homomorphism φ is determined by $\varphi(1) = \overline{f(x)}$, and then $\varphi(\overline{g}) = \overline{gf}$. So

$$\mathrm{End}_A(M) \cong k[x]/(x^n)$$

as rings, the isomorphism sending φ to $\varphi(1)$. This is a local ring with maximal ideal $\mathfrak{m} = (x)/(x^n)$: an element $\overline{f} = \overline{a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}}$ is a unit if and only if $a_0 \neq 0$ (with inverse computed by the geometric series when $a_0 = 1$), and it is nilpotent if and only if $a_0 = 0$. The non-units are precisely the nilpotents, matching theorem 2.2.2, and $\mathrm{End}_A(M)/\mathfrak{m} \cong k$ is the residue division ring (here a field, since A is commutative).

The local endomorphism ring is now used to prove the main theorem. Existence of a decomposition into indecomposables is easy, by induction on length; the content is uniqueness. The classical route to uniqueness is an *exchange argument*: given two decompositions of the same module, one shows that each indecomposable summand of the first can be swapped into the second, one at a time, without changing the ambient module, and the local endomorphism rings are exactly what make each swap possible. The theorem is due to Krull and Schmidt for finite groups and finite-length modules, with Azumaya's extension to any module that is a direct sum of submodules with local endomorphism rings.

Theorem 2.2.4: Krull-Schmidt(-Azumaya)

Let M be a nonzero A-module of finite length. Then:

1. M is a finite direct sum of indecomposable submodules,
$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_r$$
with each M_i indecomposable.
2. This decomposition is unique up to isomorphism and reordering: if
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
are two decompositions into indecomposable submodules, then $r = s$ and there is a permutation σ of $\{1, \ldots, r\}$ with $M_i \cong N_{\sigma(i)}$ for every i.

***Proof*:**

1. *Existence.* Argue by induction on the length $\ell(M)$. If M is indecomposable, then $M = M$ is already such a decomposition, with one summand. Otherwise $M = M_1 \oplus M_2$ with both M_i nonzero, and then $\ell(M_1), \ell(M_2) < \ell(M)$ because length is additive over direct sums and both summands are nonzero (hence of positive length). By the inductive hypothesis each M_i is a finite direct sum of indecomposables, and concatenating the two lists expresses M as a finite direct sum of indecomposables. The base case $\ell(M) = 1$ is a simple, hence indecomposable, module. This proves existence.
2. *Uniqueness.* Suppose
$$M = M_1 \oplus \cdots \oplus M_r = N_1 \oplus \cdots \oplus N_s$$
with all M_i and N_j indecomposable of finite length, so each $\mathrm{End}_A(M_i)$ and $\mathrm{End}_A(N_j)$ is local by theorem 2.2.2. Induct on r. The strategy is to exchange M_1 for one of the N_j, peel it off both sides, and apply the inductive hypothesis to what remains.

   Let $\iota_i : M_i \hookrightarrow M$ and $\pi_i : M \twoheadrightarrow M_i$ be the inclusion and projection for the first decomposition, and $\iota'_j : N_j \hookrightarrow M$, $\pi'_j : M \twoheadrightarrow N_j$ those for the second. These satisfy
$$\sum_{i=1}^{r} \iota_i \pi_i = \mathrm{id}_M \qquad \sum_{j=1}^{s} \iota'_j \pi'_j = \mathrm{id}_M$$
   Restricting the second identity to M_1 and projecting back to M_1 gives, in $\mathrm{End}_A(M_1)$,
$$\mathrm{id}_{M_1} = \pi_1 \Big(\sum_{j=1}^{s} \iota'_j \pi'_j \Big) \iota_1 = \sum_{j=1}^{s} (\pi_1 \iota'_j)(\pi'_j \iota_1)$$
   Now $\mathrm{End}_A(M_1)$ is local, so its non-units form an ideal and cannot sum to the unit id_{M_1}; at least one summand $(\pi_1 \iota'_j)(\pi'_j \iota_1)$ must be a unit of $\mathrm{End}_A(M_1)$. After reindexing the N_j

we may assume this happens for $j = 1$: the composite

$$\theta = (\pi_1 \iota_1')(\pi_1' \iota_1) \colon M_1 \longrightarrow M_1$$

is an automorphism of M_1, where $\pi_1' \iota_1 \colon M_1 \to N_1$ and $\pi_1 \iota_1' \colon N_1 \to M_1$.

Exchange step. Consider the maps

$$f = \pi_1' \iota_1 \colon M_1 \to N_1 \qquad g = \pi_1 \iota_1' \colon N_1 \to M_1$$

so $gf = \theta$ is an automorphism of M_1. Then f is a split monomorphism, with left inverse $\theta^{-1} g$, since $(\theta^{-1} g) f = \theta^{-1}(gf) = \mathrm{id}_{M_1}$. The endomorphism $e = f\theta^{-1} g \colon N_1 \to N_1$ is then idempotent:

$$e^2 = f\theta^{-1}(gf)\theta^{-1} g = f\theta^{-1}\theta\theta^{-1} g = f\theta^{-1} g = e$$

and its image is $\mathrm{im}(f)$ (from $ef = f\theta^{-1} gf = f$), so $N_1 = \mathrm{im}(e) \oplus \ker(e) = \mathrm{im}(f) \oplus \ker(e)$ is the splitting associated to the idempotent e. But N_1 is indecomposable and $\mathrm{im}(f) \neq 0$ (since f is injective and $M_1 \neq 0$), which forces $\mathrm{im}(f) = N_1$, making f surjective as well. Hence $f \colon M_1 \to N_1$ is an isomorphism, and in particular

$$M_1 \cong N_1$$

Peeling off the summand. It remains to remove M_1 and N_1 from the two decompositions and match the rest. Set $M' = M_2 \oplus \cdots \oplus M_r$, so $M = M_1 \oplus M'$ and M' is the kernel of the projection $\pi_1 \colon M \to M_1$. The claim is

$$M = N_1 \oplus M'$$

viewing $N_1 = \iota_1'(N_1) \subseteq M$ as the submodule it names in the second decomposition. The key observation is that the restriction of π_1 to N_1 is an isomorphism onto M_1: on N_1 this restriction is $\pi_1 \iota_1' = g$, and $g = \theta f^{-1}$ is a composite of the automorphism θ with the isomorphism f^{-1}, hence itself an isomorphism $N_1 \to M_1$. Granting that $\pi_1|_{N_1} \colon N_1 \to M_1$ is an isomorphism, the claim follows:

- *Trivial intersection.* If $y \in N_1 \cap M'$, then $\pi_1(y) = 0$ because $M' = \ker(\pi_1)$, and $\pi_1|_{N_1}$ is injective, so $y = 0$. Thus $N_1 \cap M' = 0$.
- *Spanning.* Given $z \in M$, set $a = \pi_1(z) \in M_1$. Since $\pi_1|_{N_1}$ is onto M_1, choose $y \in N_1$ with $\pi_1(y) = a$. Then $\pi_1(z - y) = a - a = 0$, so $z - y \in \ker(\pi_1) = M'$, and $z = y + (z - y) \in N_1 + M'$.

Hence $M = N_1 \oplus M'$, establishing the claim.

Induction. Quotienting $M = N_1 \oplus M'$ by the summand N_1 gives $M/N_1 \cong M'$, and quotienting the original $M = N_1 \oplus N'$ by the same submodule N_1 gives $M/N_1 \cong N'$. Therefore

$$M' \cong N'$$

Now M' carries the decomposition $M_2 \oplus \cdots \oplus M_r$ into $r - 1$ indecomposables, and the isomorphism $M' \cong N'$ transports the decomposition $N_2 \oplus \cdots \oplus N_s$ onto M' as a second decomposition into indecomposables. Applying the inductive hypothesis to M', whose first decomposition has $r - 1$ summands, yields $r - 1 = s - 1$, so $r = s$, together with a bijection matching $M_2, \ldots, M_r$ against $N_2, \ldots, N_s$ up to isomorphism. Combined with $M_1 \cong N_1$ from the exchange step, this produces the permutation σ with $M_i \cong N_{\sigma(i)}$ for all i. The base case $r = 1$ has $M = M_1$ indecomposable, forcing $s = 1$ and $N_1 = M$, completing the induction and the proof.

The theorem says that the finite-length representation theory of *any* algebra, semisimple or not, has canonical building blocks: the indecomposable modules. Over a semisimple algebra these are the simple modules and Krull-Schmidt recovers the decomposition of lemma 1.1.7, but the statement here needs no semisimplicity. What it does need is finite length, and the two places that hypothesis enters are worth marking. Finite length gave the Fitting decomposition (via the ascending and descending chain conditions), which produced the local endomorphism rings; and finite length is what let "injective endomorphism" upgrade to "automorphism" at several points. Drop finiteness and uniqueness can fail. A concrete failure: over the ring $R = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ of real polynomial functions on the 2-sphere, the module R^3 splits both as $R \oplus P$, where P is the (indecomposable, non-free) module of tangent vector fields, and as $R \oplus R \oplus R$; the two decompositions have non-isomorphic summands, so uniqueness is lost once the summands are no longer of finite length. Krull-Schmidt is thus not a formality but a theorem with content tied to its hypotheses.

The exchange argument also explains *why* the answer is unique, not just *that* it is. The reason each M_i has a unique isomorphism partner is that when the identity of M_1 is written as a sum of composites through the N_j, locality of $\mathrm{End}_A(M_1)$ forbids the sum from being "spread out" among non-units; one composite must already be invertible, and that single composite pins down the partner. Locality is doing the work that a division ring did in the simple case, and it is the exact amount of structure required.

Reading off the multiplicities gives the invariant form of the theorem that is used in practice.

Corollary 2.2.5: Uniqueness of Multiplicities

Let M be a nonzero A-module of finite length. For each isomorphism class of indecomposable module I, the number of summands isomorphic to I appearing in a decomposition of M into indecomposables is independent of the decomposition. That is, writing

$$M \cong I_1^{\oplus m_1} \oplus I_2^{\oplus m_2} \oplus \cdots \oplus I_t^{\oplus m_t}$$

with the I_j pairwise non-isomorphic indecomposables, the classes $I_1, \ldots, I_t$ and the multiplicities $m_1, \ldots, m_t$ are uniquely determined by M.

***Proof*:**
By theorem 2.2.4, any two decompositions of M into indecomposables have the same number of summands and are matched by a permutation σ respecting isomorphism type. Grouping the summands of a decomposition by isomorphism class, the multiplicity of a class I is the number of indices i with $M_i \cong I$. Since σ carries each M_i to an $N_{\sigma(i)} \cong M_i$ and is a bijection, it restricts to a bijection between the summands isomorphic to I in the first decomposition and those isomorphic to I in the second. Hence the two decompositions assign I the same multiplicity, and this holds for every isomorphism class I. The set of classes that occur and their multiplicities are therefore invariants of M, completing the proof.

The multiplicities m_j are the numbers one computes when decomposing a representation, and the corollary is what guarantees the computation has a well-defined answer regardless of the method used to obtain it. In the language of the Grothendieck-style count, M determines a formal $\mathbb{N}$-linear combination $\sum_j m_j[I_j]$ of indecomposable classes, and two modules are isomorphic if and only if these combinations agree. This is the precise sense in which the indecomposables are a basis for the finite-length modules of A.

The full classification is transparent for $A = k[x]/(x^n)$, where the indecomposables can be listed and the uniqueness of multiplicities checked against an explicit computation.

Example 2.2.6: Modules over the Truncated Polynomial Algebra $k[x]/(x^n)$

Fix $n \geq 1$ and let $A = k[x]/(x^n)$. A finite-dimensional A-module is the same datum as a finite-dimensional k-vector space V together with a k-linear operator T, the action of the class of x, satisfying $T^n = 0$, so that T is nilpotent of nilpotency index at most n. Being finite-dimensional over k, every such module has finite length, so Krull-Schmidt applies.

The indecomposables. Since A is a quotient of the principal ideal domain $k[x]$, the structure theorem for finitely generated modules over $k[x]$ describes every finite-length A-module as a direct sum of cyclic modules $k[x]/(x^m)$. Only the exponents $1 \leq m \leq n$ occur, because x^n annihilates the module. Each

$$J_m := k[x]/(x^m) \qquad 1 \leq m \leq n$$

is indecomposable: as in example 2.2.3, its submodules form the single chain $0 \subsetneq (x^{m-1}) \subsetneq \cdots \subsetneq (x) \subsetneq J_m$, which admits no splitting into two nonzero summands. So the indecomposable A-modules are exactly

$$J_1, J_2, \ldots, J_n$$

up to isomorphism, with $\dim_k J_m = m$; these are the Jordan blocks for the nilpotent operator, J_m being a single Jordan block of size m with eigenvalue 0. There are precisely n of them.

A decomposition and its uniqueness. Take $n = 3$ and the module $M = A \oplus J_1$, that is, $V = k^4$ with the operator

$$T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

a single Jordan block of size 3 in the top-left together with a 1×1 zero block. As an A-module,

$$M \cong J_3 \oplus J_1$$

with multiplicities $m_3 = 1$, $m_1 = 1$, and $m_2 = 0$. corollary 2.2.5 asserts these multiplicities are forced by M, and they can be read off invariantly from the ranks of powers of T, which do not depend on any chosen decomposition. The number of Jordan blocks of size at least j equals $\operatorname{rank}(T^{j-1}) - \operatorname{rank}(T^j)$, so the multiplicity of J_m is

$$m_m = \operatorname{rank}(T^{m-1}) - 2\operatorname{rank}(T^m) + \operatorname{rank}(T^{m+1})$$

Here $\operatorname{rank}(T^0) = \operatorname{rank}(\mathrm{id}) = 4$, $\operatorname{rank}(T) = 2$, $\operatorname{rank}(T^2) = 1$, and $\operatorname{rank}(T^3) = 0$. Substituting,

$$m_1 = 4 - 2 \cdot 2 + 1 = 1, \qquad m_2 = 2 - 2 \cdot 1 + 0 = 0, \qquad m_3 = 1 - 2 \cdot 0 + 0 = 1$$

recovering $M \cong J_3 \oplus J_1$. Since the ranks $\operatorname{rank}(T^j)$ are intrinsic to the operator T (hence to the module M), the multiplicities are intrinsic too, exactly as Krull-Schmidt predicts. Any presentation of M as a direct sum of J_m's, arrived at by any change of basis, must yield one copy of J_3, one copy of J_1, and no copy of J_2.

2.3 Idempotents, Projective Covers, and the Regular Representation

The Artin–Wedderburn theorem told us how to read off the simple modules of a semisimple algebra: they are exactly the summands of the regular module A, because a semisimple A splits as a direct sum of minimal left ideals and each minimal left ideal is simple. Once A is not semisimple this picture breaks. The regular module A still decomposes into indecomposable summands (the algebra is finite-dimensional, so a dimension count forces any module to be a finite direct sum of indecomposables), but those summands are no longer simple. They are the *indecomposable projective modules*, and the simple modules sit one level down, as their tops. This section builds the dictionary that replaces the semisimple one: the summands of A are cut out by a decomposition $1 = e_1 + \cdots + e_n$ of the identity into primitive orthogonal idempotents, each summand Ae_i is an indecomposable projective covering a single simple module, and the whole configuration is governed, through the quotient $A/\mathrm{Jac}(A)$, by the semisimple theory of chapter 1.

Throughout this section A is a finite-dimensional algebra over a field k (so in particular A is left Artinian and $\mathrm{Jac}(A)$ is nilpotent). Modules are left modules. The tool that transports information from the semisimple quotient $A/\mathrm{Jac}(A)$ back up to A is the lifting of idempotents, so that is where we begin.

The projection $\pi\colon A \to A/\mathrm{Jac}(A)$ collapses the radical. A decomposition of $A/\mathrm{Jac}(A)$ into a sum of simple summands is a statement about idempotents in $A/\mathrm{Jac}(A)$, and to pull such a decomposition back to A we need to produce idempotents in A that reduce to the given ones. The obstruction is that a preimage of an idempotent need not be idempotent. The nilpotence of $\mathrm{Jac}(A)$ removes the obstruction: a Newton-style correction converts any element that is idempotent modulo $\mathrm{Jac}(A)$ into a genuine idempotent lying over it.

Theorem 2.3.1: Lifting Idempotents Modulo the Radical

Let A be a finite-dimensional k-algebra and write $J = \mathrm{Jac}(A)$. Let $\pi\colon A \to A/J$ be the quotient map.

1. If $\bar{e} \in A/J$ satisfies $\bar{e}^2 = \bar{e}$, then there exists an idempotent $e \in A$ with $\pi(e) = \bar{e}$.
2. If $\bar{e}_1, \ldots, \bar{e}_m \in A/J$ are pairwise orthogonal idempotents with $\bar{e}_1 + \cdots + \bar{e}_m = \bar{1}$, then there exist pairwise orthogonal idempotents $e_1, \ldots, e_m \in A$ with $\pi(e_i) = \bar{e}_i$ and $e_1 + \cdots + e_m = 1$.

***Proof*:**
Since A is left Artinian, $J = \mathrm{Jac}(A)$ is nilpotent by theorem 2.1.8; fix N with $J^N = 0$.

Step 1: Lifting a single idempotent. Choose any $a \in A$ with $\pi(a) = \bar{e}$. Then $a^2 - a \in J$, since $\pi(a^2 - a) = \bar{e}^2 - \bar{e} = 0$. The idea is to correct a inside the coset $a + J$ until the error $a^2 - a$ is pushed into higher and higher powers of J, where it eventually vanishes. Set

$$e = 3a^2 - 2a^3$$

which lies in $a + J$, since $e - a = (3a^2 - 2a^3) - a = -(2a - 1)(a^2 - a) \in J$, so $\pi(e) = \bar{e}$. A direct computation gives

$$e^2 - e = (3a^2 - 2a^3)^2 - (3a^2 - 2a^3) = (a^2 - a)^2\,(4a^2 - 4a - 3)$$

Writing $u = a^2 - a \in J$, this says $e^2 - e \in (u^2) \subseteq J^2$. Thus the operation $a \mapsto 3a^2 - 2a^3$ sends an element whose defect $a^2 - a$ lies in J^r to an element whose defect lies in J^{2r}, while staying in the same coset modulo J. Starting from $a_0 = a$ with defect in J^1 and iterating,

$$a_{t+1} = 3a_t^2 - 2a_t^3$$

produces $a_t \in a + J$ whose defect $a_t^2 - a_t$ lies in J^{2^t}. Once $2^t \geq N$ we have $a_t^2 - a_t \in J^N = 0$, so $e := a_t$ is a genuine idempotent with $\pi(e) = \bar{e}$. This proves (1).

Step 2: Lifting an orthogonal family. We argue by induction on m. For $m = 1$ the statement is $e_1 = 1$, which is immediate. Suppose the result holds for families of size $m-1$, and let $\bar{e}_1, \ldots, \bar{e}_m$ be pairwise orthogonal idempotents summing to $\bar{1}$. Apply the inductive hypothesis to the shorter orthogonal family $\bar{e}_1, \ldots, \bar{e}_{m-2}, \bar{e}_{m-1} + \bar{e}_m$ (its members are pairwise orthogonal idempotents summing to $\bar{1}$, since $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal). This yields pairwise orthogonal idempotents $e_1, \ldots, e_{m-2}, f \in A$ lifting them, with $e_1 + \cdots + e_{m-2} + f = 1$ and $\pi(f) = \bar{e}_{m-1} + \bar{e}_m$.

It remains to split f. Consider the algebra fAf, which has identity element f. Its radical is $\mathrm{Jac}(fAf) = fJf$, and the reduction $fAf \to fAf/fJf$ identifies with the corner $\bar{f}(A/J)\bar{f}$ of the semisimple algebra A/J, where $\bar{f} = \bar{e}_{m-1} + \bar{e}_m$. In that corner, $\bar{e}_{m-1}$ and $\bar{e}_m$ are orthogonal idempotents summing to the identity $\bar{f}$. Applying part (1) inside the finite-dimensional algebra fAf to the idempotent $\bar{e}_{m-1} \in fAf/fJf$ produces an idempotent $e_{m-1} \in fAf$ with $\pi(e_{m-1}) = \bar{e}_{m-1}$. Set $e_m = f - e_{m-1}$. Then e_m is idempotent, since

$$e_m^2 = (f - e_{m-1})^2 = f - 2e_{m-1} + e_{m-1} = f - e_{m-1} = e_m$$

using $fe_{m-1} = e_{m-1}f = e_{m-1}$ (because $e_{m-1} \in fAf$) and $e_{m-1}^2 = e_{m-1}$; moreover $\pi(e_m) = \bar{f} - \bar{e}_{m-1} = \bar{e}_m$, and $e_{m-1}e_m = e_{m-1}f - e_{m-1}^2 = e_{m-1} - e_{m-1} = 0$, with $e_m e_{m-1} = 0$ symmetrically. Finally $e_{m-1}, e_m \in fAf$ are orthogonal to each of $e_1, \ldots, e_{m-2}$, since f is, and $e_1 + \cdots + e_{m-2} + e_{m-1} + e_m = e_1 + \cdots + e_{m-2} + f = 1$. This completes the induction and the proof.

The polynomial $3a^2 - 2a^3$ is the second Newton iterate for the equation $x^2 - x = 0$; the quadratic doubling of the exponent is exactly Newton's quadratic convergence, made to terminate by nilpotence rather than to converge by a metric. The content of the theorem is that the arithmetic of idempotents in A is completely controlled by the arithmetic of idempotents in the semisimple quotient $A/\mathrm{Jac}(A)$: any way of splitting the identity in $A/\mathrm{Jac}(A)$ lifts to a way of splitting it in A, so the summand structure of the regular module is governed from below by Artin–Wedderburn. The next result turns this into a decomposition of A itself.

Before stating it, recall the mechanism from example 1.2.10: an idempotent $e \in A$ produces a direct sum decomposition $A = Ae \oplus A(1-e)$ of the regular module, because every $a \in A$ writes uniquely as $a = ae + a(1-e)$ with $ae \in Ae$ and $a(1-e) \in A(1-e)$. More generally an orthogonal decomposition $1 = e_1 + \cdots + e_n$ gives $A = Ae_1 \oplus \cdots \oplus Ae_n$. The summand Ae_i is indecomposable precisely when e_i cannot be split further, which is the meaning of *primitive*. Pushing the splitting as far as it will go decomposes the regular module into indecomposables.

Theorem 2.3.2: Primitive Idempotent Decomposition of the Regular Module

Let A be a finite-dimensional k-algebra. Then the identity admits a decomposition

$$1 = e_1 + e_2 + \cdots + e_n$$

into pairwise orthogonal primitive idempotents, and correspondingly the regular module decomposes as

$$A = Ae_1 \oplus Ae_2 \oplus \cdots \oplus Ae_n$$

where each Ae_i is an indecomposable projective left A-module. For an idempotent $e \in A$, the summand Ae is indecomposable if and only if e is primitive, if and only if the corner algebra eAe has no idempotents other than 0 and e.

***Proof*:**
Step 1: Idempotent decompositions correspond to direct-sum decompositions. Let $e \in A$ be an idempotent. Direct summands of the module Ae correspond to idempotents of its endomorphism ring: if $Ae = P \oplus Q$, the projection onto P along Q is an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$, and conversely an idempotent $\varepsilon \in \mathrm{End}_A(Ae)$ splits $Ae = \varepsilon(Ae) \oplus (1-\varepsilon)(Ae)$. By the same computation as in lemma 1.2.3, the map $eAe \to \mathrm{End}_A(Ae)$ sending eae to right multiplication $x \mapsto x(eae)$ is a ring isomorphism

$$\mathrm{End}_A(Ae) \cong (eAe)^{\mathrm{op}}$$

The idempotents of $(eAe)^{\mathrm{op}}$ are the idempotents of eAe. So Ae decomposes nontrivially if and only if eAe has an idempotent other than 0 and $e = 1_{eAe}$. Writing such an idempotent as g and its complement $e - g$ (which is idempotent and orthogonal to g, by the computation in Step 2 of theorem 2.3.1), the decomposition $Ae = Ag \oplus A(e-g)$ corresponds to the refinement $e = g + (e-g)$ of e into orthogonal idempotents. Thus e is primitive (cannot be written as a sum of two nonzero orthogonal idempotents) if and only if eAe has no idempotent besides $0, e$, if and only if Ae is indecomposable. This proves the final three-way equivalence.

Step 2: The decomposition terminates. Start from 1. If 1 is primitive, take $n = 1$. Otherwise $1 = f + f'$ with f, f' nonzero orthogonal idempotents, giving $A = Af \oplus Af'$ with both summands nonzero. Since A is finite-dimensional and the summands Af, Af' are proper (each has strictly smaller k-dimension than A, as the other is nonzero), repeating the splitting on any summand that is still decomposable strictly decreases dimension at each step and so must stop. When it stops, every summand Ae_i is indecomposable, so by Step 1 each e_i is primitive. The resulting $e_1, \ldots, e_n$ are pairwise orthogonal (an idempotent split off from a corner Ae is orthogonal to every idempotent orthogonal to e) and sum to 1.

Step 3: Projectivity. Each Ae_i is a direct summand of the free module A, hence projective. A direct summand of a free module is projective because the splitting $A = Ae_i \oplus A(1-e_i)$ exhibits Ae_i as a retract of A, and retracts of free modules satisfy the lifting property defining projectivity; the general statement is proved in the companion volume `NateCoreAlgebra` [5] on projective modules. This gives the decomposition of A into indecomposable projectives, completing the proof.

The decomposition $A = \bigoplus_i Ae_i$ is the non-semisimple replacement for the Wedderburn splitting of a semisimple algebra into minimal left ideals. In the semisimple case each Ae_i was already simple, and the e_i were the matrix units picking out columns of the matrix blocks (proposition 1.2.5). In general each Ae_i is only indecomposable, and it will turn out to have a whole composition series rather than length one. The indecomposable summands that appear are the objects the rest of the section studies, so they deserve a name.

Definition 2.3.3: Indecomposable Projective Module

Let A be a finite-dimensional k-algebra and $1 = e_1 + \cdots + e_n$ a decomposition of the identity into primitive orthogonal idempotents. The indecomposable projective left A-modules are the summands

$$P_i = Ae_i$$

Its *radical* is $\mathrm{rad}(P_i) = \mathrm{Jac}(A)\, P_i$, and its *top* (or *head*) is the quotient

$$\mathrm{top}(P_i) = P_i/\mathrm{rad}(P_i) = Ae_i/\mathrm{Jac}(A)Ae_i$$

The module P_i is called the *projective cover* of its top; the top is a simple module.

The claim that the top is simple is the hinge of the whole correspondence, so we verify it at once. Reducing modulo $\mathrm{Jac}(A)$ turns the primitive idempotent e_i into a primitive idempotent $\bar{e}_i \in A/\mathrm{Jac}(A)$, and over the semisimple algebra $A/\mathrm{Jac}(A)$ a primitive idempotent cuts out a simple module.

Example 2.3.4: The Top of Ae_i

Let $A = k[x]/(x^3)$, a local algebra with $\mathrm{Jac}(A) = (x)/(x^3)$. Here 1 is the only nonzero idempotent (a commutative local ring has no idempotents besides 0 and 1), so the identity decomposition is trivial, $n = 1$ and $e_1 = 1$, giving a single indecomposable projective $P_1 = A$. Its radical is $\mathrm{rad}(P_1) = \mathrm{Jac}(A)\, A = (x)/(x^3)$, and its top is

$$P_1/\mathrm{rad}(P_1) = \big(k[x]/(x^3)\big)\big/\big((x)/(x^3)\big) \cong k[x]/(x) = k$$

a one-dimensional, hence simple, module. So the single projective $P_1 = A$ (of dimension 3) covers the single simple module k (of dimension 1); the two extra dimensions live in the radical $\mathrm{rad}(P_1)$, whose composition factors are again copies of k. This is the smallest example in which a projective indecomposable is strictly larger than the simple it covers.

The passage $P_i \mapsto \mathrm{top}(P_i)$ from indecomposable projectives to simple modules, and its reverse, is the organizing principle of non-semisimple representation theory. Simple modules are visible in the semisimple quotient $A/\mathrm{Jac}(A)$, where Artin–Wedderburn counts them; primitive idempotents in A are the summands of the regular module; and lifting idempotents connects the two. The following theorem assembles these into a single correspondence.

Theorem 2.3.5: Simples, Indecomposable Projectives, and Primitive Idempotents

Let A be a finite-dimensional k-algebra, and let $1 = e_1 + \cdots + e_n$ be a decomposition of the identity into primitive orthogonal idempotents. There are mutually inverse bijections among the following three finite sets:

1. isomorphism classes of simple left A-modules;
2. isomorphism classes of indecomposable projective left A-modules;
3. conjugacy classes of primitive idempotents of A, where e and e' are conjugate if $e' = ueu^{-1}$ for a unit $u \in A^\times$.

The bijection from (2) to (1) sends $P_i \mapsto S_i := P_i/\mathrm{rad}(P_i)$; the bijection from (3) to (2) sends the class of e_i to $P_i = Ae_i$. If A is split (for instance $k = \bar{k}$), the number of classes equals the number of blocks in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$, and

$$A/\mathrm{Jac}(A) \cong M_{d_1}(k) \times \cdots \times M_{d_r}(k), \qquad d_i = \dim_k S_i$$

where $S_1, \ldots, S_r$ are the simple modules and P_i appears $d_i = \dim_k S_i$ times in $A = \bigoplus_i P_i^{\oplus m_i}$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $\bar{A} = A/J$, which is semisimple by theorem 2.1.9.

Step 1: Simple modules are the same for A and $\bar{A}$. A simple A-module S is annihilated by J: the submodule $JS \subseteq S$ is either 0 or S, and $JS = S$ is impossible by Nakayama's lemma (lemma 2.1.7, applied to the cyclic module S: $JS = S$ would force $S = 0$). Hence $JS = 0$ and S is an $\bar{A}$-module; conversely every $\bar{A}$-module is an A-module through $\pi\colon A \to \bar{A}$. Since a submodule for A is the same as a submodule for $\bar{A}$ on a module killed by J, simplicity is preserved in both directions. Thus the simple A-modules are exactly the simple $\bar{A}$-modules, and by corollary 1.1.12 there are finitely many, one for each Wedderburn block of $\bar{A}$.

Step 2: $\mathrm{top}(P_i)$ is simple, and $P_i \mapsto \mathrm{top}(P_i)$ lands in (1). Reducing the decomposition $1 = e_1 + \cdots + e_n$ modulo J gives $\bar{1} = \bar{e}_1 + \cdots + \bar{e}_n$, a decomposition into orthogonal idempotents of $\bar{A}$. Applying the right-exact functor $\bar{A} \otimes_A -$ (equivalently reducing mod J) to $P_i = Ae_i$ gives

$$P_i/JP_i = Ae_i/JAe_i \cong \bar{A}\bar{e}_i$$

as $\bar{A}$-modules. Now $\bar{e}_i$ is primitive in $\bar{A}$: an orthogonal splitting $\bar{e}_i = \bar{g} + \bar{h}$ in $\bar{A}$ would lift, by theorem 2.3.1 applied inside the corner e_iAe_i, to an orthogonal splitting of e_i in A, contradicting primitivity of e_i. Over the semisimple algebra $\bar{A}$ a primitive idempotent generates a simple module, since $\bar{A}\bar{e}_i$ is an indecomposable summand of the semisimple module $\bar{A}$ and indecomposable semisimple means simple. Hence $\mathrm{top}(P_i) = P_i/\mathrm{rad}(P_i) = \bar{A}\bar{e}_i =: S_i$ is simple.

Step 3: The map (2) $\to$ (1) is a bijection. Distinct indecomposable projectives have distinct tops: if $Ae_i \cong Ae_j$ as A-modules then reducing mod J gives $\bar{A}\bar{e}_i \cong \bar{A}\bar{e}_j$, so $S_i \cong S_j$; conversely if $S_i \cong S_j$ then the two primitive idempotents $\bar{e}_i, \bar{e}_j$ generate isomorphic simple summands of $\bar{A}$, hence are conjugate in $\bar{A}$ (two primitive idempotents of a semisimple algebra generating isomorphic simples lie in the same Wedderburn block and differ by an inner automorphism), and a conjugating unit lifts to a unit of A because $\pi\colon A^\times \to \bar{A}^\times$ is surjective (an element of A mapping to a unit of $\bar{A}$ is itself a unit, since J is nilpotent: if $\bar{a}$ is invertible then a has a left and right inverse modulo the nilpotent ideal J, and $1 + J \subseteq A^\times$). Conjugate idempotents give isomorphic projectives $Ae_i \cong Ae_j$ via right multiplication by the conjugating unit. So $P_i \cong P_j$ if and only if $S_i \cong S_j$, giving a bijection between (2) and (1). Every simple module arises as some S_i: a simple $\bar{A}$-module is a summand of $\bar{A} = \bigoplus_i \bar{A}\bar{e}_i$, hence isomorphic to some $\bar{A}\bar{e}_i = S_i$.

Step 4: Conjugacy classes of primitive idempotents. By the last paragraph, $Ae_i \cong Ae_j$ as modules if and only if e_i and e_j are conjugate in A: one direction lifts a conjugation, the other reduces an isomorphism to a conjugation via $\mathrm{End}_A(A) \cong A^{\mathrm{op}}$ (lemma 1.2.3), under which a module isomorphism $Ae_i \to Ae_j$ is right multiplication by an element u with $ue_iu^{-1} = e_j$. Thus the class of e_i under conjugacy corresponds to the isomorphism class of P_i, giving the bijection between (3) and (2).

Step 5: The split case. If A is split, Artin–Wedderburn over the semisimple $\bar{A}$ (corollary 1.2.20) gives $\bar{A} \cong \prod_{i=1}^r M_{d_i}(k)$ with r the number of simple modules and $d_i = \dim_k S_i$, since the simple module of the block $M_{d_i}(k)$ is k^{d_i}. The multiplicity of P_i in $A = \bigoplus_j Ae_j$ equals the number of j with e_j conjugate to e_i, which reducing mod J equals the multiplicity of the column module k^{d_i} inside the block $M_{d_i}(k)$, namely d_i; so $m_i = d_i$. This gives the stated dimension bookkeeping, as we sought to show.

The correspondence is the reason the count of simple modules is finite and computable: it equals the number of Wedderburn blocks of $A/\mathrm{Jac}(A)$, which Artin–Wedderburn hands us. In the semisimple case the three sets coincide with the columns of the matrix blocks and the map $P_i \mapsto S_i$ is the identity; the passage to a general finite-dimensional algebra keeps the same indexing set but separates the projective P_i from the simple S_i it covers. The word "cover" is precise, and the next definition makes it so. A cover should be the most economical projective surjecting onto a module, with no wasted projective sitting over the radical.

Definition 2.3.6: Superfluous Submodule and Projective Cover

Let A be a ring and M an A-module. A submodule $L \subseteq M$ is *superfluous* (or *small*) if for every submodule $N \subseteq M$ the equality $L + N = M$ forces $N = M$. A surjection $f\colon P \to M$ is *essential* if $\ker f$ is superfluous in P.
A *projective cover* of M is a projective module P together with an essential surjection $f\colon P \to M$; that is, f is surjective, P is projective, and $\ker f$ is superfluous in P.

The condition that $\ker f$ be superfluous is what pins the cover down. Any module has a surjection from a projective (a free module surjects onto it), but without the superfluous condition one could pad the source with a free summand mapping to zero and never have a canonical choice. Superfluousness forbids exactly this padding: no proper submodule of P maps onto M, so P is as small a projective as can surject onto M. Over a finite-dimensional algebra the radical supplies a ready source of superfluous submodules, and this makes covers exist and be unique.

Theorem 2.3.7: Existence of Projective Covers

Let A be a finite-dimensional k-algebra and M a finitely generated (equivalently, finite-length) left A-module. Then M has a projective cover $f\colon P \to M$, unique up to isomorphism over M. For a simple module $S = S_i$, the projective cover is the indecomposable projective $P_i = Ae_i$ with its quotient map $P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$.

***Proof*:**
Write $J = \mathrm{Jac}(A)$.

Step 1: JP is superfluous in a finitely generated P. Let P be finitely generated and suppose $JP + N = P$ for a submodule $N \subseteq P$. Then $J \cdot (P/N) = (JP + N)/N = P/N$, so the finitely generated module P/N satisfies $J \cdot (P/N) = P/N$. Nakayama's lemma (the module form, lemma 2.1.7; concretely, using the nilpotence of J: if $J^N = 0$ and $JQ = Q$ then $Q = JQ = J^2Q = \cdots = J^NQ = 0$) forces $P/N = 0$, i.e. $N = P$. Hence JP is superfluous in P.

Step 2: Construction of the cover. The top $\bar{M} = M/JM$ is a module over the semisimple algebra $\bar{A} = A/J$, hence semisimple: $\bar{M} \cong \bigoplus_i S_i^{\oplus c_i}$ for multiplicities $c_i \geq 0$. Set

$$P = \bigoplus_i P_i^{\oplus c_i}$$

with each $P_i = Ae_i$ the indecomposable projective covering S_i. Then $P/JP = \bigoplus_i (P_i/JP_i)^{\oplus c_i} = \bigoplus_i S_i^{\oplus c_i} = \bar{M}$, so the projection $P \to P/JP = \bar{M}$ agrees with $M \to \bar{M}$ on tops. Because P is projective, the surjection $M \twoheadrightarrow \bar{M}$ lifts through $P \to \bar{M}$ to a homomorphism $f\colon P \to M$ with $M \to \bar{M}$ equal to $P \xrightarrow{f} M \to \bar{M}$. The image $f(P)$ satisfies $f(P) + JM = M$ (their images in $\bar{M}$ agree and equal all of $\bar{M}$), and JM is superfluous in M by Step 1 applied to M; hence $f(P) = M$ and f is surjective.

Step 3: f is essential. Let $K = \ker f$. Suppose $K + N = P$ for a submodule N. Then $f(N) = f(P) = M$, so $N + JP \supseteq N + \ker(P \to \bar{M})$ surjects onto $\bar{M} = P/JP$, giving $N + JP = P$; since JP is superfluous (Step 1), $N = P$. Thus K is superfluous and $f\colon P \to M$ is a projective cover.

Step 4: Uniqueness. Let $f\colon P \to M$ and $f'\colon P' \to M$ be two projective covers. Projectivity of P' lifts f' through f to $g\colon P' \to P$ with $fg = f'$. Then $f(g(P')) = f'(P') = M$, so $g(P') + \ker f = P$; since $\ker f$ is superfluous, $g(P') = P$ and g is surjective. As P' is projective, the surjection g splits, $P' = \ker g \oplus P''$ with $P'' \cong P$; but $\ker g \subseteq \ker f' = \ker(fg)$ is superfluous in P' and a superfluous direct summand is zero (a summand L with complement C satisfies $L + C = P'$, forcing $C = P'$ hence $L = 0$). So g is an isomorphism over M.

Step 5: The simple case. For $M = S_i$ the top $\bar{M} = S_i$ has $c_i = 1$ and all other multiplicities 0, so the construction returns $P = P_i$ with $f\colon P_i \twoheadrightarrow P_i/\mathrm{rad}(P_i) = S_i$, the stated cover, completing the proof.

The theorem certifies the terminology in definition 2.3.3: $P_i \to S_i$ really is *the* projective cover, and every finite-length module is covered by an explicit direct sum of the P_i read off from its top. Uniqueness makes the assignment $M \mapsto P(M)$ well defined and turns the correspondence of theorem 2.3.5 into a computational tool: to find the projective cover of any module, compute its top $M/\mathrm{Jac}(A)M$, decompose that semisimple module into simples, and sum the corresponding P_i. What the projective P_i contributes beyond the simple top S_i is recorded in its composition series, and tabulating those multiplicities across all i produces the central numerical invariant of the algebra.

Definition 2.3.8: Cartan Matrix of an Algebra

Let A be a finite-dimensional k-algebra with simple modules $S_1, \ldots, S_r$ and indecomposable projectives $P_1, \ldots, P_r$, indexed so that $P_i/\mathrm{rad}(P_i) = S_i$. The *Cartan matrix* of A is the $r \times r$ matrix $C = (C_{ij})$ of nonnegative integers

$$C_{ij} = [P_i : S_j]$$

the multiplicity of the simple module S_j among the composition factors of a composition series of P_i.

The multiplicities $[P_i : S_j]$ are well defined by the Jordan–Hölder theorem, which guarantees the composition factors of a finite-length module are independent of the chosen composition series. The Cartan matrix measures how far A is from semisimple in a single array: when A is semisimple every $P_i = S_i$ has length one, so $C_{ij} = \delta_{ij}$ is the identity matrix, and any off-diagonal or large diagonal entry records simples fused together by the radical. The diagonal entry $C_{ii} = [P_i : S_i]$ is at least 1 (the top contributes one copy of S_i), and $C_{ii} > 1$ signals a self-extension living in the radical. The next example computes the whole apparatus of this section for the smallest non-semisimple algebra with two non-isomorphic simple modules, so that the Cartan matrix is not diagonal.

Before the example, one further point for completeness: the entire theory has a dual half. Everything above was built from the regular module A as a source of projectives; dualizing over k turns projectives into injectives and covers into envelopes.

2.3.9 Remark: The Dual Theory of Injective Envelopes Dual to a projective cover is an *injective envelope*: an injective module E with an essential injection $M \hookrightarrow E$, meaning every nonzero submodule of E meets the image of M nontrivially. Over a finite-dimensional algebra A these exist and are unique, and the duality $D = \mathrm{Hom}_k(-, k)$ carries left A-modules to right A-modules (equivalently left A^{op}-modules), exchanging projective covers of A-modules with injective envelopes of A^{op}-modules. The indecomposable injective A-modules are thus $D(e_i A)$, and each has a simple *socle* (its largest semisimple submodule) rather than a simple top; the socle plays for injectives the role the top plays for projectives. The general theory of injective and projective modules over a ring, including existence of injective envelopes via injective hulls, is developed in the companion volume `NateCoreAlgebra` [5]; the present section uses only the projective side.

Example 2.3.10: Projective Covers over the Upper-Triangular Algebra $T_2(k)$

Let

$$A = T_2(k) = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} : a, b, c \in k \right\}$$

the algebra of 2×2 upper-triangular matrices over a field k, a three-dimensional k-algebra. We work out its idempotents, indecomposable projectives, simple modules, projective covers, and Cartan matrix.

Idempotents. The matrix units

$$e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

are orthogonal idempotents with $e_1+e_2 = 1$. Each is primitive: the corner algebra $e_1Ae_1 = k\,e_1 \cong k$ has no idempotents but $0, e_1$, and likewise $e_2Ae_2 \cong k$, so by theorem 2.3.2 both Ae_1 and Ae_2 are indecomposable. Thus $1 = e_1 + e_2$ is a primitive orthogonal decomposition and there are two indecomposable projectives.

The radical. The Jacobson radical is the strictly upper-triangular part,

$$\mathrm{Jac}(A) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix}$$

a one-dimensional nilpotent ideal with $\mathrm{Jac}(A)^2 = 0$, and the quotient is

$$A/\mathrm{Jac}(A) \cong \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \cong k \times k$$

a product of two copies of k, confirming via theorem 2.3.5 that there are exactly two simple modules, both one-dimensional.

The indecomposable projectives. Left multiplication by A on e_i gives, with $E_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$,

$$P_1 = Ae_1 = \begin{pmatrix} k & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_2 = Ae_2 = \begin{pmatrix} 0 & k \\ 0 & k \end{pmatrix}$$

so $\dim_k P_1 = 1$ and $\dim_k P_2 = 2$, and indeed $\dim_k A = 1 + 2 = 3 = \dim_k P_1 + \dim_k P_2$, consistent with $A = P_1 \oplus P_2$.

The simple modules and covers. The tops are

$$S_1 = P_1/\mathrm{rad}(P_1) = P_1/\mathrm{Jac}(A)P_1 = P_1 \cong k, \qquad S_2 = P_2/\mathrm{rad}(P_2) = P_2/\mathrm{Jac}(A)P_2 \cong k$$

Here $\mathrm{Jac}(A)P_1 = 0$, so $P_1 = S_1$ is already simple and is its own projective cover, an essential surjection $P_1 \xrightarrow{\cong} S_1$. For P_2, the radical $\mathrm{Jac}(A)P_2 = \left(\begin{smallmatrix} 0 & k \\ 0 & 0 \end{smallmatrix}\right)$ is one-dimensional, and

$$\mathrm{rad}(P_2) = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix} \cong S_1, \qquad P_2/\mathrm{rad}(P_2) \cong S_2$$

So P_2 is a length-two module with a nonsplit exact sequence

$$0 \to S_1 \to P_2 \to S_2 \to 0$$

and the quotient map $P_2 \twoheadrightarrow S_2$ is the projective cover of S_2, with kernel $\mathrm{rad}(P_2) = S_1$ superfluous in P_2 (it lies inside $\mathrm{Jac}(A)P_2$). The two simple modules S_1 and S_2 are not isomorphic, since e_1 and e_2 act as 1 on one and 0 on the other; the algebra is not semisimple, as $\mathrm{Jac}(A) \neq 0$.

The Cartan matrix. Reading off composition factors: P_1 has the single factor S_1, while P_2 has factors S_2 (its top) and S_1 (its radical). Hence

$$[P_1 : S_1] = 1, \quad [P_1 : S_2] = 0, \quad [P_2 : S_1] = 1, \quad [P_2 : S_2] = 1$$

and the Cartan matrix is the upper-triangular integer matrix

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Its determinant is 1 and its off-diagonal entry $C_{21} = 1$ records the single copy of S_1 that the radical of P_2 contributes: the exact numerical trace of the failure of $T_2(k)$ to be semisimple, where the Cartan matrix would have been the identity.

3

Representation of Finite Groups

All modules will be left R-modules for a not necessarily commutative ring R. Furthermore, all actions will be non-trivial actions unless specified otherwise.

3.1 Representations of Finite Groups

Similarly to how we studied groups and by how they acted on themselves through group-actions (ex. how a group acted on itself by conjugation gave us Sylow's Theorem), we will now try to study groups using *linear algebra*. Many really difficult results in classical group theory will become much easier when studied through this new "representation" of groups (and hence why it's called *represenation theory*). Furthermore, vector-spaces afford many more structures than a module structures (ex. a complex vector-space is the starting point for the Nullstellensatz, Complex Geometry, Analysis, and much more!) Hence, one large advantage of Representation theory will be the intersection of many mathematical diciplines which contribute towards our study (an they all feedback on each other)

In this part of the group, G will always be a finite group[1], k will always be a field (usually $\mathbb{C}$, but can be any field of any character), $\mathrm{Aut}_{k\text{-}\mathbf{Vect}}(V) = \mathrm{GL}(V)$ represents all automorphisms from the k-vector space V to itself, which when $\dim(V) < \infty$ is equivalent to the group of all nonsingular linear transformations from V to itself $\mathrm{GL}_n(k)$ where $n = \dim(V)$.

Definition 3.1.1: Representation

Let G be a finite group and V a vector space over k. Then a *linear representation* is any homomorphism $\varphi : G \to \mathrm{GL}(V)$. The *degree* of the representation is the dimension of V. A *matrix representation* of G is any homomorphism $\varphi : G \to \mathrm{GL}_n(k)$

[1]Not sure about where to address infinite groups

Usually, when we say representation we will mean *linear representation*. If a vector-space V has a representation of G, we will often call it a G-representation. To be as rigorous as possible we would say we have a representation of G on V over k, if the context requires it. Furthermore, most of the time we will have map from G to $\mathrm{GL}(V)$ to be *injective*; non-injective representations will be more common when representing infinite groups that are "trivial" on some subgroup (see [3]). Just like for group actions, A representation is *faithful* if it is injective.

As usual, we can curry or uncurry an action[2]: we can take $\varphi : G \to \mathrm{GL}(V)$ and make it $G \times V \to V$ where $g : V \to V$ with $v \mapsto g \cdot v$ is linear over k for all g. It is left for the curious to see that this is indeed the proper uncurrying of this action.

Using the definition of $\varphi : G \to \mathrm{GL}(V)$, it is easy to define a representation homomorphism mirroring the definition of group-actions and modules (which are ring actions on abelian groups):

Definition 3.1.2: Representation Homomorphism

Let V, W be two vectors spaces over k with a linear representation by G. Then $\varphi : V \to W$ a *representation homomorphism* if φ k-linear and for any $g \in G$:

$$\varphi(g \cdot v) = g \cdot \varphi(v)$$

such a homomorphism will be called G-linear.

From this perspective, it might be hard to see how our previous discussion on semi-simple rings might give us any insight. We thus introduce a perspective better suited to the results of chapter 1: every representation of G on V over k is in bijective correspondence with all $k[G]$-modules over V, where $k[G]$ represents the *group ring* as seen in section **??**. This second perspective will also make it easier to define a couple of examples, and so we will delay giving examples till the correspondence has been established. As a reminder, a group ring $k[G]$ over a ring k (which in our case will always be a field) is the set of all formal sums:

$$\sum_{g_i \in G} a_i g_i$$

where addition and multiplication are defined in the natural way[3]:

$$\sum_i^n a_i g_i + \sum b_i g_i = \sum_i (a_i + b_i) g_i$$

$$\left(\sum_i^n a_i g_i\right)\left(\sum b_i g_i\right) = \sum_k^n \left(\sum_{\substack{i,j \\ g_i g_j = g_k}} a_i b_j\right) g_k$$

Another common notation for groups rings is kG, where the brackets are dropped. A couple of important acts about group rings should be recalled (or the excises ref:HERE can be completed)

1. The group G naturally injects into $k[G]$ by identifying g_i with $1g_i$ which is the formal sum $\sum_j \delta_{ij} g_i$ (where δ_{ij} is the Kronecker delta).

[2]See remark **??**

[3]A word on this really being a natural way?

2. The field k naturally injects in $k[G]$ by corresponding an element $a \in k$ with $ag_1 = ae$ where $g_1 = e$ is the identity. If it is clear from context, we will sometimes simply write $a \in k[G]$ if it is understood $a = ae = ag_1$.

3. There is a natural k action on $k[G]$ defined by

$$b \cdot \left(\sum_i a_i g_i \right) = \sum_i (ba_i) g_i$$

due to this, $k[G]$ is a vector space over k with the elements of G as a basis, and hence is finite dimensional if $|G| < \infty$.

4. By definition, The elements of k identified as kg_1 commute with all the elements of $k[G]$, hence it is in the center of $k[G]$: $k \subseteq Z(k[G])$. Thus, $k[G]$ is in fact a k-algebra.

5. A group ring the adjoint given by the forgetful functor from **Ring** to **Grp**, hence is free with respect to groups, justifying the name "formal sum" for it's elements.

Using these, we show how representations of G on V over k correspond to $k[G]$-modules: let $\varphi : G \to \mathrm{GL}(V)$ be a representation, with $G = \{g_1, g_2, ..., g_n\}$. Thus, each $\varphi(g_i)$ is linear transformation (in particular an automorphism). Define the $k[G]$ action on V by defining:

$$\left(\sum_i^n a_i g_i \right) \cdot v = \left(\sum_i^n a_i \varphi(g_i) \right)(v) = \sum_i^n a_i \varphi(g_i)(v)$$

Then this defines a $k[G]$-module (as should be verified if it doesn't seem like it). A particular reminder for those who are checking that $\varphi(g_i g_j) = \varphi(g_i) \circ \varphi(g_j)$, since φ is a group homomorphism onto $\mathrm{GL}(V)$ under composition. Notice how this action in fact extends the action of k on $k[G]$ since $a \cdot v = av$ (similarly to how the $F[x]$ action extended the action of F on V in section **??**).

Conversely, let's say V is an $k[G]$-module. Seeking to make V a vector space over k and finding a group homomorphism $\varphi : G \to \mathrm{GL}(V)$, the first thing is that V is already a K vector-space by limiting to the action of $k \subseteq k[G]$. Furthermore, for each $g \in G$, we can obtain a map $\varphi : V \to V$ by defining:

$$\varphi(g)(v) = g \cdot v$$

Then it is easy to check that $\varphi(g)$ is a linear transformation: for $\alpha, \beta \in k$, $v, w \in V$

$$\begin{aligned} \varphi(g)(\alpha v + \beta w) &= g \cdot (\alpha v + \beta w) \\ &= \alpha(g \cdot v) + \beta(g \cdot w) \\ &= \alpha\varphi(g)(v) + \beta\varphi(g)(w) \end{aligned}$$

and by the associativity axiom of modules, we get

$$\varphi(g_i g_j)(v) = (\varphi(g_i) \circ \varphi(g_j))(v)$$

Thus, $\varphi : G \to \mathrm{GL}(V)$ is indeed a group homomorphism, and we have:

$$\left\{ V \text{ an } k[G]\text{-module} \right\} \rightleftarrows \left\{ \begin{matrix} V \text{ a vector space over } k \\ \text{and} \\ \varphi : G \to \mathrm{GL}(V) \text{ a representation} \end{matrix} \right\}$$

Note that this also means that this also shows that linear representations also have kernel objects, since modules have kernels. Hence, the kernel of a G-representation homomorphism between V and W is G-invariant.

Definition 3.1.3: Affording Representation

Let V be a $k[G]$-module. Then given the previous discussion, we say that V *affords* the representation φ of G.

Given this new perspective, we can also characterize sub-actions. If we have a $k[G]$-module V, then a $k[G]$-submodule $W \subseteq V$ would have to be a k-submodule, and hence a subspace, but it also has to be closed under the action of g on W, i.e. $g \cdot W \subseteq W$. Hence, any $k[G]$-submodule is a G-stable subspace of V. Conversely, if $W \subseteq V$ is G-stable, then it is closed under the action of G and is a subspace, and hence a $k[G]$-submodule. Hence

$$\left\{ W \subseteq V \text{ a } k[G]\text{-submodule} \right\} \rightleftarrows \{W \subseteq V \text{ a } G\text{-stable subspace}\}$$

Next, we should know when two representations are equivalent. Recall that when studying a linear transformation T on V to itself, a basis for V can be found that gives a "canonical form" to the matrix representation of T. Changing basis for V kept V the same, but changed the matrix representation of T up to similarity (i.e. changed the isomorphism from $\mathrm{GL}(V)$ to $\mathrm{GL}_n(k)$). The following is the analogous term:

Definition 3.1.4: Equivalent Representation

Let φ and ψ be two representations of G over k. Then we say these two represetnations are *equivalent* (or *similar*) if the $k[G]$-module affording them are isomorphic as modules. If two representations are not equivalent, they are called *inequivalent* (or *not similar*).

When the field is known[a], if V and W are equivalent (i.e. isomorphic as $k[G]$-modules), then we write $V \cong_G W$.

[a]In most cases, the field will be $\mathbb{C}$

Note that two different representations can mean two different vector-spaces over k. If $\varphi : G \to \mathrm{GL}(V)$ and $\psi : G \to \mathrm{GL}(W)$ are equivalent representations, it would make sense if there is a relation between V and W. If $T : V \to W$ is the $k[G]$-module isomorphism between them, It is a k-module isomorphism, and hence $V \cong W$ as vector spaces, implying $\dim(V) = \dim(W)$. Furthermore, by definition of T, $T(g \cdot v) = g \cdot T(v)$. By definition of the action, we have that $T(\varphi(g)(v)) = \psi(g)(T(v))$, i.e.

$$T \circ \varphi(g) = \psi(g) \circ T \quad \Leftrightarrow \quad T \circ \varphi(g) \circ T^{-1} = \psi(g) \qquad \forall g \in G$$

That is, T is the *simultaneous change of basis* for all $\varphi(g)$, $g \in G$. In terms of matrices, there must exist a fixed invertible matrix P such that for all $g \in G$,

$$P\varphi(g)P^{-1} = \psi(g)$$

The function T (or matrix P) are said to *intertwine* φ and ψ (it gives the rules on how to relate φ and ψ).

Example 3.1.5: Representations

1. Let G be any finite group, V any 1 dimensional vector-space over k. Define the $k[G]$-module structure on V by
$$g \cdot v = v$$
for all $g \in G$, $v \in V$. This called the *trivial representation* of G. This module affords the representation $\varphi : G \to \mathrm{GL}(V)$ by $\varphi(g) = I$ (the corresponding matrix representation from G to $\mathrm{GL}_1(k)$ sends $g \mapsto I$).

2. Let G be any finite group and k any field. Let $V = k[G]$ and consider it as a $k[G]$-module over itself. Then $\dim_k(V) = |G| = n$, with the basis being the elements of G. Then we see that each $g \in G$ permute the basis elements:
$$g \cdot g_i = gg_i = g_j \in G$$
Thus, the matrix representation for the action by the element g is an $n \times n$ matrix with 1 in the ij component if $g \cdot g_j = g_i$ (hence n 1's) and zero everywhere else. This representation of g is called the *regular representation* of G. Note how the underlying field didn't matter: the coefficients in the matrices can only ever by 1 or 0, and we would simply say "Let φ be the regular representation of G" without specifying which field the group-ring is.

3. Let $G = S_n$ ($n \in \mathbb{N}_{>0}$) and k be any field. Then let V be a vector-space over k of dimension n, $(e_1, e_2, ..., e_n)$ a basis for V, and define the action:
$$\sigma \cdot e_i = e_{\sigma(i)}$$
that is, σ permute the basis of V. Through this, we see that we afford a faithful representation $\varphi : S_n \hookrightarrow \mathrm{GL}(V)$. Furthermore, since any finite group is a subgroup of S_n for an appropriate n, we see that a group of order n we can define $\varphi : G \to S_n \to \mathrm{GL}(V)$, giving another perspective on how to embed any finite group into a matrix group. Notice too how we made a $k[G]$-module given the action of a $k[S_n]$-module. This is called the *permutation representation* of G.

   Note that this means that *any* finite groups acts on a vector space, and by using this construction we see that any finite group G of size n can be embedded into $\mathrm{GL}_n(k)$ showing how much structure the matrix groups embody.

4. Generlizing the last result, if $\psi : H \to \mathrm{GL}(V)$ is any representation and $\varphi : G \to H$ is a group homomorphism, then $\psi \circ \varphi$ is a representation of G on V.

5. Recall that in section **??** we defined a *group character* to be a homomorphism from G to $L^\times$. This is in fact a 1-dimensional representation of G, since $\mathrm{GL}_1(L) = L^\times$. For a concrete example, let $G = C_n$ be the cyclic group of order n and $\zeta_n \in \mathbb{C}^\times$ the nth root of unity. Then the map:
$$\chi : G \to \mathbb{C}^\times \qquad a^i \mapsto \zeta_n^i$$
is a faithful representation of G.

6. Let $G = D_8$. We already have the regular represenation of dimension 8 and premutation represetntation of dimension 8, but can we do better? In fact we can. In order to see this, let's first write out D_8 in terms of generators and relations:
$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$$

Let $k = \mathbb{R}$ and consider $\mathrm{GL}_2(\mathbb{R})$. If we can find matrices that satisfy these relations, than we would have a faithful representation of degree 2. This can be laborious, but with some work we can find:

$$r \mapsto \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \qquad s \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Let $G = Q_8$. We will show how different fiels can produce different represetnations. First, recall the following two generator and relation set for Q_8:

$$Q_8 = \langle i, j \mid i^4 = j^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

$$Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = 1 \rangle$$

If $k = \mathbb{C}$, and we work over $\mathrm{GL}_2(\mathbb{C})$ we can use the first representation an find:

$$i \mapsto \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \qquad j \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

can you find one if $k = \mathbb{R}$ and $\mathrm{GL}_4(\mathbb{R})$? (Hint: think of the Hamiltonian's $\mathbb{H}$)

7. Let $H \trianglelefteq G$ which is further an abelian p-group for some prime p. Then $V = H$ is a vector-space over $\mathbb{F}_p$ with the action of $a \cdot v = v^a$. Then define the action of G on H by conjugation: $g \cdot h = ghg^{-1}$. This action is well-defined since $gh^ag^{-1} = (ghg^{-1})^a$. The kernel of this action is $C_G(H)$. Such an action is called a *module representation.*

8. Let V be the $k[S_n]$-module which affords the permutation representation. We will find 2 S_n-invariant subspaces: Let $N \subseteq V$ be the subspace defined by:

   $$N = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\}$$

   Then this a subspace that is also S_n-invariant. As practice, show that if $n \geq 3$, then N is the unique 1-dimension subspace which is S_n-stable. This submodule is called the *trace* submodule of V.

   Another natural S_n-stable subspace is the following: take $I \subseteq V$ such that

   $$I = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\}$$

   notice that $I = \ker(\varphi)$ where $\varphi : V \to \mathbb{C}$, $\varphi(\alpha_1 e_1 \alpha_2 e_2 \cdots \alpha_n e_n) = \sum_i \alpha_i$, and so I is $n-1$ dimensional. Since φ is called the *augmentation map* (was in section 7.3 of D&F), I is called the *augmentation module.*

9. If V is a regular representation of G so that $V = k[G]$, then V has the same two natural G-invariant subspaces:

   $$N = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 = \cdots = \alpha_n\}$$
   $$I = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_1 + \cdots + \alpha_n = 0\}$$

   these are both in fact *two-sided ideals*, even if G is not abelian. Since they appear naturally, they are usually given a name: N is called the *trace ideal* and I is called the *augment ideal.*

Corollary 3.1.6: Schur's Lemma For Representations

Let V, W be irreducible G-representations (over $\mathbb{C}$) with represetnations ρ_V and ρ_W. Then:

1. If $V \not\cong_G W$, then there does not exist a non-trivial G-linear map between V and W
2. Let's say $V = W$, are finite dimensional vecor-spaces with $\rho_V = \rho_W$, then the only non-trivial G-linear maps from V to W are scalars: $\lambda I : V \to W$, $\lambda I(v) = \lambda v$.

***Proof*:**

1. Since V and W are irreducible G-representations, they are simple $k[G]$-modules, and hence we apply the original Schur's lemma.
2. Let $V = W$ of the same dimension with matching actions $\rho_V = \rho_W$, and let $f : V \to W$. Since f is a linear transformation over $\mathbb{C}$, there always exists an eigenvalue, say λ so that $f(x) = \lambda x$. Consider:
$$\varphi = f - \lambda I$$
so that $\varphi(x) = 0$. Since the difference of two G-linear maps is G-linear, then $\ker(\varphi)$ is a G-invariant subspace. But V is irreducible, and $\ker(\varphi)$ is non-empty, and hence must be all of V, and so $\varphi(V) = 0$, telling us that $0 = f(x) - \lambda I(x)$, i.e.
$$f(x) = \lambda x$$
completing the proof.

Recall that if V is a simple R-module over a semi-simple ring R, then $\text{End}_R(V) \cong_{\mathbf{Ring}} D^{\text{op}}$ is a division ring, and that any finite-dimensional division $\mathbb{C}$-algebra is in fact equal to $\mathbb{C}$. This lemma now gives an explicit isomorphism between these two!

Since we are working over a vector-space, it is often convenient when possible to bring in an inner product that give further relations between our objects. The key the following inner product is that it is invariant on G-actions:

Definition 3.1.7: G-Invariant Inner Product

Let V be $k[G]$-module. Then a *G-invariant inner product* is an inner product $\langle \cdot, \cdot \rangle$ along with the additional property of:
$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle$$

Instead of giving examples of this inner product, we will soon show if V is a semi-simple $k[G]$-module, we can transform any inner product into a G-invariant inner product (hint: can you see how to use the "averaging" idea in Machke's Theorem to do so? The answer is given in corollary 3.1.12, page 77).

Now, the study of linear representations in themseves can difficult; we are essentially studying all possible $\mathbb{C}[G]$-module. We would rather be able to decompose them into smaller pieces, and show that any represetnation of G is some how build form these simpler pieces (similarly to our exploration of semi-simple modules). The rest of this chapter is dedicated to seeing if we can bring

Artin-Wedderburn's theorem and all of it's consequences to understanding linear representations of finite groups:

Definition 3.1.8: Decomposing Representations

A representation is called *simple, reducible, indecomposable, decomposable*, or *semisimple* if the $k[G]$-module affording it has the corresponding property

Recall that indecomposable in general does not imply irreducible. The example under definition 1.1.1 could is in fact a representation of $\mathbb{Z}$ defined by:

$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

which is reducible (since it is in block upper-triangular form) but not indecomposable (since it's not in block-diagonal form). Notice that any 1-dimensional representation is automatically an irreducible, since any $k[G]$-submodule of k would be a k-subspace of k, but k is already irreducible as vector-space over k.

To get an idea of how $k[G]$-submodules interact with the definitions above, let V be an n-dimensional representation of G affording the represetnation ρ, and $W \subseteq V$ an m-dimensional $k[G]$-submodule (i.e. m-dimensional G-invariant subspace) with $0 \neq m \neq n$. Then choosing a basis $\{w_1, w_2, ..., w_m\}$ for W, we can extend it to a basis $\{w_1, w_2, ..., w_m, v_{m+1}, ..., v_n\}$. Since W is G-invariant, for any $w \in W$, $\rho(g)(w) \in W$. In matrix represetntation, $\varphi(g)(w^i) = \rho^i_j w^j$, and the coefficeints of $v^{m+1}, ..., v_n$ are zero. Hence we get the matrix representation of $\rho(g)$ as:

$$\begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix}$$

Notice too that:

$$\begin{aligned}
\begin{pmatrix} \alpha(gh) & \beta(gh) \\ 0 & \gamma(gh) \end{pmatrix} &= \rho(gh) \\
&= \rho(g)\rho(h) \\
&= \begin{pmatrix} \alpha(g) & \beta(g) \\ 0 & \gamma(g) \end{pmatrix} \begin{pmatrix} \alpha(h) & \beta(h) \\ 0 & \gamma(h) \end{pmatrix} \\
&= \begin{pmatrix} \alpha(g)\alpha(h) & \alpha(g)\beta(h) + \beta(g)\gamma(h) \\ 0 & \gamma(g)\gamma(h) \end{pmatrix}
\end{aligned}$$

which shows the the converse is also true: if we find a basis so that the matrices are block upper triangular, then the subspace spanned by the first m are G-invariant. As an example of this, let V be the permutation representation of S_3. Then the set of all vectors that are invariant to the action of $G = S_3$ is $V^G = \text{span}(e_1 + e_2 + e_3)$. Let $W = \text{span}(e_1 + e_2 + e_3)$. Since $G \curvearrowright W$ is of the form $g \cdot w = w$, we see that W is in fact the *trivial representation*. Since V is semi-simple, we have that $V = V^G \oplus V^\perp$. Making some new basis for V, say $(e_1 + e_2 + e_3, e_2, e_3)$, we see that $S_3 \curvearrowright V$ will have matrix representations that are of the above form:

$$\begin{pmatrix} 1 & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$$

If V were irreducible, then no such basis would exist. If $V = U \oplus W$, then we would get a block diagonal form. It turns out that all representations of groups over either a character zero ring or fields where $|G| \nmid \mathrm{char}(k)$ are block-diagonal, i.e. are semisimple!

Theorem 3.1.9: Maschkle's Theorem

Let G be a finite group and let k be a field whose characteristic does not divide $|G|$ or $\mathrm{char}(k) = 0$. Then if V is a $k[G]$-module, it is a semi-simple $k[G]$-module.

The condition that $|G| \nmid \mathrm{ch}(k)$ is equivalent to $|G| \in k^{\times}$. Note how this theorem is "surprising" as a finite group G certainly cannot be reduced to a direct sum of irreducible (or simple) groups, but as extensions of simple groups, but furthermore (as of writing this book) we do not know how to construct all finite group via extensions of simple groups. Hence, having a very computable decomposition into irreducible representations provides a very potent tool to analyze the structure of finite groups.

***Proof*:**
We will use lemma 1.1.7[3] to show that for any $U \subseteq V$, there exists a compliment W such that $V = U \oplus W$

We will find a projection onto U: $\pi : V \to U$. Thus, we need to show it has the following properties:

1. $\pi(u) = u$ for all $u \in U$
2. $\pi(\pi(v)) = \pi(v)$ for all $v \in V$[a]

As a reminder on why this is sufficient[b], let π be such that the function defined on the $k[G]$-module that satisfies these properties. Let $W = \ker(\pi)$. Then $W \subseteq V$ is a submodule, and W is the comliment of U. First, let $v \in U \cap W$. Then since $v \in U$, $\pi(v) = v$ and since $v \in W$, $\pi(v) = 0$, hence $v = 0$, meaning $U \cap W = 0$. Next, let $v \in V$ be an arbitrary element, and write $v = \pi(v) + (v - \pi(v))$. By definition, $\pi(v) \in U$, and by the 2nd property:

$$\pi(v - \pi(v)) = \pi(v) - \pi(\pi(v)) = \pi(v) - \pi(v) = 0$$

and hence $v - \pi(v) \in W$. Hence $v \in U + W$, and so $V = U + W$, hence $V \cong U \oplus W$. Hence, it suffices to find an $k[G]$-module projection π.

Let $U \subseteq V$ be a $k[G]$-submodule. Then it is already a k-subspace, and so it has a vector-space direct sum compliment $W_0 \subseteq V$ (take a basis β for U, extend it to a basis $\mathcal{B}$ for V, then take $\mathcal{B} - \beta$) Thus, $V = U \oplus W_0$ as a vector-space. However, W_0 need not be G-stable (i.e. a $k[G]$-submodule), for example $G = C_2$ on $\mathbb{R}^2$ with the refection across $y = x$ which has the G-subspace $y = x$ and compliment spanned by $(1, 0)$ which is certainly not G-invariant. We will show how to find such the appropriate $k[G]$-module given W_0. First, let $\pi_0 : V \to U$ be the vector-space projection map:

$$\pi_0(u + w_0) = u$$

Here is the key to finding a proper compliment: we will "average" π_0 over G as follows:

$$g\pi_0 g^{-1} : V \to U \qquad g\pi_0 g^{-1}(v) = g \cdot \pi_0(g^{-1} \cdot v)$$

Since π_0 maps V to U, and U is a $k[G]$-submodule, U is stable under this G-action, hence $g\pi_0 g^{-1}$ is well-defined. Note that both g and g^{-1} act as k-linear transformations, so $g\pi_0 g^{-1]}$

is a k-linear transformation. Furthermore, if $u \in U$, then so must $g^{-1}u$, so by definition of π_0, $\pi_0(g^{-1}u) = g^{-1}u$. Thus, for all $g \in G$:

$$g\pi_0 g^{-1}(u) = u$$

hence, $g\pi_0 g^{-1}$ is also a vector-space projection of V onto U!

Let $n = |G|$, and viewing n as an element of k $(n = 1 + \cdots + 1)$, since $\mathrm{ch}(k) \nmid |G|$ $n \neq 0$, i.e. $n \in k^\times$. Thus, n has an inverse in k, meaning we can define:

$$\pi = \frac{1}{n}\sum_{g \in G} g\pi_i g^{-1}$$

We can immediately deduce that π has the following properties:

1. By definition, π is still k-linear from V to U.
2. Restricting π to U would restrict each summand in the definition of π, and so as we've pointed out earlier each summand would be the identity map. Hence, we get $1/n$ times n copes of u, meaning $\pi|_U = \mathrm{id}$
3. π is well-defined, hence $\pi(v) \in U$, so $\pi(v)^2 = \pi(v)$

It remains to show that π is in fact a $k[G]$-module homomorphism (i.e. is $k[G]$ linear). It suffices to show that for any $h \in G$, $\pi(hv) = h\pi(v)$. This is simply a computation:

$$\begin{aligned}
\pi(hv) &= \frac{1}{n}\sum_{g \in G} g\pi_0(g^{-1}hv) \\
&= \frac{1}{n}\sum_{g \in G} h(h^{-1}g)\pi_0(g^{-1}hv) \\
&= \frac{1}{n}\sum_{\substack{k=h^{-1}g \\ g \in G}} h(k\pi_0(k^{-1}v)) \\
&\overset{!}{=} h\frac{1}{n}\sum_{\substack{k=h^{-1}g \\ g \in G}} (k\pi_0(k^{-1}v)) \\
&= h\pi(v)
\end{aligned}$$

where $\overset{!}{=}$ comes form the fact that as g goes over all elements of G, then so will $k = h^{-1}g$ (i.e. groups acting on themselves are always transitive actions). Hence, π is indeed a $k[G]$-homomorphism, and so it is a projection map, completing the proof.

[a] In fact, this is implied by 1

[b] see ref:HERE for the original development of this

Thus, if we take any representation V over k of G, and consider the $k[G]$-module, then we know through $k[G]$ and G being finite and k a field (so V is a vector-space) that there are only *finitely many* simple $k[G]$-modules (i.e. finitely many irreducible representations of G), meaning we only have a finite number of "building blocks" from which we can build) any representation for G! Even better, the way we put together irreducible representation is through direct sums the simplest of ways of putting together algebraic objects! Then through the Artin-Wedderburn decomposition of V, we can find all *irreducible* representations that make up the representation V of G, giving us a simple finite list of irreducible representations per finite group! Note that if $p = \text{ch}(k) > 0$ and $p \big| |G|$, then $k[G]$ is *not* semisimple, or if G is infinite, it need not be semi-simple

Example 3.1.10: Non-Simple Group-Ring

For the infinite case, we've presented $\mathbb{Z} \curvearrowright \mathbb{Z}^2$ above. For the finite case where $p \big| |G|$,

1. let $G = C_n$ be the cyclic group of order n. Show that $k[C_p] \cong k[a]/(a^n - 1)$
2. Show that if $\text{char} k \nmid |G|$, then $k[C_n]$ is isomorphic product of fields. Since $M_1(k) = k$, by the uniqueness of Artin Wedderburn, this is the Artin-Wedderburn decomposition. Conclude that since $k[C_n]$ is abelian, any Artin-Wedderburn decomposition *has* to be a product of fields.
3. Show that if $\text{char}(k) = p$, then $k[C_p]$ has nilpotent elements, and hence cannot be a product of fields.

If $k = \mathbb{C}$, then Maschkle's Theorem always applies, and we can further use Artin Wedderburn over closed k-algebras (corollary 1.2.20) to deduce a relatively simple structure! Hence, from now on, we will always have $k = \mathbb{C}$, meaning all of our representations of finite groups will be over the complex vector-field of finite dimension.

Corollary 3.1.11: Matrix Representation Consequences

Let $\rho : G \to \text{GL}(V)$ be a representation over a field k where $\text{char}(k) \nmid |G|$. Then there exists a basis for V such that for all $g \in G$, the matrix $\rho(g)$ is of block-diagonal form:

$$\begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \ddots & \\ & & & \rho_m \end{pmatrix}$$

where ρ_i is an irreducible representation of G, $1 \leq i \leq m$.

***Proof*:**

Let V be the $k[G]$-module representing ρ. By Maschkle's Theorem, we may write $V = U_1 \oplus \cdots \oplus U_m$. where U_i is simple $k[G]$-module. For each U_i, choose a basis β_i. Let $\beta = \bigcup_i^m \beta_i$. then each matrix representation of $\rho(g)$ is of the form in the corollary, with $\rho(g)|_{U_i} = \rho_i(g)$, completing the proof.

Corollary 3.1.12: Defining G-Invariant Inner Product

Let V be a $k[G]$-module and $\langle \cdot, \cdot \rangle$ be an inner product. Then this inner product induces a G-invariant inner product

***Proof*:**
If $\langle \cdot, \cdot \rangle$ is an inner product, then:

$$\langle v, w \rangle_G = \frac{1}{|G|} \sum_{g \in G} \langle gv, gw \rangle$$

is a well-defined inner product (as can be checked), and for any $h \in G$

$$\begin{aligned} \langle hv, hw \rangle_G &= \frac{1}{|G|} \sum_{g \in G} \langle ghv, ghw \rangle \\ &= \frac{1}{|G|} \sum_{k = gh \in G} \langle kv, kw \rangle \\ &= \langle v, w \rangle_G \end{aligned}$$

In other words, there is always an inner product on our space in which the group actions keeps all elements of the same length and the same angle between them, showing how we can interpret a group as somehow rotating and reflecting an object in a vector-space. Now that we know any represetnation of G is composed in some way by finitely many irreducible representations, it would be useful to be able to quickly deduce the Artin-Wedderburn decomposition of $\mathbb{C}[G]$:

Proposition 3.1.13: Properties Of Representations

Let $k = \mathbb{C}$ and G be finite. Let $S_1, S_2, ..., S_r$ be the irreducible G-representations (up to isomorphism). Then:

1. the number of irreducible G-representations is equal to the number of conjugacy classes in G
2. For every representation V of G,
$$V \cong S_1^{\oplus \ell_1} \oplus \cdots \oplus S_r^{\ell_r}$$
where $\ell_i > 0$ and is unique
3. $\mathbb{C}[G] \cong \bigoplus_{i=1}^n \left(S_i^{\oplus \dim_k S_i} \right)$ as regular representation
4. $|G| = \sum (\dim_k S_i)^2$

***Proof*:**
part (3) and (4) come from corollary 1.2.20.

For part (1), we first use Artin-Wedderburn to conclude that $Z(\mathbb{C}[G]) = \prod_i^r \mathbb{C}$, so $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = r$, so the dimension of the center is equal to the number of irreducible G-representations.

Now, let $\mathcal{K}_1, ..., \mathcal{K}_s$ represent the conjugacy classes (recall they partition G). Define:

$$\sigma_i = \sum_{g \in \mathcal{K}_i} g$$

Note that any pair of elements σ_i and σ_j have no common terms since the conjugacy classes partition G. Furthermore, by definition, for any $g \in G$, $g\sigma_i g^{-1} = \sigma_i$, and so $\sigma_i \in Z(\mathbb{C}[G])$. If the σ_i's form a basis for $Z(\mathbb{C}[G])$, then we would have $s = t$. As a $\mathbb{C}$-vector-space, they are already linearly independent since the group elements from a basis, so it suffices to show they span. If $\sum_{g\in G} \lambda_g a_g \in Z(\mathbb{C}[G])$. Then by definition for all $h \in G$,

$$\begin{aligned}
\Leftrightarrow h\left(\sum_{g\in G}\lambda_g g\right) &= \left(\sum_{g\in G}\lambda_g g\right) h \\
\Leftrightarrow \left(\sum_{g\in G}\lambda_g g\right) &= h^{-1}\left(\sum_{g\in G}\lambda_g g\right) h \\
\Leftrightarrow \sum_{g\in G}\lambda_g g &= \left(\sum_{g\in G}\lambda_g (h^{-1}gh)\right) \\
\overset{!}{\Leftrightarrow} \sum_{g\in G}\lambda_g g &= \sum_{g\in G}\lambda_{hgh^{-1}} g \\
\Leftrightarrow \lambda_{hgh^{-1}} &= \lambda_g \qquad \forall g \in G
\end{aligned}$$

since h arbitrary, we see that the coefficients of elements in the same conjugacy class remain with an element that's in that conjugacy class. Equivalently, the function $G \to \mathbb{C}$, $g \mapsto \lambda_g$ is constant on conjugacy classes. Thus:

$$Z(\mathbb{C}[G]) = \operatorname{span}\left\{\sum_{g\in G} f([g])g\right\}$$

where f is a function from the conjugacy classes of G to $\mathbb{C}$. Hence $s = t$.

for (2), Since $\mathbb{C}[G]$ is a semi-simple ring, V is a semi-simple $\mathbb{C}[G]$-module, giving us existence of the decomposition by AW. For uniqueness, first note that:

$$\operatorname{End}_{\mathbb{C}[G]}(S_i) = \mathbb{C}$$

since $\mathbb{C}$ is the only [finite-dimensional?] division $\mathbb{C}$-algebra. Then, we get by Schur's lemma:

$$\operatorname{Hom}_{\mathbb{C}[G]}(S_i, V) \overset{\text{Schur}}{\cong} \operatorname{Hom}_{\mathbb{C}[G]}(S_i, S_i)^{\oplus l_i} \cong \mathbb{C}^{l_i}$$

Corollary 3.1.14: Abelian Groups And Degree 1 Decomposition

1. If A is an abelian group, then every irreducible complex representation of A is 1-dimensional (i.e. a homomorphism from A to $\mathbb{C}^\times$), and A has $|A|$ inequivalent irreducible complex representations.
2. The number of inequivalent (irreducible) 1-dimensional complex representations of any finite group G is equal to $|G/G'|$

***Proof*:**

1. Since A is abelian, so is $\mathbb{C}[A]$. By the Artin-Wedderburn decomposition, it must be a product of matrices, which since $\mathbb{C}[A]$ abelian is so must each matrix ring, which is only true if and only if $n_i = 1$. Then, by corollary 1.2.20, $\mathbb{C}[A]$ is a product of fields. It follows that any irreducible representation (i.e. simple $\mathbb{C}[A]$-module) is 1-dimensional. Conversely, if A were not abelian, $\mathbb{C}[A]$ cannot be a product of fields, hence one of the dimensions of the matrix rings *must* be greater than 1

2. Let G be an arbitrary group. We want to show that there is a bijective correspondence between the 1-dimension representations of G and the 1-dimensional representations of G/G'. $\rho : G/G' \to \mathbb{C}^\times$ is a 1-dimensional representations of G/G', then we have the natural map $G \to G/G'$, which shows that each 1-dimensional representations of G/G' is also a 1-dimensional representations of G. Convexly, if f $\rho : G \to \mathbb{C}^\times$ is a 1-dimensional representations of G, then by the universal property, this factors through:

$$\begin{tikzcd} G/G' \arrow[r] & \mathbb{C}^\times \\ G \arrow[u] \arrow[ur] & \end{tikzcd}$$

   showing that any 1-dimensional representation of G is associated to a 1-dimensional representations of G/G', showing there can't be more 1-dimensional representations of G than there are of G/G', completing the proof.

A more general technique can be adopted from this proof: If $H \trianglelefteq G$ and V is an irreducible representation of G/H, then V is an irreducible representation of G.

Example 3.1.15: Decomposition Of Modules

1. DnF p. 847, and a bit about $\dim(V) = 1$ example right under definition

Finally, for those who read chapter **??**, the following results are a glimpse at how representation theory is a "generalization" of number theory. First, if $H \leqslant G$ is a subgroup and (ρ, V) is an irreducible representation, $\rho|_H$ breaks up into the direct sum of representations, not necessarily of the same size or multiplicity. This is similar to how for a prime $\mathfrak{p} \subseteq F$ it may split to multiple primes with varying inertia degree and ramification numbers in a finite separable extension L/F. If L/F is furthermore normal (and hence Galois) then all the inertia degree's and ramification numbers match. In a similar vein:

Theorem 3.1.16: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \to \mathrm{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

***Proof*:**
Let W be an irreducible subrepresentation of $V|_H$. Since H is normal in G, for $g \in G$, H acts irreducibly on gW by $(h, gw) \mapsto hgh^{-1}hw$. The subspace $\sum_{g \in G} gW$ is a nonzero subrepresentation of V. Since V is irreducible, it is equal to V. Since a representation which is a sum of irreducible sub-representations is semisimple, $V|_H$ is semisimple. A similar argument follows for the 2nd assertion, completing the proof.

Exercise 3.1.1

1. If χ_1 and χ_2 are both 1 dimensional representations, then $\chi_1 \cong \chi_2 \Leftrightarrow \chi_1 = \chi_2$. Use the property of equivalent representations to see this.
2. Let V be a non-trivial irreducible G-representation (i.e. not $\chi(g) = 1$ for all $g \in G$). Show that there are $\dim_{\mathbb{C}}(V^G) = 0$
3. By Caley's Theorem, all finite groups are linear. Do there exists infinite non-linear groups? (Hard: Do there exists finitely generated non-linear groups?)
4. Show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v, N acts trivially on all of V.

3.2 Character Theory

We know now we can decompose a G-representation V into the direct sum of r different types of irreducible G-modules. How do we know proceed to find these? These can be quite unwiedly: if we have a 10 or 100-dimensional represenation of a group G, then we have a matrix with 100 and $10'000$ entries! We'd then have to multiply these and figure out what relation's they have between each other, quite a hassle! Furthermore, how do we do this in such way that is independent of equivalent representations? A convinient basis might be nice, but to find a nice basis for every $\rho(g) \in \rho(G)$, we'd have to find a basis that can *simultaneously* diagonalize every element in $\rho(G)$! If there was some invariant we can find on a representation that we can use to keep track of the decomposition information between representations that is invariant of representation, that would be useful to this endevaour. The following does exactly this (Move the discussion of trace here to build-up to it? Give some of D&F examples to demonstrate it does get quite ugly sometimes?)

Definition 3.2.1: Class Function and Character

Let V be a representation of G over an arbitrary field F. Then

1. A function $f : G \to F$ that is constant on conjugacy classes is called a *class function* of G
2. If φ is a representation of G afforded by the $F[G]$-module V, then a function:

$$\chi_V : G \to \mathbb{C} \qquad g \mapsto \mathrm{tr}(g : V \to V) = \mathrm{tr}(\varphi(g))$$

is called the *character* of φ, where $\mathrm{tr}(\varphi(g))$ is the trace of φ with respect to some basis for V. A character is called *irreducible* if φ is irreducible (and reducible oherwise). The *degree* of a character is the degree of any representation affording it.

For ease of reference, recall these properties of the trace:

1. $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$ an $\mathrm{tr}(cA) = c\mathrm{tr}(A)$
2. $\mathrm{tr}(A) = \mathrm{tr}(A^t)$
3. $\mathrm{tr}(AB) = \mathrm{tr}(BA)$

Note that in general, a character is *not* a homomorphism from G into the additive or multiplicative structure of k. Since $\mathrm{tr}(AB) = \mathrm{tr}(BA)$, $\mathrm{tr}(ABA^{-1}) = \mathrm{tr}(B)$, we see that the trace (and hence the character) is independent of choice of basis, and so equivalent representations have the same character, meaning the character is indeed an invariant. Furthermore, for any representation φ of G,

$$\varphi(g^{-1}xg) = \varphi(g)^{-1}\varphi(x)\varphi(g)$$

hence, we can take the trace and use the 3rd property above to show that characters an an example of *class functions*. Furthermore, since the trace is linear, we can extend the domain of a character to be the trace over any element in $\mathbb{C}[G]$. If $F = \mathbb{C}$, then the collection of class function's form a complex vector-space, since the sum of two class functions is a class function, and multiplying by a scalar $z \in \mathbb{C}$ does not change that a class function is invariant.

Example 3.2.2: Character

1. Given the trivial representation of G (i.e. any 1-dim vector space where $g \mapsto I$), then if φ is the representating affording the trivial representation, it has $\chi_\varphi(g) = 1$ for all g. This is called the *trivial character*.
2. If we have a degree 1-representation, then the character and the representation are the same. Thus for abelian group, complex representations and their characters are the same. For example, take the representation $\varphi : C_p \to \mathbb{C}^\times$, $\varphi(a^i) = \zeta_p^i$. Then:

$$\chi(a^i) = \zeta_p^i$$

3. Let $\Pi : G \to S_n$ be a permutation representation of G and φ the linear of G of degree n on V where:

$$\varphi(g)(e_i) = e_{\Pi(g)(i)}$$

with respect to the given basis, $\varphi(g)$ has 1 in the ii-entry if $\Phi(g)$ fixes i, and zero otherwise. Thus, we can say that the character χ is:

$$\chi(g) = \#\{\text{number of fixed points of } g \text{ on } \{1, 2, ..., n\}\}$$

If in particular the permuation representation of G into S_n is by left multiplication, then the character is called the *pemutation character*.

4. Consider any regular representation of G, φ. Then the character is:

$$\chi(g) = \begin{cases} 0 & g \neq 1 \\ |G| & g = 1 \end{cases}$$

This character is called the *regular character* of G (note how this character is not a group homomorphism into k or $k^\times$)

5. Take the representation $\varphi : D_n \to \mathrm{GL}_2(\mathbb{R})$ from example 3.1.5. Then:

$$\chi(r) = 2\cos(2\pi/n) \qquad \chi(s) = 0 \qquad \chi(1) = 2$$

where the last one is since $1 \in G$ must map to the identity matrix $I \in \mathrm{GL}_2(\mathbb{R})$.

From now on, all our representation will be complex.

Lemma 3.2.3: Properties Of Character

Let $\rho_V : G \to \mathrm{GL}(V)$ be a complex representation. Then:

1. $\chi_V(1) = \dim_{\mathbb{C}}(V)$
2. $\rho_V(g)$ is diagonalizable for all $g \in G$ and $\chi(g)$ is integral over the integers: $\chi(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$
3. If $g \in G$ has order n, then $\chi_V(g)$ is a sum of $\dim(V)$ number of nth roots of 1
4. $|\chi_V(g)| \leq \chi_V(1)$ for all $g \in G$
5. $\chi_V(g^{-1}) = \overline{\chi_V(g)}$

***Proof*:**

1. We showed this in the above examples.
2. The criterion we need from linear algebra is that $A \in M_n(\mathbb{C})$ is diagonlizable if and only if the minimal polynomial has no repeated roots, which can be implied if $M^k = I$ for some k, which it certainly is in our case since every element is finite.
3. If $g^n = 1$, then $\rho_V(g)^n = 1$, so by (2)

$$\rho_V(g) \sim \begin{pmatrix} z_1 & & \\ & \ddots & \\ & & z_k \end{pmatrix}$$

where $z_i^n = 1$ for all $1 \leq i \leq k$. Furthermore, this shows that these trace values in the matrix representation are always integral over the integers, being the sum of integral elements.

4. $$|\chi_V(g)| = \left|\sum z_i\right| \leq \sum |z_i| = \dim(\chi_V(1))$$

5. Since $\rho_V(g^{-1}) = \rho_V(g)^{-1}$ and

$$\rho_V(g)^{-1} \sim \begin{pmatrix} z_1^{-1} & & \\ & \ddots & \\ & & z_k^{-1} \end{pmatrix}$$

$\chi_V(g^{-1}) = \sum_i z_i^{-1}$, but $\|z_i\| = 1$, so $z_i^{-1} = \overline{z_i}$, so

$$\chi_V(g^{-1}) = \sum_i z_i^{-1} = \sum_i \overline{z_i} = \overline{\chi_V(g)}$$

Now, if we have some G-representation V, it has some character χ. In the following, we will show how we can always break-down a character to be the sums of characterse on the irreducible G-representations:

Proposition 3.2.4: Linearly Independent Characters

Let G be a finite group and V a semi-simple $k[G]$-module, and φ the representation that V affords. Then:

1. The character of V is the sum of the characters of the componens appearing in the direct sum decomposition.
2. The irreducible characters of G are linearly independent set in the set of class functions. Furthermore, the irreducible characters of G form a basis of the vector-space of class functions.

***Proof*:**
For the 1st point, By the Artin-Wedderburn decoposition, we have:

$$V \cong (S_1)^{n_1} \oplus \cdots \oplus (S_k)^{n_k}$$

where S_i are simple $k[G]$-modules (irreducible G-representations). If we focus on the case where $M = S_1 \oplus S_2$, then we can choose a basis for S_1 and S_2 which will give us:

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & 0 \\ 0 & \varphi_2(g) \end{pmatrix}$$

where φ_1, φ_2 are the representations afforded by M_1 and M_2 respectively. Then if ψ_1 and ψ_2 are the characters of φ_1 and φ_2 an χ is the character of φ then we get:

$$\chi = \psi_1 + \psi_2$$

using induction get's us the result.

For the 2nd point, first notice that the class functions defined by being 1 on a conjugacy class and 0 otherise forms a basis for class functions of dimension r. It thus suffices to sho that the r irreducible characters are independent. To that end, define:

$$\widetilde{\chi_V} : \mathbb{C}[G] \to G \qquad x \mapsto \mathrm{tr}(x : V \to V)$$

Note that this map is $\mathbb{C}$-linear since $\mathbb{C}$ commutes with $\mathbb{C}[G]$, and $\widetilde{\chi_V}|_G = \chi_V$. Now, say $S_1, ..., S_r$ are simple $\mathbb{C}[G]$-modules and:

$$\sum_i^r \lambda_i \chi_{S_i} = 0 \qquad \lambda_i \in \mathbb{C}$$

Then this is equivalent to:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(g) = 0 \qquad \forall g \in G$$

and since the maps are $\mathbb{C}$-linear, by linearity:

$$\sum_i^r \lambda_i \widetilde{\chi_{S_i}}(x) = 0 \qquad \forall x \in \mathbb{C}[G]$$

which by plugging into the definition we get:

$$\sum_i^r \lambda_i \mathrm{tr}(x : S_i \to S_i) = 0 \qquad \forall x \in \mathbb{C}[G]$$

Next, by Artin Wederburn, we have the decomposition:

$$C[G] \cong M_{n_1}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$$

where each $M_{n_i}(\mathbb{C}) \cong S_i^{\oplus n_i}$, with $S_i = \mathbb{C}^{\oplus n_i}$. Using this, take the element:

$$x_i = (0, ..., I, ..., 0)$$

where I is the identity matrix[a]. Then $x_i : S_j \to S_j$ is 0 if $i \neq j$, and the identity marix if $i = j$. Thus, the trace is:

$$\mathrm{tr}(x_j : S_j \to S_j) = \begin{cases} 0 & i \neq j \\ n_i & i = j \end{cases}$$

and since n_j is at least 1, we must have that

$$\lambda_1 = \cdots = \lambda_r = 0$$

where we took advantage of the fact we are in characeristic 0

[a] If the field had finite characteristic, it is better to take the elementary matrix E_{11}

3.2.5 Remark We can prove this directly without using Artin-Wedderburn, which we will

Corollary 3.2.6: Detecting Isomorphic Representations Using Characters

Let V_1, V_2 be representations that are semi-simple modules.

1. Every character of G is uniquely of the form

$$\sum_i^r \lambda_i \chi_i \qquad \lambda_i \geq 0$$

where χ_i are irreducible characters.

2. the representations V_1 and V_2 are isomorphic if and only if $\chi_{V_1} = \chi_{V_2}$

***Proof*:**

1. This comes straight from the fact that irreducible characters form a basis, and that any G-representation V will be some direct sum of a finite list of known simple G-modules

2. If $V_1 \cong_G V_2$, there is a matrix P that simultaneously diagonalizes so that if A and B are basis matrices, $A = PBP^{-1}$. But the trace is independent of conjugation, and so the characters are the same. For the othe way around, treating a representation as a $k[G]$-module, then by Machke's theorem, we *know* the list simple $k[G]$-modules and the actions of $k[G]$ on them. Since any $k[G]$-module can be broken down into a direct summand of simlpe $k[G]$-modules, and the two chracters are the same, the list of $k[G]$-modules and their multiplicity are the same.

Thus, we have translated the problem of finding the irreducible representations of a group G to finding the irreducible characters of G! Since a character of a representation is determined by the irreducible characters, we give a way to systematize the tracking of irreducible characters:

Definition 3.2.7: Character Table

Let $\varphi : G \to \mathrm{GL}(V)$ be a representaiton of G an χ_V it's character. Then we call the *character table* a table with the rows being irreducible characters and columns being an element in the conjugacy class. Common practice is to make the first column the identity conjugacy class and the first row the trivial representation

Example 3.2.8: Character Table

Consider $G = S_3$ Then we can start creating it's character table:

	e	(1 2)	(1 2 3)
1	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since G has 3 conjugacy classes, exactly 3 irreducible characters of G. How would we go about finding it? The comning propositions work towards this.

To be able to find the missing character, and more generally find new characters for a character table, we may look to construct new character given old one, and hope that we break down those characters into a linear combination of irreducible characters. In a sense, we are finding a way to create represetnations without doing it "by hand". We will do this in two way: the first is by taking advatange of group-actions, since are relatively easy to define and their representations are matrices with 1's and 0's. Since all groups embedd into a symmetric group, this can be done systematically. We then see how to work directly with known representation to create new represenations.

Definition 3.2.9: Precaution Representation

Let G act on a finite set X. Define $\mathbb{C}X$ to be the set of formal sums:

$$\mathbb{C}X = \left\{ \sum_{x \in X} \lambda_x x \;\middle|\; \lambda_x \in \mathbb{C} \right\}$$

So that we get a $\mathbb{C}$-algebra with $\dim_{\mathbb{C}}(\mathbb{C}X) = |X|$. Then we can make this a G-represetnation by defining:

$$g \cdot \left(\sum \lambda_x x \right) = \sum \lambda_x (g \cdot x)$$

The 3 usual premutation representations is $G \curvearrowright G$ by left multiplication (which gives the regular representation) or conjugation, and $G \curvearrowright \{*\}$ trivially (which gives the trivial representation).

Lemma 3.2.10: Character Of Permutation Representation

Let G act on X, $G \curvearrowright X$. Then

$$\chi_{\mathbb{C}X}(g) = \#\{x \in X \mid gx = x\}$$

***Proof*:**
FIx $g \in G$. Then the set of x's is a basis for the representation. To make the computation simpler, we will find a convinient order for them. In particular, by choosing any $x_1 \in X$, we can find the orbit of the element:

$$\begin{tikzcd} x_1 \arrow[r, "g"] & x_2 \arrow[r] & \cdots \arrow[r] & x_k \arrow[lll, bend left] & x_{k+1} \arrow[r, "g"] & \cdots \end{tikzcd}$$

so the basis representation will look like:

$$\begin{pmatrix} 0 & & & \\ 1 & 0 & & \\ 0 & 1 & \ddots & \\ & & 1 & 0 \end{pmatrix}$$

so we get that the trace is 0 if the order of the orbit is greater than 1, hence:

$$\chi_{\mathbb{C}X}(g) = \#\{g\text{-orbits of size } 1\}$$

An immediate consequence of this is that we have another way of computing the character of the regular representation

$$\chi_{\mathbb{C}G}(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

which makes sense, since the $\dim_{\mathbb{C}}(\mathbb{C}G) = |G|$.

Next, we turn towards a more systematic way of constructing characters by creating them out of pervious ones, that is, given some representations V and W of a group G, we can make new representations by creating new representations from V and G:

Definition 3.2.11: Constructing Representations from smaller ones

Let V, W be G-representations. Then

1. we can make a $V \oplus W$ representation by:
$$g \cdot (v, w) = (g \cdot v, g \cdot w) \; [(\rho_V(g)v, \rho_W(g)w)]$$
2. we can make $V \otimes_{\mathbb{C}} W$ a representations by:
$$g \cdot (v \otimes w) \mapsto (gv) \otimes (gw)$$
3. we can make $\mathrm{Hom}_{\mathbb{C}}(V, W)$ a representations by:
$$g \cdot f : V \to W \quad v \mapsto g \cdot f(g^{-1} \cdot v)$$
4. we can make $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$ a representaiton by:
$$g \cdot f : V \to \mathbb{C}, \; v \mapsto f(g^{-1} \cdot v)$$

Note that the last construction is the same as the 3rd if we take $\mathbb{C}$ as the trivial representation. Before we show what the character's of these new representations are, we first find a relatively simpler way of representing the hom representation of V and W:

Lemma 3.2.12: Hom-Tensor Representation Dual

Let V, W be G representations. Then:

$$\mathrm{Hom}_{\mathbb{C}}(V, W) \cong_G V^* \otimes_{\mathbb{C}} W$$

as G-representations.

Note that this is not true in general, even for vector-spaces. It is important that the spaces are finite-dimensional. In this way, this is a stronger result then the hom-tensor adjoint presented in **??** for the case of finite-dimensional vector-spaces.

***Proof*:**
By the universal property:

$$\theta : V^* \otimes_{\mathbb{C}} W \to \mathrm{Hom}_{\mathbb{C}}(V, W) \qquad \lambda \otimes w \mapsto (v \mapsto \lambda(v)w)$$

you should check that this is well-define (note that it is bilinear).

Now, pick basis $e_1, e_2, ..., e_n$ for V, $f_1, f_2, ..., f_m$ for W, and the natural dual $e_1^*, ..., e_n^*$ for V^*. Then we get a basis for $V^* \otimes W$:

$$(e_i^* \otimes f_j) \qquad 1 \leq i \leq n,\ 1 \leq j \leq m$$

Now, notice that:

$$\theta(e_i^* \otimes f_j) \qquad v \mapsto e_i^*(v) f_j \sim \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & 0 \\ 0 & & 1 & & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} = E_{ij}$$

which is a matrix with 1 in the ij position and zero evywehre else. This is clearly a basis, hence θ sends a basis to a basis, and hence θ is a vector-space isomorphism. We next check that θ is G-linear:

$$\begin{aligned} \theta(g \cdot (\lambda \otimes w)) &= \theta(g\lambda \otimes gw) \\ &\equiv v \mapsto (g\lambda)(v) gw \\ &= v \mapsto (g\lambda)(g^{-1}v) w \\ &= g(\lambda(g^{-1}v)w) \\ &\equiv g(\theta(\lambda \otimes w) g^{-1} v) \\ &= g \cdot \theta(\lambda \otimes w) \end{aligned}$$

Lemma 3.2.13: Creating New Characters

1. $\chi_{V \oplus W} = \chi_V + \chi_W$
2. $\chi_{V \otimes W} = \chi_V \cdot \chi_W$
3. $\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \chi_W$
4. $\chi_{V^*} = \overline{\chi_V}$

***Proof*:**

1. obvious

2. It is enough to show that if $f : V \to V$ and $g : W \to W$ are linear functions, then:

$$\mathrm{tr}(f \otimes h) = \mathrm{tr}(f)\mathrm{tr}(h)$$

Now if we pick basis (e_i) for V and (f_j) for W, then $f(e_i) = \sum a_{ij}e_j$ and $h(f_k) = \sum b_{lk}f_l$. Then:

$$(f \otimes h)(e_i \otimes f_j) = \left(\sum_j a_{ij}e_j\right) \otimes \left(\sum_l b_{kl}f_l\right) = \sum_{j,l} a_{ij}b_{lk}(e_j \otimes f_l)$$

Notice the cofficient of $e_i \otimes f_j$ is $\sum_{i,k} a_{ii}b_{kk} = (\sum_i a_{ii})(\sum_k b_{kk}) = \mathrm{tr}(f)\mathrm{tr}(h)$

3. Recall that for $f \in V^*$,

$$(gf)(v) = f(g^{-1} \cdot v)$$

that is

$$\rho_{V^*}(g)(f) = f \circ \rho_V(g^{-1})$$

that is, it is the transpose:

$$\rho_{V^*}(g) = (\rho_V(g^{-1}))^t$$

So, since the transpose matrix has the same trace, we have

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}$$

we then apply lemma 3.2.12 and part (3)to get:

$$\chi_{\mathrm{Hom}_{\mathbb{C}}(V,W)} = \overline{\chi_V} \cdot \chi_W$$

4. $\chi_{V^*} = \chi_{\mathrm{Hom}(V,\mathbb{C})} = \overline{\chi_V} \cdot 1 = \overline{\chi_V}$.

With this, we can create new characters! However, we have yet to establish how we may find the irreducible components of these chracters. For example, if we take $V \otimes V$, we may find that $\chi_{V\otimes V}$ is a linear combination of some characters already present in our table, but then we have a left-over part. How would we determine if the left-over is irreducible? We will show this by defining an *inner product* on the vector-space of class equations, and show that the irreducible form an *orthogonal basis*!

Definition 3.2.14: Inner Product On Class functions

Let $\chi_1, \chi_2 : G \to \mathbb{C}$ be class functions. Then define:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_1(g)\overline{\chi_2(g)} \in \mathbb{C}$$

Then this is a [Hermitian] *inner product*.

It should be verified that this is indeeed an inner product. Since class functions are by definition constant on conjugacy classes, we only need to run through representatives of conjgacy classe times the order of each conjugacy class:

$$\langle \chi_1, \chi_2 \rangle = \frac{1}{|G|} \sum_{\substack{\mathcal{K} \in \mathrm{conj}(G) \\ g \in \mathcal{K}}} |\mathcal{K}| \chi_1(g) \overline{\chi_2(g)} \in \mathbb{C}$$

where we pick a single represetnative $g \in \mathcal{K}$ and sum over the number of conjugacy classes, i.e. $|\mathrm{conj}(G)|$. Looking a the inner product like this is desirable to make computation easier (ex. if taking the inner product of a character with itself, we would simply take the squrae of each element in a row multiplied by the size of it's conjugcay class). Notice how this inner product mirrors the G-invariant inner product we induced in corollary 3.1.12 if we let $\langle, \cdot, \cdot \rangle$ be the usual conjugate dot product on $\mathbb{C}^n$. (TBD)

Lemma 3.2.15: Inner Product With Trivial Character

Let V be a G-representation. Then if $\mathbb{1}$ is the trivial representation,

$$\langle \chi_V, \chi_{\mathbb{1}} \rangle = \dim_{\mathbb{C}}(V^G)$$

where $V^G = \{v \in V \mid g \cdot v = v \ \forall g \in G\}$ is the subspace fixed by G.

***Proof*:**
Define the linear map $\pi : V \to V$ where

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} g \cdot v$$

then:

$$\pi|_{V^G} = \mathrm{id}$$

and

$$\mathrm{im}(\pi) \subseteq V^G$$

which implies $\pi^2 = \pi$, i.e. a projection. By linear linear algebra:

$$V = \ker(\pi) \oplus \mathrm{im}(\pi)$$

where on the kernel, $\pi = 0$, and on the image $\pi = \mathrm{id}$. Thus,

$$\frac{1}{|G|} \sum_g \chi_V(g) = \frac{1}{|G|} \sum_g \mathrm{tr}(g : V \to V) = \mathrm{tr}(\pi) = \dim_{\mathbb{C}}(V^G)$$

but the left hand side is $\langle \chi_V, \chi_{\mathbb{1}} \rangle$, completing the proof.

Corollary 3.2.16: Row Orthogonality

If V and W are any representations of G, then

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\hom_{\mathbb{C}[G]}(W, V))$$

In particualr, if V an W are irreducible, then

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

that is, irreducible character's are orthogonal to each other.

***Proof*:**
We reduce to the previous lemma:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}(g) \\ &= \langle \chi_{\mathrm{Hom}_{\mathbb{C}}(W,V)}, 1 \rangle \\ &= \dim_{\mathbb{C}}(\mathrm{Hom}_{\mathbb{C}}(W, V)^G) \end{aligned}$$

Note that $f \in \mathrm{Hom}_{\mathbb{C}}(W, V)$ is G-invariant, so $g \cdot f = gfg^{-1} = f$ for all $g \in G$ (recall the definition of $g \cdot f$ on $\mathrm{Hom}_{\mathbb{C}}(W, V)$). Thus this is true iff

$$gf = fg$$

Thus

$$gf(w) = f(gw) \qquad \forall g \forall w \in W$$

so f is $\mathbb{C}[G]$-linear so f is in the $\mathrm{Hom}_{\mathbb{C}[G]}(W, V)$.

There is also a more concrete proof worth remarking: since characters are linear combination's of irreducible characters, the direct product of irreducible modules translates to the sum of an irreducible character with the multiplicity of the irreducible module afording the representation being the multiplicity of the ireducibile characters. Hence:

$$\begin{aligned} \langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle \end{aligned}$$

where we have now reduced it to $\langle \chi^{(i)}, \chi^{(j)} \rangle$. Expanding the inner product and using the hom-set definition, we can apply Schur's Lemma and get:

$$\langle \chi^{(i)}, \chi^{(j)} \rangle = \delta_{ij}$$

Thus, our formula becomes:

$$\begin{aligned}\langle \chi_V, \chi_W \rangle &= \langle \sum_i m_i \chi^{(i)}, \sum_j m_j \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \langle \chi^{(i)}, \chi^{(j)} \rangle \\ &= \sum_{i,j} m_i m_j \delta_{ij}\end{aligned}$$

To find $\langle \chi^{(i)}, \chi^{(i)} \rangle$ for an irreducible character, seeing it through multiplicites, then we know that an irreducible character corresponds to an irreducible representation, which necessarily has to then have multiplicity one, so if in particular:

$$\langle \chi^{(i)}, \chi^{(i)} \rangle = m_i^2 = 1 \cdot 1 = 1$$

Thus, the irreducible character produce and orthogonal basis of class functions!

Corollary 3.2.17: Consequence of Orthogonality

Let $V = \bigoplus_i^r S_i^{l_i}$ where each S_i irreducible and $S_i \ncong S_j$. Then

1. the multipliity of l_i of S_i is: $l_i = \langle \chi_V, \chi_{S_i} \rangle$
2. $\langle \chi_V, \chi_V \rangle = \sum_i l_i^2$
3. $\hom_G(W, V) \cong \hom_G(V, W)^*$

***Proof*:**

1. Use the Artin-Wedderburn decomposition and the fact that direct sums of modules becomes sums of characters
2. this is just taking advantage of the summing formula
3. This is taking advantage of the fact that $\langle \chi, \psi \rangle = \overline{\langle \psi, \chi \rangle}$.

The second line is useful enough to be given it's own name:

Definition 3.2.18: Character Norm

Let V be a G-representation and let $\langle \cdot, \cdot \rangle$ be the inner product on the class functions of G. Then for any class function $\psi = \sum_i \lambda_i \chi_i$,

$$\|\psi\| = \left(\sum_i \lambda_i^2 \right)^{1/2}$$

We will use all this informaiton to deduce new characters:

Example 3.2.19: Orthogonality Relation

Let's go back to our table for S_3:

	e	(1 2)	(1 2 3)
$\mathbb{1}$	1	1	1
χ_{sgn}	1	-1	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Since S_3 has 3 conjugacy classes, it can at most have one more irreducible character. What can this character be? We may start by looking at the regular represetnation of S_3 and take it's character χ_V. Then:

$$\langle \chi_V, \mathbb{1} \rangle = 1$$
$$\langle \chi_V, \chi_{\text{sng}} \rangle = 0$$

showing the regular character χ_V has one copy of the trivial representation (which makes sense, since $\chi_V(e) = 1$). Since

$$\chi_V(e) = 3 \qquad \chi_V((1\ 2)) = 1 \qquad \chi_V((1\ 2\ 3)) = 0$$

then we can take a "compliment" character by subtracting the trivial character, which defines a character which gives:

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}(1\ 2) = 0 \qquad \chi^{\perp}(1\ 2\ 3) = -1$$

So, if there were to be another irreducible character, it would be of this form. But does there exist a degree 2 irreducible representation with this character? In fact, there has to: we saw that the number of irreducible representations is equal to the number of conjugacy classes, meaning the number of characters must be equal to the number of irreducible representations.

More generally, there can be a few approaches to finding it. One is to tensor two characters we know and see if it is irreducible, or one of it's linear components is irreducible. Another is to find the compliment of a represetnation of an existing character (since $k[G]$-modules are semi-simple). This compliment method is in general less desirable, and usually harder, but is managable if working with permutation representations.To demonstrate, let V be the degree three permutation representation of S_3[a] with basis $\{e_1, e_2, e_3\}$. Then if $\mathbb{1}$ is the trivial action, $V^{\mathbb{1}} = \text{span}(e_1 + e_2 + e_3)$, Thus, by Machke's theorem, $V^{\mathbb{1}}$ has a compliment. To see it, we simply have to extend $e_1 + e_2 + e_3$ to an orthogonal basis for V, for example, choose $(e_1 + e_2 + e_3, e_2 - e_1, e_3 - e_1)$. Then we can calculate what the action ρ does on this basis. For example:

$$\begin{aligned} \rho(1\ 2)(e_1 + e_2 + e_3) &= e_1 + e_2 + e_3 \\ \rho(1\ 2)(e_2 - e_1) &= e_1 - e_2 = -(e_2 - e_1) \\ \rho(1\ 2)(e_3 - e_1) &= e_3 - e_2 = -(e_2 - e_1) + (e_3 - e_1) \end{aligned}$$

If we write out ρ for an element in each conjugacy class, we will find that it's trace will be

$$\chi^{\perp}(e) = 2 \qquad \chi^{\perp}((1\ 2)) = 0 \quad \chi^{\perp}((1\ 2\ 3)) = -1$$

showing we have found the desired irreducible representation! Finally, by Artin-Wedderburn, we have that $\mathbb{C}S_3 \cong \mathbb{1} \oplus \text{sgn} \oplus V^{\perp}$

Now that we know the character table for S_3, we can construct any other character of representations of S_3, and hence any other representation of S_3. For example, we know that $V^\perp \otimes V^\perp$ has as character $\chi_{V^\perp} \cdot \chi_{V^\perp}$. Using this, we can compute that:

$$\langle V^\perp \otimes V^\perp, \mathbb{1} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, \mathrm{sgn} \rangle = 1$$
$$\langle V^\perp \otimes V^\perp, V^\perp \rangle = 1$$

showing that

$$\chi_{V^\perp \otimes V^\perp} = \chi_{\mathbb{1}} + \chi_{\mathrm{sgn}} + \chi_{V^\perp}$$

[a]Which in this case, is just the regular representation

Recall from linear algebra that if the rows are orthogonal in a square matrix, then so would be the column (and vice-versa). We bring to light this relation for the character table. As in linear algebra, it comes down to proper normalizing of a different basis: recall that the class equations of the form $\delta_{\mathcal{K}} : G \to \mathbb{C}^\times$ where:

$$\delta_{\mathcal{K}}(g) = \begin{cases} 1 & g \in \mathcal{K} \\ 0 & \text{otherwise} \end{cases}$$

this naturally forms a basis for the class equations too. The following theorem will also show how to go between this basis and the irreducible character basis:

Theorem 3.2.20: Column Orthogonality

If $\chi_1, \chi_2, ..., \chi_r$ are the irreducible characters of G and $g_1, g_2, ..., g_r$ represent the conjugacy classes of G, then

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st}\frac{|G|}{|\mathrm{conj}(g_s)|}$$

which by Orbt-stabalizer,

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \delta_{st}|C_G(g_s)|$$

If $g_s, g_t \in \mathrm{conj}(g)$, then we get that

$$\sum_i \chi_i(g_s)\overline{\chi_i(g_t)} = \sum_i |\chi_i(g_s)|^2 = |C_G(g_s)|$$

in particular, we must have:

$$\sum_i |\chi_i(e)|^2 = |G|$$

***Proof*:**
Define $f_s : G \to \mathbb{C}$,

$$g \mapsto \begin{cases} 1 & g \in \mathrm{conj}(g_s) \\ 0 & \text{otherwise} \end{cases}$$

which is natural basis for the class functions. since it's a class function, we know we can write it as the sum of orhtogonal ireducibile characters:

$$f_s = \sum_i^r \lambda_{is}\chi_i \qquad \lambda_{is} \in \mathbb{C}$$

since these are orthornormal, we know that:

$$\lambda_{is} = \langle f_s, \chi_i \rangle$$

so computing:

$$\begin{aligned}\lambda_{is} &= \langle f_s, \chi_i \rangle \\ &= \frac{1}{|G|}\sum_{g\in G} f_s(g)\overline{\chi_i(g)} \\ &= \frac{1}{|G|}\sum_{g\in \mathrm{conj}(g_s)} \overline{\chi_i(g)} \\ &= \frac{|\mathrm{conj}(g_s)|}{|G|}\overline{\chi_i(g_s)}\end{aligned}$$

so

$$f_s = \frac{\mathrm{conj}(g_s)|}{|G|}\sum_i^r \overline{\chi_i(g_s)}\chi_i$$

we know evaluate at g_t.

Since both the rows and columns of a character table are all orthogonal to each other, the character table is in fact invertible!

Example 3.2.21: Using Column Orthogonality

Let $G = S_4$. We want to find it's character table. We first pick what conjugacy classes to index the columns by, say:

conj. class	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
size	1	6	8	3	6

Next, we can immediately deduce that $\chi_{\mathbb{1}}$ and χ_{sgn} are irreducible characters, giving us the current begining of a character table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1

To find more characters, we can first take the pemutation character of $\mathbb{C}X$. The character of each conjugacy class is the number of fixed points, giving us:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{C}X}$	4	2	1	0	0

We can see that the trivial character is part of linear decomposition of $\chi_{\mathbb{C}X}$, since:

$$\langle \chi_{\mathbb{C}X}, \chi_{\mathbb{1}} \rangle = (1(4 \cdot 1) + 6(2 \cdot 1) + 8(1 \cdot 1))/24 = 24/2 = 1$$

and so $\chi_{\mathbb{C}X} - \chi_{\mathbb{1}}$ is a 3 dimensional character, let's label this χ_3. Then $\|\chi_3\|^2 = 1$, showing that it's irreducible, hence we expand our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
$\chi_{\mathbb{C}X} - \chi_{\mathbb{1}} = \chi_3$	3	2	0	-1	-1

To find the next character, we can multiply χ_{sgn} and χ_3 (note we can do χ_3 with itself, but it will be 9 dimensional, which we'll show in the next proposition cannot be an irreducible representation). Let$\chi_4 = \chi_{\text{sgn}} \cdot \chi_3$. Then $\|\chi_4\|^2 = 1$, meaning it's irreducible, and so we add another row to our table:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
$\chi_{\text{sng}} \cdot \chi_3 = \chi_4$	3	-1	0	-1	1

Finally, since $|G| = \sum_i \chi_i(e)^2$, we can deduce that $\chi_5(e) = 2$. We can't do this same trick exactly for the other columns since we will get the value up to a complex number z with $\|z\| = 1$. To find the rest of the character values, we can either use column orthogonality For example, to find $\chi_5(1\ 2)$, noticet that the inner product of the first and second column is:

$$1 \cdot 1 + (-1) \cdot 1 + 3 \cdot 1 + 3 \cdot (-1) = 0$$

and so $\chi_5(2)$ must be 0. For the next column, we'd get :

$$1 \cdot 1 + 1 \cdot 1 + 3 \cdot 0 + 3 \cdot 0 = 2$$

and so, since $2 \cdot -1 = -2$, we get that $\chi_5(1\ 2) = -1$. Continuing this way, we get that the rest of the table will be:

	1	6	8	3	6
	1	(1 2)	(1 2 3)	(1 2)(3 4)	(1 2 3 4)
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

We can verify this is indeed irreducible by taking $\|\chi_5\|^2 = 1$.

Another method would be to use the regular representation, where we get:

$$\chi_{\mathbb{C}[G]} = \sum_i \chi_i(1)\chi_i$$

These can be verified by taking advantage of the fact that the "column norm" should give you $C_G(g)$ (i.e. the number on top of the table).

As a fun aside, it can be shown that the represetned by χ_4 is the action of the rotation of a cube that is, if $C \subseteq \mathbb{R}^3$ are the vertex points of the cube, then χ_3 is the character of $S_4 \curvearrowright \mathrm{GL}_3(\mathbb{R})$ of all rotations C.

Notice how the dimension of each irreducible representation divided the order of the group. This is in fact true in general. It is not a trivial fact, the only "order" information we've shown so far is that the sums of the squares of the dimensions of the ireducibile representation sum to $|G|$. It is also relatively eays to show that if $W \subseteq V$ is an irreducible $k[G]$-module then the dimension of W is $< |G|$. The following is thus a much stronger result:

Proposition 3.2.22: Dimension Theorem of Irreducible Representations

Let V be an irreducible G-representation. Then:

$$\dim_{\mathbb{C}}(V) \bigm| |G|$$

that is, the dimension of V divides the order of G.

***Proof*:**
Let $g_1, g_2, ..., g_r$ be representatives of the conjugacy class. Recall the following trick from proposition 3.1.13:

$$e_i = \left(\sum_{g \in \mathrm{conj}(g)} g \right) \in Z(\mathbb{C}[G])$$

Note that e_i in fact lies in $e_i \in Z(\mathbb{Z}[G])$, since $g e_i g^{-1} = e_i$ (by definition), and so extending linearly all $x e_i x^{-1} = e_i$. Now we have a commutative ring that is a finitely generated $\mathbb{Z}$-module (one way to see this is as we've done in proposition 3.1.13, another way is to notice that $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$-module and $\mathbb{Z}$ is Noetherian, so the center is finitely generated)

Thus, by **??**, $\mathbb{Z} \subseteq Z(\mathbb{Z}[G])$ is an *integral extension*. Next, we have a ring homomorphism:

$$\varphi : Z(\mathbb{C}[G]) \to \mathrm{End}_{\mathbb{C}[G]}(V) \overset{!}{\cong} \mathbb{C} \qquad z \mapsto (z : V \to V)$$

where $\overset{!}{=}$ comes from corollary 1.2.19 and Schur's lemma, and $z(v) = zv$. Hence, each element of $\varphi(Z(\mathbb{C}[G]))$ is $\mathbb{C}$-linear, and must be invertible, and so φ is in fact a map from $Z(\mathbb{C}[G]) \to \mathrm{Aut}_{\mathbb{C}}(V) = \mathrm{GL}(V)$, which is simply a restriction of the origional G-action ρ to $Z(\mathbb{C}[G])$. Hence $\rho(e_i) = \lambda_i I$ for appropriate $\lambda_i = \varphi(e_i)$, meaning $\chi(e_i) = \mathrm{tr}(e_i : V \to V) = n\varphi(e_i)$ where $n = \dim(V)$ since by what we've just shown, $\varphi(e_i)$ acts as scalar multiplication on the representation V.

Now, since $e_i \in Z(\mathbb{C}[G])$ is integral over $\mathbb{Z}$, $\varphi(e_i) \in \mathbb{C}$ is integral over $\mathbb{Z}$, that is $\varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}} \subseteq \mathbb{C}$ (i.e. $\varphi(e_i)$ is algebraic). Next, we can express $\chi(e_i)$ in a different way:

$$\sum_{g \in \mathrm{conj}(g_i)} \chi_V(g) = |\mathrm{conj}(g_i)| \cdot \chi_V(g_i) = \chi(e_i)$$

so:

$$\frac{\mathrm{conj}(g_i)}{n} \chi_V(g_i) = \varphi(e_i) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

which is telling us that if we divide it by n, we still get an algebraic integer! Taking a nice linear combination and multiplying by $\overline{\chi_V(g_i)}$ we get:

$$\sum_i^r \frac{\text{conj}(g_i)}{n}\chi_V(g_i)\overline{\chi_V(g_i)}$$

Then we've just shown that $\frac{\text{conj}(g_i)}{n}\chi_V(g_i)$ is an algebraic integer, and $\overline{\chi_V(g_i)}$ is an algebraic integer (being the sum of roots of unity). Now, notice this is the form of the inner product, giving us:

$$\frac{|G|}{n}\langle\chi_V,\chi_V\rangle = \frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

as a V irreducible. Thus, we have that $\frac{|G|}{n}$ is some algebraic integer. We further have that it is certainly a rational number ($|G|, n \in \mathbb{N}$), but then we have:

$$\frac{|G|}{n} \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} \stackrel{!}{=} \mathbb{Z}$$

where $\stackrel{!}{=}$ comes from the fact that if R is a Unique Factorization Domain, $\alpha \in \text{Frac}(R)$, α is integral over R if and only if $\alpha \in R$, completing the proof.

This will make the computation of characters of abelian groups paticularly simple! (see rep theory of finite groups: An introductory Appraoch p. 47!)

Exercise 3.2.1

1. Let G act on the n-dimensional vector-space V by basis permutation. Show that $\dim(V^G)$ equals the number of orbits of G.

3.2.1 More Character Tables

In this section, we go over a few more exmaples of character tables in order to see some techniques used to find them. For any abelian, group, the character table is straight forward. As a really simple example, consider $\mathbb{Z}/3\mathbb{Z}$:

	1	1	1
	1	2	3
χ_1	1	1	1
χ_2	1	ζ_3	ζ_3^2
χ_3	1	ζ_3^2	ζ_3

If we take $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, we will get a 9 by 9 table. The table bellow fills in a few of the characters (note that these were found by taking $\chi_i(g_i)\chi_j(g_j)$)

$\mathbb{Z}/3\mathbb{Z}\times\mathbb{Z}/3\mathbb{Z}$	1 $(1,1)$	1 $(2,1)$	1 $(3,1)$	1 $(1,2)$	1 $(2,2)$	1 $(3,2)$	1 $(1,3)$	1 $(2,3)$	1 $(3,3)$
$\chi_{(1,1)}$	1	1	1	1	1	1	1	1	1
$\chi_{(2,1)}$	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2	1	ζ_3	ζ_3^2
$\chi_{(2,2)}$	1	ζ_3	ζ_3^2	ζ_3	ζ_3^2	1	ζ_3^2	1	ζ_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Note that the root ζ_3 might change if we take the direct product of cyclic groups of differnet orders, for example the character table of $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/3\mathbb{Z}$ has roots of unity of order 3 and 6 (as you can verify).

More generally, if G is abelian, we break G down into a product of cyclic groups and construct the table like above. Thus, we focus on non-abelian groups. We have already found the character table for S_3:

S_3	e	$(1\ 2)$	$(1\ 2\ 3)$
$\chi_{\mathbb{1}}$	1	1	1
χ_{sgn}	1	-1	1
χ_3	2	0	-1

and the character table for S_4:

S_4	1 1	6 $(1\ 2)$	8 $(1\ 2\ 3)$	3 $(1\ 2)(3\ 4)$	6 $(1\ 2\ 3\ 4)$
$\chi_{\mathbb{1}}$	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_3	3	2	0	-1	-1
χ_4	3	-1	0	-1	1
χ_5	2	0	-1	2	0

Next, let $G = D_4$. Then: $D' = \{e, r^2\}$, so G has 4 degree 1-irreducible representations. We also found in example 3.1.5 that D_4 has a degree 2 irreducible representation. These are all of the representations, sinc $1+1+1+1+2^2 = 8$. To find the 1-dimensional irreducible representations, notice that $D_4/D_4' \cong \mathbb{Z}_2\times\mathbb{Z}_2$, meaning we can find the irreducible representations of $\mathbb{Z}/2\mathbb{Z}$, which are:

$\mathbb{Z}/2\mathbb{Z}$	0	1
χ_0	1	1
χ_1	1	-1

and take the tensor product to create:

$\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
$\chi_{(0,0)}$	1	1	1	1
$\chi_{(1,0)}$	1	-1	1	-1
$\chi_{(0,1)}$	1	-1	-1	1
$\chi_{(1,1)}$	1	1	-1	-1

and using this, we can complete the table for D_4:

D_4	e	r	r^2	σ	σr
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

More generally, for D_n, the character table will conssite of characters of degree 1 and 2.

Let's now let $G = Q_8$. Taking the presentation:

$$Q_8 = \langle i, j | i^4 = 1,\ i^2 = j^2,\ i^{-1}ji = j^{-1} \rangle$$

with $k = ij$ and $i^2 = -1$, we get the conjugacy classes can be represented by $1, -1, i, j, k$, with sizes 1, 1, 2, 2, 2. Since $Q_8/Q_8' \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, there are 4 degree 1 characters. The final irreducible character thus *must* have degree 2; let's label if χ_5. We have in fact explicitely found this character before in example 3.1.5, but we can find it without actually manually finding a representation. Let φ be an irreducible representation of degree 2. Since $-1 \in Z(Q_8)$, by Schur's lemma $\varphi(-1)$ is a 2×2 scalar matrix of order 2, hence it is $\pm I$, and so $\chi_5(-1) = \pm 2$. Next, if $\chi_5(i) = a$, $\chi_5(j) = b$, and $\chi_5(k) = c$, then since χ_5 is orthogonal, we have that:

$$1 = \langle \chi_5, \chi_5 \rangle = \frac{1}{8}(2^2 + (\pm 2)^2 + 2a\overline{a} + 2b\overline{b} + 2c\overline{c})$$

and since $a\overline{a}, b\overline{b}, c\overline{c} \in \mathbb{R}^+$, they must all be zero. Furthermore, we must have that:

$$0 = \langle \chi_1, \chi_5 \rangle = \frac{1}{8}(2(\pm 2) + 0 + 0 + 0 + 0)$$

implying that $\chi_5(-1) = -2$. Thus, we have that the character table is:

Q_8	1	i	-1	j	k
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	1	1	-1	1
χ_5	2	0	-2	0	0

Notice that this character table is identical to the character table of D_8. Hence, identical character tables does not imply isomorphic groups! Finally, we could have found the last row of the above table by applying column orthogonality to find the missing values, once we knew $\chi_5(1) = 2$.

Next, let $G = A_4$. Since we found the character table for S_4, we can use it it to find the character table of A_4. We first note that A_4 has 4 conjugacy classes (represented by 1, (1 2)(3 4), (1 2 3) and (1 3 2)), and $[A_4 : A_4'] = 3$, meaning A_4 has 3 chacacters of degree 1, and the only other character must be of degree 3. Next, using the orthogonality relation, notice that χ_3 in the table for S_4 when restricted to A_4 is still irreducible (this is certainly not generally the case, for ex. the degree 2 representaiton of S_3 will become reducible since any proper subgroup of S_3 is abelian). Overall, we get:

A_4	1	(1 2)(3 4)	(1 2 3)	(1 3 2)
χ_1	1	1	1	1
χ_2	1	1	ζ_3	ζ_3^3
χ_3	1	1	ζ_3^2	ζ_3
χ_5	3	-1	0	0

Finally, we will do the character table for S_5:

S_5	1	(1 2)	(1 2 3)	(1 2 3 4)	(1 2 3 4 5)	(1 2)(3 4)	(1 2)(3 4 5)
χ_1	1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	1	-1
χ_3	4	2	1	0	-1	0	-1
χ_4	4	-2	1	0	-1	0	1
χ_5	5	-1	-1	1	0	1	-1
χ_6	5	1	-1	-1	0	1	1
χ_7	6	0	0	0	1	-2	0

TBD

3.2.2 Burnside's Theorem

We now move towards Burnside's Theorem, which shows that any group of order $p^a q^b$ for primes p, q and numbers $a, b \in \mathbb{N}$ is a solvable group! This proof is relatively simple in the context of representation theory, but it took over 70 years before a proof was found using just group theory! This is a good example of the power perspective: simply switching to the context of representation theory and seeing groups as embedded into $GL(V)$ let us prove a claim that from another perspective is much much harder!

Lemma 3.2.23: Burnside build-up Lemma

Suppose that V is an irreducible representation is of dimensin n (i.e. $\chi_1(V) = n$). If $\gcd(n, |\text{conj}(g)|) = 1$, then:

$$|\chi_V(g)| = 0 \text{ or } n$$

***Proof*:**
By the above proof, we have:

$$\frac{\text{conj}(g)}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

where $\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$ as well, so by Bézout ($1 = am + b|\text{conj}(g)|$)

$$\frac{1}{n}\chi_V(g) \in \overline{\mathbb{Z}}^{\mathbb{C}}$$

Let $\alpha = \frac{1}{n}\chi_V(g)$. Furthermore, notice that:

$$|\alpha| = \left|\frac{\chi_V(g)}{\chi_V(1)}\right| \leq 1$$

Now, let N be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ so that $G = \text{Gal}(N/\mathbb{Q})$ is the glaois group of N. Let

$$\beta = \prod_{\sigma \in G} \sigma(\alpha)$$

Then certainly β is in the fixed field of the galois group, $\beta \in N^G = \mathbb{Q}$. Notice that each $\sigma(\alpha)$ satisfies the same monic polynomials as that of α, and so $\sigma(\alpha) \in \overline{\mathbb{Z}}^{\mathbb{C}}$, and so $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}}$, but then $\beta \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$.

Now, recall that:

$$\alpha = \frac{\chi_V(g)}{n} = \frac{\zeta_1 + \zeta_2 + ... + \zeta_n}{n}$$

which are roots of unity, which means that:

$$\sigma(\alpha) = \frac{\sigma(\zeta_1) + \cdots + \sigma(\zeta_n)}{n}$$

so $|\sigma(\zeta_1)| \leq 1$, so $|\beta| \leq 1$. Since β is an integer, we either get that $|\beta| \in \{0, 1\}$, If $|\beta| = 0$, then $\alpha = 0$, and if $|\beta| = 1$, then $|\alpha| = 1$, and so plugging back into our original equation we get that:

$$\chi_V \in \{0, n\}$$

as we sought to show.

We can also treat this lemma as another check that we have given the correct character table (verify it for the character table of S_4 we found in example 3.2.21).

Theorem 3.2.24: Burnside's Theorem

Let $p \neq q$ be prime numbers, $a, b \in \mathbb{N}$. If $|G| = p^a q^b$, then G is solvable.

Notice how this immedaitely implies that the smallest possible order of a non-abelian simple group is 30, since $30 = 2 \cdot 3 \cdot 5$, which matches with the fact that the smallest non-abelian group is A_5 which has order 60! Recall also that if $|G| = p^a$, then we would need that $a = 1$ (namely, any p-group has a non-trivial center)

***Proof*:**
Take the composition series:

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$$

where G_i/G_{i-1}. Then we'll show that if G is simple, then G is *abelian*. For the sake of contradiction, suppos that G is non-abelian and simple with order $p^a q^b$ (where we may as well assume $a, b \in \mathbb{N}_{>0}$). Let $\chi_1, \chi_2, ..., \chi_r$ be the irreducible representation for the group G, and $\chi_1 = \chi_{\mathbb{1}}$. Let $Z_i = \{g \in G \mid |\chi_i(g)| = \chi_i(1),\ \forall i\}$. The condition in side doesn't happen so often, we can equivanelty say that $\rho_i(g)$ is a scalar by using the triangle inequality. Thus,

$$Z_i \trianglelefteq G$$

If $Z_i = G$, then G acts by scalars on V_i. But then every subspace of V_i is a subrepresentation, so the dimension of all the irreducible is 1. And so each character is of the form:

$$\chi_i : G \to \mathbb{C}^\times$$

Since G is simple, each each i is injective or trivial. If each i is injective, then G is abelian, a contradiction, and if it is trivial we get $i = 1$ (the trivial representation). Thus, it must be that:

$$Z_i = 1 \qquad \forall i > 1$$

Now, take the sylow q-subgroup Q: (notice we are using that $a, b > 0$)

$$1 \subsetneq Q \subsetneq G$$

So Q is a p-group, and so has a non-trivial center: $Z(Q) \neq 1$. Take $h \in Z(Q) \setminus \{e\}$. Hence $Q \leqslant C_G(h)$. , so $[G : C_G(h)] \big| [G : Q] = p^a$, where $[G : C_G(h)] = |\text{conj}(h)$. Now recall we've shown that:

$$\chi_1 \big| |G| = p^a q^b$$

Now, if $p \nmid \chi_i(1)$, then by the previous line we have that $\chi_1(1) \big| q^b$. So, $\gcd(\chi_i(1), |\text{conj}(h)) = 1$, and so by lemma 3.2.23

$$\chi_i(h) = 0 \qquad \text{or} \qquad \chi_i(1)$$

But if $|\chi_i(h)| = \chi_i(1)$, $h \in Z_i$, and so $i = 1$

So, if $p \nmid \chi_i(1)$ and $i > 0$, then $\chi_i(h) = 0$. Using this, we can get our contradiction: as h is not the trivial element, we get:

$$\begin{aligned} 0 &= \sum_i \overline{\chi_i(h)} \chi_i(1) \\ &= \sum_i \chi_i(h) \chi_i(1) \\ &\stackrel{!}{=} 1 + \sum_{\substack{i>1 \\ p \mid \chi_i(1)}} \underbrace{\chi_i(1)}_{\in p\mathbb{Z}} \underbrace{\chi_i(h)}_{\in \overline{\mathbb{Z}}^{\mathbb{C}}} \\ &= 1 + p\alpha && \text{some } \alpha \in \overline{\mathbb{Z}}^{\mathbb{C}} \end{aligned}$$

where the one in the $\stackrel{!}{=}$ equality came from when $i = 1$. Thus,

$$a = -1/p \in \overline{\mathbb{Z}}^{\mathbb{C}} \cap \mathbb{Q} = \mathbb{Z}$$

a contradiction, completing the proof.

(huge list of character tables: commented)

3.3 Induced Representation

This is another way of constructing representations. We saw that if $H \leqslant G$ and V is a G representation, then we can make V an H-representation through the inclusion map $H \hookrightarrow G \to \text{GL}(V)$. We may ask if we may go the other way: given an H representation W, can we find a G-representation? If φ is the representations afforded by H, can we find a Φ such that $\Phi|_H = \varphi$? It turns out we cannot: take $A_3 \subseteq S_3$ and consider a faithful representation of degree 1 (recall it's a cyclic group). Then notice that any 1 dimensional representations of S_3 will have to have A_3 in it's kernel, meaning we can't extend the action of A_3 to an action of S_3. A little more generally, if we let G be a simple group, $H \leqslant G$, and φ a representation of H where $\ker(\varphi)$ is a proper nontrivial subgroup of H, then if we somehow extend the representation φ of H to Φ of G, then $\ker(\Phi)$ would have a proper kernel too, contradicting that G is simple. So instead, we will not be extending representation from H to G, but *inducing* a representation in a natural way. This method of inducing a representation will work even in the two above cases, meaning an induced representation will not always be an extension.

Let $H \leqslant G$, and consider some H-representation V, which can be represented as a $k[H]$-module V.

Recall that we can "extend scalars" and define a new $k[G]$-module:

$$k[G] \otimes_{k[H]} V$$

This is (the outer tensor product for representatives, which I haven't talked about yet, but probably should have it here to motivate the idea)

Definition 3.3.1: Induced Representation

Let $H \leqslant G$ of a finite group G and let V be a $k[H]$-module affording the representation φ. Then the $k[G]$-module $k[G] \otimes_{k[H]} V$ is called the *induced module* of V, the representation is called the *induced representation*, and if χ is the character of φ, then the character of the induced representation is called the *induced character*, and is written as $\text{Ind}_H^G(\psi)$.

Another intuition behind defining the induced module is the following: If W is an H-representation, then let our new module be:

$$\text{Ind}_H^G(W) = \{f \in \text{Hom}_{\textbf{Set}}(G, W) \mid f(hg) = h \cdot f(g),\ \forall g \in G, h \in H\}$$

Then $\text{Ind}_H^G(W)$ is clearly a complex vector space under pointwise addition and scalar multiplication, and the G-action on this space would be:

$$\gamma \cdot f : G \to W \qquad g \mapsto f(g\gamma)$$

It should be verified that this is indeed G-linear. In this case, we get that:

$$\text{Ind}_H^G(W) = \text{Hom}_{\mathbb{C}[H]}(\mathbb{C}[G], W)$$

where we may use a variant of lemma 3.2.12 to show that this is the same thing as we've just defined. It is also easy to see the dimension of $\text{Ind}_H^G(W)$, namely:

$$\dim_{\mathbb{C}}(\text{Ind}_H^G(W)) = [G : H] \dim_{\mathbb{C}}(W)$$

with the $\mathbb{C}$-vector space isomorphism being:

$$\text{Ind}_H^G(W) \to W^{\oplus [G:H]} \qquad f \mapsto f(g_1), ..., f(g_{[G:H]})$$

where we treat $G = \coprod_i Hg_i$ for the cosets Hg_i. By letting $g_1 = e$ being the identity, there is a natural canonical map:

$$\text{Ind}_H^G(W) \to W \qquad f \mapsto f(e)$$

This motivates the following adjoint of the restriction functor:

Theorem 3.3.2: Frobenius Reciprocity

Let $H \leqslant G$. Then if V is a G-representation and W and H-representation, then:

$$\text{Hom}_G(V, \text{Ind}_H^G W) \cong_{\mathbb{C}} \text{Hom}_H(V|_H, W)$$

where $V|_H$ is the restriction of the action to H

***Proof*:**
We will explicitly construct inverse bijections Φ from left to right and Θ from right to left.
Let $\alpha : \mathrm{Ind}_H^G(W) \to W$, $f \mapsto f(1)$. Then α is H-linear. Then let

$$\Phi(f) = \alpha \circ f$$

Then $\Phi(f)$ is H-linear (check $\mathbb{C}$-linearity of Φ). For the other way, let $\psi : V|_H \to W$ be H-linear,

$$\Theta(\psi) : V \to \mathrm{Ind}_H^G(W) \qquad v \mapsto (g \mapsto \psi(gv))$$

Then this is well-defined. What's left is to check they are inverses of each other

3.3.3 Left Vs Right Adjoint This proof shows that induction is right-adjoint to restriction. It can be shown to also be *left adjoint*. A pair of functors (F, G) where the functor F is both left and right adjoint to a functor G (and hence G is also right/left adjoint to F) is sometimes called a *Frobenius pairing*. The functor Ind is usually only right-adjoint to the restriction restriction functor when G is infinite (The *compact induction* functor is usually left-adjoint to the restriction functor, see [3]).

Corollary 3.3.4: Frobenius Reciprocity On Characters

Let $H \leqslant G$, χ be the character of a G representation and ψ a character of an H-representation. Then:

$$\langle \chi|_H, \psi \rangle_H = \langle \chi, \mathrm{Ind}_H^G(\psi) \rangle_G$$

***Proof*:**
https://unapologetic.wordpress.com/2010/12/03/dual-frobenius-reciprocity/

Example 3.3.5: Induced Representations

1. $\mathrm{Ind}_G^G(W) = W$
2. $\mathrm{Ind}_{\mathbb{1}}^G(\mathbb{C}) = \mathbb{C}[G]$
3. $\mathrm{Ind}_H^G(\mathbb{1}) = \mathbb{C}[G/H]$

Another way of thinking of this result is that if V is an irreducible representation of G and W is an irreducible representation of W, then the number of times W occurs in $\mathrm{Ind}_H^G(W)$ is equal to the number of times W occurs in $V|_H$.

Proposition 3.3.6: Properties Of Induced Representation

Let $K \leqslant H \leqslant G$, W be an K-representation, $\mathrm{Ind}_K^H(W)$ the induced H representation, $\mathrm{Ind}_H^G(\mathrm{Ind}_K^H(W))$ the induced G-representation. Then:

1. If χ, χ' are two characters of W,
$$\mathrm{Ind}_K^H(\chi + \chi') = \mathrm{Ind}_K^H(\chi) + \mathrm{Ind}_K^H(\chi')$$
2. Induction is transitive:
$$\mathrm{Ind}_H^G(\mathrm{Ind}_K^H(W)) = \mathrm{Ind}_K^G(W)$$

***Proof*:**

Theorem 3.3.7: Computing Induced Representation

Let $H \leqslant G$ of a finite group G, and $g_1, g_2, ..., g_m$ be representatives for distinct left cosets of H in G. Let V be an H-representation of degree n. Then the induced G-representation $W = k[G] \otimes_{k[H]} V$ is an nm degree representation, and if Φ repersented the induced representation, then there exists a basis for W such that W affords the matrix representation defined for each $g \in G$ by:
$$\Phi(g) = \begin{pmatrix} \varphi(g_1^{-1}gg_1) & \cdots & \varphi(g_1^{-1}gg_m) \\ \vdots & \vdots & \cdots \\ \varphi(g_m^{-1}gg_1) & \cdots & \varphi(g_m^{-1}gg_m) \end{pmatrix}$$
where each $\varphi(g_i^{-1}gg_j)$ is an $n \times n$ block, with $\varphi(g_i^{-1}gg_j) = 0$ if $g_i^{-1}gg_j \notin H$.

***Proof*:**
First, $k[G]$ is a free right $k[H]$-module with decomposition:
$$k[G] = g_1k[H] \oplus \cdots \oplus g_mk[H]$$
Since tensor products commute with direct sums, we get:
$$W = k[G] \otimes_{k[H]} V \cong (g_1 \otimes V) \oplus \cdots \oplus (g_m \otimes V)$$
Since k is in the center of $k[G]$, the above isomorphism is also a k-vector space isomorphism. Thus, if $v_1, v_2, ..., v_n$ is a basis for V, then $\{g_i \otimes v_j \mid 1 \leq i \leq m,\ 1 \leq j \leq m\}$ is a basis for W, and hence $\dim_k(W) = nm$.

To show $\Phi(g)$ is of the given form, order the basis into m of size n as:
$$g_1 \otimes v_1, g_1 \otimes v_2, ..., g_1 \otimes v_n, g_2 \otimes v_1, ..., g_m \otimes v_n$$
Using this order, we can compute the matrix $\varphi(g)$ for each g acting on W given this basis. Fixing j and g, let $g_j = g_ih$ for some index i and $h \in H$. Then for every k:
$$g(g_j \otimes v_k) = (gg_j \otimes v_k) = g_i \otimes v_k = \sum_{t=1}^{n} a_{tk}(h)(g_i \otimes v_t)$$

where a_{tk} is the t, k coefficient of the matrix h acting on V with respect to the basis $v_1, v_2, ..., v_n$. Notice that the action of g on W maps the jth block of n basis vectors of W to the ith block of basis vectors, and then has the matrix $\varphi(h)$ as that block. Since $h = g_i^{-1} g g_j$, this fully describes $\Phi(g)$, as we sought to show.

Corollary 3.3.8: Computing The Induced Character

Let $H \leqslant G$ where G is a finite group, V be an H-representation with character χ, and $\mathrm{Ind}_H^G(\chi)$ the induced character. Then:

$$\mathrm{Ind}_H^G(\chi) = \sum_{i=1}^{m} \chi(g_i^{-1} g g_i)$$

where $\chi(g_i^{-1} g g_i) := 0$ if $g_i^{-1} g g_i \notin H$. Hence, $\mathrm{Ind}_H^G(\chi)(g) = 0$ if g is not conjugate of some element in H in G, in particular if $H \trianglelefteq G$ then $\mathrm{Ind}_H^G(\chi)$ is zero on all elements $G - H$.

Proposition 3.3.9: Induced Index Two Induced Representation

Let $H \leqslant G$ where $[G : H]$ has index two[a] (ex. $A_n \leqslant S_n$ or $C_n \leqslant D_{2n}$). Then if V is a finite-dimensional irreducible representation of G, either $V|_H$ is irreducible or $V|_H = W_1 \oplus W_2$ where $W_1 \ncong W_2$, are both ireducibile, and have the same dimension. Moreover, the second option implies that $V \cong V \otimes \chi$ where $\chi : G \twoheadrightarrow G/H \cong \{\pm 1\}$ (i.e. a 1-dimensional representation)

[a]Recall that this makes H a normal subgroup

In the case where $V|_H$ is not irreducible, we can imagine it splitting into a diagram like so:

$$\begin{tikzcd}
 & W_1 \\
V \arrow[ur, dash] \arrow[dr, dash] & \\
 & W_2
\end{tikzcd}$$

while in the case where $V|_H$ is irreducible, what we actually have (and didn't have time to prove), is that we have *two* irreducible representation's of G (V and V') that both get restricted to W:

$$\begin{tikzcd}
V \arrow[dr, dash] & \\
 & W \\
V' \arrow[ur, dash] &
\end{tikzcd}$$

and then the induced representation of W is the direct sum of V and V'.

***Proof*:**
Let $W \subseteq V|_H$ be an irreducible H-representation. Then Since $H \trianglelefteq G$, gW is is an H-stable space: gW is some subspace of V (it satisfies the subspace axioms since the action is linear), and for any $h \in H$, $h \cdot gW = hgW = g'hW$ that is $gW = hg'h^{-1}W$, and since $hg'h^{-1} \in H$, we have that $h \cdot gW = gW$. Now, gW is irreducible: since W is irreducible and $\varphi_g(W) = gW$ is a G-linear isomorphism (with inverse$\varphi_{g^{-1}}(gW) = W$), gW is irreducible. Finally, the structure of gW only depends on that of the coset gH

Thus, fix $\gamma \in G \setminus H$ so that

$$\sum_{g \in G} gW = W + \gamma W \subseteq V$$

where the $\subseteq$ comes from the fact that this is a sum of two subspaces. Now, notice that the left hand side is a G-stable space, but V is irreducible meaning it does not have a G-stable subspace, and therefore, we actually have the equality:

$$W + \gamma W = V$$

meaning the $H + gH = G$ is parallelled in the action! We know have two possibilities: either $W = \gamma W$, and so $V|_H = W$, and we have $\dim V = \dim W$ (which is case (a)).

on the otherhand, if $W \neq \gamma W$, then $W \cap \gamma W = (0)$ (since they are both irrreducible), and so $V = W \oplus \gamma W$ (i.e. direct sum as H-representations), and so V is the direct sum of two irreducible H-representations.

Now, by Frobeinus reciprocity, we get:

$$\langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G$$

Since W occurs in $V|_H$ by definition, we get that $1 \leq \langle W, V|_H \rangle_H$, while since V is irreducible, meaning by a dimension argument we get $\langle \mathrm{Ind}_H^G(W), V \rangle_G \leq 1$, and hence:

$$1 = \langle W, V|_H \rangle_H = \langle \mathrm{Ind}_H^G(W), V \rangle_G = 1$$

Thus, from the first one, we get that $W \ncong \gamma W$ (or else the left hand side would be equal to 2), and we must have $V = \mathrm{Ind}_H^G(W)$ (or else the right hand side would be equal to something greater than 1)

If part (b) holds, then

$$V \oplus \chi|_H \cong V|_H$$

and

$$V \cong \mathrm{Ind}_H^G(W) \cong V \oplus \chi$$

Conversely, if $V \cong V \oplus \chi$, then $\chi_V(g) = \chi_V(g)\chi(g)$ for all $g \in G$. Thus, by simple arithmetics we see that $\chi(g) = 0$ for all $g \notin H$. And so:

$$\|\chi_{V|_H}\|_H^2 = 2\|\chi_V\|_G^2$$

and so it is a direct sum of irreducibles, as we sought to show.

3.4 *Application to Harmonic Analysis

In this section, we take a brief moment to explore the representation of locally compact abelian groups, such as $\mathbb{R}$. This section will assume some understanding of topological groups, and is mainly to motivate for the reader the connection between the representation theory of this chapter with *Fourier analysis* and more broadly *Harmonic analysis*.

Every group G is always a topological group with the discrete topology (see [9, the appendix on topological groups]). The group G is finite iff it is a compact discrete group. Many groups of interest are either compact (like the torus or compact lie groups), or locally compact (like the real line). The representation theory of these groups rely on the correct framing of the representation in the finite case. For the case of infinite dimension *non-abelian* groups, the build-up is a good deal too complicated for this expository section[4], however many of the major ideas for the abelian are simple enough to exposit, and even better naturally motivate the Fourier transform and abstract harmonic analysis through the lens of representation theory.

Let G be a finite abelian group so that all complex irreducible representations are 1-dimensional, and hence characters and necessarily a homomorphism. As G is finite, say $n = |G|$, then $\chi(g^n) = \chi(e) = 1$ for all $g \in G$, and so $\chi^n = \text{id}$, showing that $\chi : G \to \mathbb{C}^*$ restricts to a map $\chi : S^1$. This shows that all characters are *unitary representations*. All characters for a finite group are necessarily unitary representations, while for non-finite groups there can be non-unitary representations[5]. Such groups which have a (general) representation theory that doesn't generalize as well as infinite unitary representations (many representation are indecomposable but not irreducible, for the details see the companion Lie groups notes). Thus, for this sections we stick to maps to S^1 with the natural multiplication structure as the subgroup of $\mathbb{C}^*$: label this collection $\hat{G} = \text{Hom}_{\mathbf{Ab}}(G, S^1)$. As the codomain is a group, $\hom_{\mathbf{Ab}}(G, S^1)$ has a group structure given by point-wise multiplication.

Definition 3.4.1: Pontryagin Group

Let G be locally compact abelian group. Then the *Pontryagin dual* of G is the group:

$$\text{Hom}_{\text{Top}}(G, S^1)$$

with the operation being point-wise multiplication.

This is just another way of representing the characters of the group. Let us stick to the special case where G is a finite abelian group and re-frame the previous theory using the Pontryagin group:

[4] see [6] for the EYNTKA series on geometric representation theory
[5] $\text{SL}_2(\mathbb{R})$ acting on $\mathbb{R}^2$ need not be unitary

Proposition 3.4.2: Pontryagin Dual

Let G be a finite abelian group, $\hat{G} = \mathrm{Hom}_{\mathbf{Ab}}(G, S^1)$ the group of characters under point-wise multiplication. Then there is an isomorphism:

$$G \cong \widehat{G}$$

Furthermore, there is a natural isomorphism:

$$G \cong \hat{\hat{G}}$$

If G is infinite, it is rarely the case that $G \cong \widehat{G}$ (think $\widehat{L^p} \cong L^q$). Even when G is compact this G need not be isomorphic to $\widehat{G}$. What is most important is that $\widehat{G}$ contains the necessary information to recover G, as the double-dual recovers G. We can think of $\widehat{(-)}$ as an operation that goes back and forth from a group to a unitary representation, and as we are working with unitary representations this collection also forms a group[6]. We can thus imagine going back and forth between these two:

$$\text{Group} \overset{\widehat{(-)}}{\rightleftarrows} \text{Unitary Rep.}$$

***Proof*:**
As G is finite abelian,

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}/ \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

where $n_1 \big| n_2 \big| \cdots \big| n_k$. Choose generators, $a_1, ..., a_k$. Define the indicator characters on the generators:

$$\chi_i(a_j) = \begin{cases} e^{\frac{2\pi i}{n_i}} & i = j \\ 1 & i \neq j \end{cases}$$

These certainly generate $\hat{G}$. Define the map on the generators:

$$\Phi(a_i) = \chi_i$$

This is an isomorphism.

For the double-dual, take the natural map:

$$\Phi(g_i) = \mathrm{ev}_{g_i}$$

where as usual $\mathrm{ev}_{g_i}(\chi) = \chi(g_i)$. This is easy to check that is an isomorphism, completing the proof.

Now, when studying spaces we study all (continuous) functions on a space. In our case, our space is finite abelian groups. Let G have the discrete topology so that the set of continuous functions from G to $\mathbb{C}$ is all set functions. We want a $\mathbb{C}$-algebra isomorphism between $\mathbb{C}[G]$ and $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$.

[6] the collection of representations also form a ring called the *representation ring* and has fascinating ties to K-theory and intersection theory, but it is again too much for this section and the connections are discussed in [1, Part II] of the EYTNKA series

These two sets are clearly bijective via the natural mapping:

$$\sum_i a_i g_i \mapsto (g_i \mapsto a_i)$$

And this map preserves addition if addition is defined point-wise in the hom-set:

$$\begin{aligned}
&\Phi\left(\sum_i a_i g_i + \sum_i b_i g_i\right) \\
&= \Phi\left(\sum_i (a_i + b_i) g_i\right) \\
&\mapsto (g_i \mapsto a_i + b_i) \\
&= (g_i \mapsto a_i) + (g_i \mapsto b_i) \\
&= \Phi\left(\sum_i a_i g_i\right) + \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

But it does *not* preserve multiplication if the operation is pointwise multiplication:

$$\begin{aligned}
&\Phi\left(\sum_{g\in G} a_i g \sum_{g\in G} b_i g\right) \\
&= \Phi\left(\sum_{g\in G} \sum_{g_k g_\ell = g} (a_k b_\ell) g\right) \\
&\mapsto (g_i \mapsto \sum_{g_k g_\ell = g_i} (a_k b_\ell)) \\
&\neq (g_i \mapsto a_i b_i) \\
&\neq (g_i \mapsto a_i) \cdot (g_i \mapsto b_i) \\
&= \Phi\left(\sum_i a_i g_i\right) \cdot \Phi\left(\sum_i b_i g_i\right)
\end{aligned}$$

This is because the points g_i act on each other. There is a natural multiplication function defined on $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ called *convolution*:

Definition 3.4.3: Convolution

Let $p, q \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Then their *convolution* is define pointwise as:

$$(p * q)(g) = \sum_{g_i g_j = g} p(g_i) q(g_j) = \sum_{h\in G} p(h) q(h^{-1} g)$$

Lemma 3.4.4: Isomorphism With Convolution

Let G be a finite abelian group. Then:

$$(\mathbb{C}[G], +, \cdot) \cong_{k\text{-}\mathbf{Alg}} (\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$$

where the left hand side is the usual group ring and the right hand side has convolution as multiplication

***Proof*:**
refer to the above calculations.

Now, $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ is a $|G|$-dimensional $\mathbb{C}$ vector-space with the natural basis given by the indicator function:

$$\delta_h(g) = \begin{cases} 1 & g = h \\ 0 & g \neq h \end{cases}$$

Namely for any $f : G \to \mathbb{C}$ there is a unique decomposition given the functionals defined by each $g \in G$:

$$f = \sum_{h \in G} f(h)\delta_h \qquad \text{so that} \qquad f(g) = \sum_{h \in G} f(h)\delta_h(g) = f(g) \cdot 1 = f(g)$$

The key feature of this basis is that it acts like evaluation when taking the inner product. Define:

$$\langle f, g \rangle = \sum_{h \in G} f(h)\overline{g(h)}$$

Then $\{\delta_g\}_{g \in G}$ is an orthogonal basis and:

$$\langle f, \delta_{g_i} \rangle = f(g_i) \cdot \overline{\delta_{g_i}(g_i)} = f(g_i) \cdot 1 = f(g_i) = a_i$$

This is a special case of the Riesz representation theorem: every linear functional here is inner-product representable using these δ_{g_i}. This inner product can be thought as expressly chosen so that the collection δ_{g_i} forms an orthonormal basis, or conversely given this standard inner product, the indicator function will always be an orthonormal basis. By Pontryagin duality, $G \cong \hat{G}$, but this group isomorphism does not *a-priori* preserve the orthogonality of the basis given by G. The key insight of Fourier analysis is that it does!

Proposition 3.4.5: Character Form Orthogonal Basis: Fourier Basis

The characters $\widehat{G}$ form an orthogonal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$. Normalizing by $1/\sqrt{|G|}$ forms an orthonormal basis.

***Proof*:**
If $i = j$:

$$\begin{aligned}\langle \chi_i, \chi_i \rangle &= \sum_{g \in G} \chi_i(g)\overline{\chi_i(g)} \\ &= \sum_{g \in G} |\chi_i(g)|^2 \\ &= \sum_{g \in G} 1^2 \\ &= |G|\end{aligned}$$

if $i \neq j$, Then as $\overline{\chi_j} = \chi_j^{-1}$ (as we are mapping to the unit circle), we get that $\chi' = \chi_i \chi_j^{-1}$ is a non-trivial character, so there is some h such that $\chi'(h) \neq 1$. Let

$$\langle \chi_i, \chi_j \rangle = S = \sum_{g \in G} \chi'(g)$$

Then:

$$\chi'(h)S = \chi'(h) \sum_{g \in G} \chi'(g) = \sum_{g \in G} \chi'(h)\chi'(g) = \sum_{g \in G} \chi'(h + g)$$

As $h + G = G$, this is just a re-arranging of terms, hence:

$$\chi'(h)S = \sum_{g \in G} \chi'(g) = S \qquad \text{so} \qquad S(\chi'(h) - 1) = 0$$

Now, by assumption $\chi'(h) \neq 1$, so $\chi'(h) - 1 \neq 0$, so as $\mathbb{C}$ is an integral domain it must be that $S = 0$, implying:

$$\langle \chi_i, \chi_j \rangle = 0$$

as we sought to show.

This inner product can be thought of as extracting the "phase" information.

Example 3.4.6: Phase-Space Basis

Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{(0,0), (1,0), (0,1), (1,1)\}$. A character $\chi : G \to \mathbb{C}^*$ is determined by its values on the generators $g_1 = (1,0)$ and $g_2 = (0,1)$. Since both generators have order 2, we must have $\chi(g_1)^2 = \chi(g_1^2) = \chi(0,0) = 1$ and likewise for g_2. Thus, the character's values on the generators must be either 1 or -1. This gives $2 \times 2 = 4$ possible characters, which we can index by pairs $(k_1, k_2) \in \{0,1\}^2$.

The four characters χ_{k_1,k_2} are defined by their values on the generators:

$$\chi_{k_1,k_2}(1,0) = (-1)^{k_1} \qquad \text{and} \qquad \chi_{k_1,k_2}(0,1) = (-1)^{k_2}$$

From this, the value for any element $(a,b) \in G$ is $\chi_{k_1,k_2}(a,b) = (-1)^{k_1 a + k_2 b}$.

This gives the following character table, which explicitly lists the functions that form our new "basis":

$\hat{G}$	**(0,0)**	**(1,0)**	**(0,1)**	**(1,1)**
χ_{00}	1	1	1	1
χ_{10}	1	-1	1	-1
χ_{01}	1	1	-1	-1
χ_{11}	1	-1	-1	1

We can verify the orthogonality relation from proposition 3.4.5 directly. Let us compute $\langle \chi_{10}, \chi_{11} \rangle$:

$$\begin{aligned}
\langle \chi_{10}, \chi_{11} \rangle &= \sum_{g \in G} \chi_{10}(g)\overline{\chi_{11}(g)} \\
&= \chi_{10}(0,0)\chi_{11}(0,0) + \chi_{10}(1,0)\chi_{11}(1,0) + \chi_{10}(0,1)\chi_{11}(0,1) + \chi_{10}(1,1)\chi_{11}(1,1) \\
&= (1)(1) + (-1)(-1) + (1)(-1) + (-1)(1) \\
&= 1 + 1 - 1 - 1 = 0
\end{aligned}$$

Using the above example, let us change perspectives on how to interpret this change of basis. The function $f : G \to \mathbb{C}$ takes the "points" $g \in G$ and maps them (continuously) to the complex numbers, and those points are evaluated by the Riesz representation of f against the indicator functions that form an orthonormal basis. But we have just found another orthonormal basis for $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ against which we can take the inner product. We may thus think of the characters $\chi \in \hat{G}$ as points in their own right: points carrying "phase" information rather than the "location" information that the indicator functions carry. This motivates the following definition:

Definition 3.4.7: Fourier Transform (Gelfand Transform)

Let G be a finite abelian group with the discrete topology. Then the *Fourier transform* is:

$$\widehat{(-)} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \mapsto \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C}) \qquad f \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

where the image of f is often written as:

$$\hat{f}(\chi) = \langle f, \chi \rangle = \sum_{g \in G} f(g)\overline{\chi(g)}$$

The name *Gelfand transform* comes from the fact that the Fourier transform is traditionally about real periodic functions, or Legesgue measurable functions. It wasn't until later where mathematicians like Folland and Gelfand generalized it to the context of locally compact abelian groups, where the above is the special case where G is finite.

Now, certainly as vector spaces $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \cong_{\mathbb{C}} \mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$ mapping:

$$(g \mapsto \langle f, \delta_g \rangle) \mapsto (\chi \mapsto \langle f, \chi \rangle)$$

But does this map preserve the inner product, and more so can the convolution product of $\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})$ be preserved given the appropriate choice of product on $\mathrm{Hom}_{\mathbf{Top}}(\hat{G}, \mathbb{C})$? The answer is yes, and can be thought of as the corner-stone result of harmonic analysis on discrete abelian groups:

Theorem 3.4.8: Fourier Reciprocity

The map given in definition 3.4.7 is a unitary $\mathbb{C}$-algebra isomorphism,

$$(\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}),+,*) \cong (\mathrm{Hom}_{\mathbf{Top}}(\hat{G},\mathbb{C}),+,\cdot)$$

with $\cdot$ in $(\mathrm{Hom}_{\mathbf{Top}}(\hat{G},\mathbb{C}),+,\cdot)$ being point-wise multiplication.

Recall that unitary means $\langle \hat{f},\hat{g}\rangle = \langle f,g\rangle$. In a conventional real analysis course:

- the $\mathbb{C}$-algebra homomorphism is called the *Convolution Theorem*,
- the unitary property is called *Plancherel's Theorem*,
- invertibility (i.e. bijectivity) is called the *Fourier Inversion Theorem*.

***Proof*:**

1. *1. Linearity:* Let $f,h \in \mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C})$ and $a,b \in \mathbb{C}$.

$$\begin{aligned}
\widehat{af+bh}(\chi) &= \langle af+bh,\chi\rangle && \text{by definition}\\
&= \sum_{g\in G}(af+bh)(g)\overline{\chi(g)}\\
&= a\sum_{g\in G} f(g)\overline{\chi(g)} + b\sum_{g\in G} h(g)\overline{\chi(g)} && \text{by linearity of sums}\\
&= a\langle f,\chi\rangle + b\langle h,\chi\rangle\\
&= a\hat{f}(\chi) + b\hat{h}(\chi)
\end{aligned}$$

Since this holds for all $\chi \in \hat{G}$, the transform is a linear map.

2. $\mathbb{C}$-algebra Homomorphism (The Convolution Theorem): We must show that the Fourier transform converts convolution into pointwise multiplication.

$$\begin{aligned}
\widehat{f*h}(\chi) &= \langle f*h,\chi\rangle\\
&= \sum_{k\in G}(f*h)(k)\overline{\chi(k)} && \text{by definition of inner product}\\
&= \sum_{k\in G}\left(\sum_{g\in G} f(g)h(g^{-1}k)\right)\overline{\chi(k)} && \text{by definition of convolution}\\
&= \sum_{g\in G} f(g)\sum_{k\in G} h(g^{-1}k)\overline{\chi(k)} && \text{swapping order of summation}
\end{aligned}$$

Let $\ell = g^{-1}k$, so $k = g\ell$. As k runs over G, so does ℓ. The inner sum becomes:

$$\sum_{\ell\in G} h(\ell)\overline{\chi(g\ell)} = \sum_{\ell\in G} h(\ell)\overline{\chi(g)\chi(\ell)} = \overline{\chi(g)}\sum_{\ell\in G} h(\ell)\overline{\chi(\ell)} = \overline{\chi(g)}\hat{h}(\chi)$$

Substituting this back into the main expression:

$$\begin{aligned}\widehat{f*h}(\chi) &= \sum_{g\in G} f(g)\left(\overline{\chi(g)}\hat{h}(\chi)\right)\\ &= \hat{h}(\chi)\sum_{g\in G} f(g)\overline{\chi(g)}\\ &= \hat{h}(\chi)\hat{f}(\chi)\end{aligned}$$

Thus, $\widehat{f*h} = \hat{f}\cdot\hat{h}$, proving the map is an algebra homomorphism.

3. Isomorphism (The Fourier Inversion Theorem): We show the map is invertible by constructing its inverse. Let $\mathcal{F}(f) = \hat{f}$. We claim the inverse is given by $\mathcal{F}^{-1}(\hat{f})(g) = \frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g)$.

$$\begin{aligned}\frac{1}{|G|}\sum_{\chi\in\hat{G}}\hat{f}(\chi)\chi(g) &= \frac{1}{|G|}\sum_{\chi\in\hat{G}}\left(\sum_{h\in G} f(h)\overline{\chi(h)}\right)\chi(g) && \text{by definition of } \hat{f}\\ &= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(g)\overline{\chi(h)} && \text{swapping order of summation}\\ &= \frac{1}{|G|}\sum_{h\in G} f(h)\sum_{\chi\in\hat{G}}\chi(gh^{-1})\end{aligned}$$

The inner sum is an example of the *Second Orthogonality Relation* and is a standard exercise the reader should do. In particular, the sum of all characters evaluated at a group element (giving all powers of a certain root of unity) is $|G|$ if $x = e$ and 0 otherwise. Thus $\sum_{\chi\in\hat{G}}\chi(gh^{-1}) = |G|\delta_{g,h}$.

$$\frac{1}{|G|}\sum_{h\in G} f(h)(|G|\delta_{g,h}) = \sum_{h\in G} f(h)\delta_{g,h} = f(g)$$

Since we can recover f uniquely from $\hat{f}$, the map is an isomorphism.

4. Unitary Property (Plancherel's Theorem): Note that here is where it is important for the inner product to be normalized. Let us calculate the inner product for the given definition.

$$\begin{aligned}
\langle \hat{f}, \hat{h} \rangle_{\hat{G}} &= \sum_{\chi \in \hat{G}} \hat{f}(\chi)\overline{\hat{h}(\chi)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g)\overline{\chi(g)} \right) \overline{\left(\sum_{k \in G} h(k)\overline{\chi(k)} \right)} \\
&= \sum_{\chi \in \hat{G}} \left(\sum_{g \in G} f(g)\overline{\chi(g)} \right) \left(\sum_{k \in G} \overline{h(k)}\chi(k) \right) \\
&= \sum_{g \in G} \sum_{k \in G} f(g)\overline{h(k)} \sum_{\chi \in \hat{G}} \chi(k)\overline{\chi(g)} && \text{re-arranging terms} \\
&= \sum_{g \in G} \sum_{k \in G} f(g)\overline{h(k)}(|G|\delta_{g,k}) && \text{by Second Orthogonality} \\
&= |G| \sum_{g \in G} f(g)\overline{h(g)} \\
&= |G|\langle f, h \rangle_G
\end{aligned}$$

This proves *Plancherel's Theorem*. To make the map strictly unitary, one must normalize the transform. If we define the unitary transform $\mathcal{U}(f) = \frac{1}{\sqrt{|G|}}\hat{f}$, then it follows immediately that $\langle \mathcal{U}(f), \mathcal{U}(h) \rangle = \langle f, h \rangle$, completing the proof.

Example 3.4.9: Fourier Transform

Take $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ form example 3.4.6. Let us now compute the Fourier transform of a concrete function. Let $f = \delta_{(1,0)}$, so that $f(1,0) = 1$ and is zero elsewhere. This is a basis vector in the standard "position" basis. We find its coordinates in the character basis by computing $\hat{f}(\chi) = \langle f, \chi \rangle$ for each character.

$$\begin{aligned}
\hat{f}(\chi) = \langle \delta_{(1,0)}, \chi \rangle &= \sum_{g \in G} \delta_{(1,0)}(g)\overline{\chi(g)} \\
&= 1 \cdot \overline{\chi(1,0)} + 0 + 0 + 0 = \chi(1,0)
\end{aligned}$$

Using the character table, we can compute this value for each χ:

- $\hat{f}(\chi_{00}) = \chi_{00}(1,0) = 1$
- $\hat{f}(\chi_{10}) = \chi_{10}(1,0) = -1$
- $\hat{f}(\chi_{01}) = \chi_{01}(1,0) = 1$
- $\hat{f}(\chi_{11}) = \chi_{11}(1,0) = -1$

Thus, the original function $f = \delta_{(1,0)}$, which is a spike at one point in the "position" basis, has been transformed into a new function $\hat{f}$ on $\hat{G}$ with values $(1, -1, 1, -1)$.

Recall the inversion formula from the proof of theorem 3.4.8: $f(g) = \frac{1}{|G|}\sum_{\chi\in\hat{G}} \hat{f}(\chi)\chi(g)$ (here $|G| = 4$). Let's check the value at $g = (1,0)$:

$$\begin{aligned} f(1,0) &= \frac{1}{4}\Big(\hat{f}(\chi_{00})\chi_{00}(1,0) + \hat{f}(\chi_{10})\chi_{10}(1,0) + \hat{f}(\chi_{01})\chi_{01}(1,0) + \hat{f}(\chi_{11})\chi_{11}(1,0)\Big) \\ &= \frac{1}{4}\left((1)(1) + (-1)(-1) + (1)(1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1+1+1+1) = 1 \end{aligned}$$

Now let's check the value at $g = (0,1)$, where we expect 0:

$$\begin{aligned} f(0,1) &= \frac{1}{4}\Big(\hat{f}(\chi_{00})\chi_{00}(0,1) + \hat{f}(\chi_{10})\chi_{10}(0,1) + \hat{f}(\chi_{01})\chi_{01}(0,1) + \hat{f}(\chi_{11})\chi_{11}(0,1)\Big) \\ &= \frac{1}{4}\left((1)(1) + (-1)(1) + (1)(-1) + (-1)(-1)\right) \\ &= \frac{1}{4}(1-1-1+1) = 0 \end{aligned}$$

Let us now connect this to Representation Theory. So far, we have established that for a finite abelian group G, the Fourier transform is a unitary $\mathbb{C}$-algebra isomorphism between the algebra of functions on G with convolution, $(\mathrm{Hom}_{\mathbf{Top}}(G,\mathbb{C}), *)$, and the algebra of functions on the dual group $\hat{G}$ with pointwise multiplication. The representation theory perspective shows that Fourier analysis is the simplest case of a Artin-Wedderburn decomposition. The key is to view the $\mathbb{C}$-group algebra $\mathbb{C}[G]$ with the natural G-action given by left multiplication (convolution), namely $\rho : G \to \mathrm{GL}(\mathbb{C}[G])$ given by:

$$\rho(g) : f \mapsto g * f$$

From this viewpoint, the study of the $\mathbb{C}$-algebra $(\mathbb{C}[G], +, *)$ and the $\mathbb{C}[G]$-module $\mathbb{C}[G]$ is the same thing that is the study of the regular representation of G. Now, this representation can be broken down into the direct sum of 1-dimensional irreducible representations. Namely, for a finite abelian group, the set of irreducible representations is exactly the dual group: $\mathrm{Irr}(G) = \widehat{G}$ and so we get:

$$\mathbb{C}[G] \cong \bigoplus_{\chi\in\widehat{G}} V_\chi$$

where each V_χ is a 1-dimensional subspace corresponding to the character χ.

This decomposition is made concrete by the Fourier transform. The Fourier Inversion Formula, $f = \frac{1}{|G|}\sum_{\chi\in\hat{G}} \hat{f}(\chi)\chi$, is the literal expression of this decomposition. It writes the function f as a sum of components, where each term $\hat{f}(\chi)\chi$ is the projection of f onto the 1-dimensional irreducible subspace V_χ. Then the fourier coefficient $\hat{f}(\chi)$ is the coordinate of f in the irreducible component corresponding to χ.

Note that we can define projection functions $p_\chi = \frac{1}{|G|}\sum_{g\in G}\overline{\chi(g)}g$ that we can construct in $\mathbb{C}[G]$ that project onto the irreducible subspace V_χ. Convolving any function f with p_χ annihilates all other "frequency components", leaving the part of f that lies in V_χ.

Hence, the Fourier basis *the* unique basis that diagonalizes the natural action of the group on its own algebra of functions.

If G is a *non-abelian* finite group, the concept of a dual group $\hat{G}$ and a simple transform onto it ceases to be useful (it is now a set with no natural group structure). However, the representation theory perspective remains: the generalized Fourier transform (given by the *Peter-Weyl Theorem*) decomposes a function f on the group into a sum of *matrix coefficients* corresponding to these higher-dimensional matrix representations.

3.4.1 Generalization to Locally Compact Abelian Groups

What happens when we wish to analyze functions on abelian groups like $(\mathbb{R}, +)$, $(\mathbb{Z}, +)$, or the circle group $(S^1, \cdot)$? These are examples of *Locally Compact Abelian (LCA)* groups. Moving from the finite to the infinite setting requires us to replace our finite sums with integrals, but to integrate on an arbitrary group, we first need a notion of "volume" or "measure". The shape of the answer is already visible in the finite case: read the delta functions δ_h as *measures*, and every f becomes a sum of weighted point masses. What we want, then, is a measure on the group that the group operation cannot see. The essential new ingredient is the *Haar measure*. For any LCA group G, there exists a measure μ that gives a consistent way to integrate functions on the group.

Definition 3.4.10: Haar Measure

A *Haar measure* on an LCA group G is a measure μ on the Borel σ-algebra of G that is left-translation-invariant. That is, for any measurable set $E \subseteq G$ and any element $g \in G$:

$$\mu(gE) = \mu(E)$$

This measure is unique up to a positive scaling constant. For abelian groups, it is also right-translation-invariant.

See [4] for details on how to construct such a measure on any LCA that is unique up to scaling. This measure is the direct generalization of taking the order of the group.

Example 3.4.11: Haar Measure

1. For a finite group with the discrete topology, the Haar measure is the counting measure, and $\mu(G) = |G|$.
2. For $(\mathbb{R}, +)$, the Haar measure is the standard Lebesgue measure dx.
3. For $(S^1, \cdot)$, the Haar measure can be taken as $d\theta$.
4. For $(\mathbb{Z}, +)$, the Haar measure is again the counting measure.

Note that in the infinite case we can no longer represent points by indicator functions; the *Dirac delta distributions* take on that role instead. See [8] for the theory of distributions.

With the Haar measure, we can redefine our main objects of study by replacing every sum over G with an integral with respect to $d\mu$.

1. *The Group Algebra $L^1(G)$:* The space of all functions becomes too large. The natural replacement for the convolution algebra $\mathbb{C}[G]$ is the space of absolutely integrable functions, $L^1(G, d\mu)$. For

two functions $f, h \in L^1(G)$, their convolution is defined by the integral:

$$(f * h)(g) = \int_G f(x)h(x^{-1}g)d\mu(x)$$

This space $(L^1(G), +, *)$ forms a Banach algebra.

2. *The Dual Group* $\widehat{G}$*:* It was proven that the correct codomain should be S^1. The dual group is the set of all *continuous* group homomorphisms into S^1.

$$\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$$

The dual group $\widehat{G}$ is itself an LCA group. It is crucial to note that, unlike the finite case, it is rarely true that $G \cong \widehat{G}$. For example:

$$\widehat{\mathbb{R}} \cong \mathbb{R}, \qquad \widehat{\mathbb{Z}} \cong S^1, \qquad \widehat{S^1} \cong \mathbb{Z}$$

This highlights why it is essential to treat $\hat{G}$ as a distinct space. The theory that governs this relationship is *Pontryagin Duality*, which states that the double-dual is canonically isomorphic to the original group: $\widehat{\widehat{G}} \cong G$.

3. *The Fourier Transform:* The definition of the Fourier transform is the direct analogue of the finite case, replacing the sum with an integral. For a function $f \in L^1(G) \cap L^2(G)$, its Fourier transform is a function $\hat{f}$ on the dual group $\hat{G}$:

$$\hat{f}(\chi) = \int_G f(g)\overline{\chi(g)}d\mu(g)$$

Similarly, the inner product that defines the Hilbert space structure on $L^2(G)$ is:

$$\langle f, h \rangle = \int_G f(g)\overline{h(g)}d\mu(g)$$

With these definitions, the entire suite of theorems proven in the finite case carries over:

Theorem 3.4.12: Fourier Reciprocity on LCA Groups

Let G be an LCA group with Haar measure μ. Then there exists a unique Haar measure ν on the dual group $\widehat{G}$ such that the Fourier transform, appropriately normalized, is a unitary $\mathbb{C}$-algebra isomorphism. The main results are:

1. *The Fourier Inversion Theorem:* For a sufficiently well-behaved function f (e.g., $f \in L^1 \cap L^2$ such that $\hat{f} \in L^1$), the original function can be recovered by an integral over the dual group:
$$f(g) = \int_{\widehat{G}} \hat{f}(\chi)\chi(g)d\nu(\chi)$$
2. *Plancherel's Theorem:* The Fourier transform extends uniquely from $L^1 \cap L^2$ to a unitary isomorphism of Hilbert spaces, $\mathcal{F} : L^2(G, \mu) \to L^2(\widehat{G}, \nu)$. This means it preserves the inner product:
$$\langle \hat{f}, \hat{h} \rangle_{L^2(\widehat{G})} = \langle f, h \rangle_{L^2(G)}$$
3. *The Convolution Theorem:* For $f, h \in L^1(G)$, the Fourier transform maps convolution to pointwise multiplication:
$$\widehat{f * h}(\chi) = \hat{f}(\chi) \cdot \hat{h}(\chi)$$

***Proof*:**
This is the Pontryagin duality theorem; a full proof via the Gelfand transform belongs to the harmonic analysis of locally compact abelian groups and is not reproduced in this starred section.

The correspondence between the finite and general LCA settings can be summarized as follows:

Finite Abelian Case	**LCA Case**
Group G	LCA Group G
Order $\|G\|$	Haar Measure μ
Group Algebra $\mathbb{C}[G]$	Banach Algebra $L^1(G, \mu)$
Hilbert Space $\mathrm{Hom}(G, \mathbb{C})$	Hilbert Space $L^2(G, \mu)$
Summation $\sum_{g \in G}$	Integration $\int_G d\mu(g)$
Dual Group $\widehat{G} = \mathrm{Hom}(G, \mathbb{C}^\times)$	Dual Group $\widehat{G} = \mathrm{Hom}_{\mathbf{Top}}(G, S^1)$
Inner Product $\sum f(g)\overline{h(g)}$	Inner Product $\int f(g)\overline{h(g)}d\mu(g)$
Fourier Transform $\sum f(g)\overline{\chi(g)}$	Fourier Transform $\int f(g)\overline{\chi(g)}d\mu(g)$

Further interesting facts can be found here: https://mathoverflow.net/questions/37021/is-fourier-analysis-a-special-case-of-representation-theory-or-an-analogue

3.4.2 The Algebraic-Geometric Meaning of the Dual Group

In algebraic geometry, a fundamental principle is that the geometry of a space can be recovered from its algebra of functions. For an affine space k^n, the corresponding algebra is the polynomial ring $k[x_1, \ldots, x_n]$. The points of the space correspond to the maximal ideals of this algebra. Specifically, for any point $p = (a_1, \ldots, a_n) \in k^n$, the evaluation map:
$$\mathrm{ev}_p : k[x_1, \ldots, x_n] \to k \qquad f \mapsto f(p)$$

is an algebra homomorphism whose kernel, $m_p = (x_1 - a_1, \ldots, x_n - a_n)$, is a maximal ideal. The quotient $k[x_1, \ldots, x_n]/m_p \cong k$ recovers the value of the function at that point.

We have established a similar setup for a finite abelian group G, with the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), *, +)$ taking the role of the function algebra. A natural question arises: can we recover the "points" of our space from the maximal ideals of this algebra? Let's first attempt a direct analogy. We might assume the points are the elements of the group G itself. For a point $g_i \in G$, we can define the point-evaluation map, which is a linear map:

$$\mathrm{ev}_{g_i} : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto f(g_i)$$

The kernel of this map is the vector subspace of functions that are zero at g_i. However, for this subspace to be an ideal, it must be closed under convolution. Let $f \in \ker(\mathrm{ev}_{g_i})$ (so $f(g_i) = 0$) and let h be any function in our algebra. Consider their convolution:

$$(h * f)(g_i) = \sum_{k \in G} h(k) f(k^{-1} g_i)$$

Just because $f(g_i) = 0$ does not imply that the sum on the right-hand side is zero. The value of f at other points, $f(k^{-1} g_i)$, can be non-zero. Therefore, the kernel of the point-evaluation map is *not* an ideal with respect to convolution. This implies the points of G do not correspond to the maximal ideals of its convolution algebra.

The key insight is that the correct "geometric points" for the convolution algebra are not the elements of G, but the characters of the dual group $\widehat{G}$.

Proposition 3.4.13: Characters and Maximal Ideals

Let G be a finite abelian group. The maximal ideals of the convolution algebra $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ are in bijective correspondence with the characters $\chi \in \widehat{G}$.

***Proof*:**

For each character $\chi \in \widehat{G}$, define a Fourier evaluation map:

$$\widehat{\mathrm{ev}}_\chi : \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \to \mathbb{C} \qquad f \mapsto \hat{f}(\chi)$$

This map takes a function and returns its Fourier coefficient at the frequency χ. By 3.4.8 $\widehat{\mathrm{ev}}_\chi$ is a $\mathbb{C}$-algebra homomorphism. Since $\widehat{\mathrm{ev}}_\chi$ is a surjective algebra homomorphism onto a field ($\mathbb{C}$), its kernel, which we denote by I_χ, is a maximal ideal:

$$I_\chi = \ker(\widehat{\mathrm{ev}}_\chi) = \{f \in \mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}) \big| \hat{f}(\chi) = 0\}$$

By the First Isomorphism Theorem for rings, we have:

$$\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C})/I_\chi \cong \mathbb{C}$$

This shows that every character $\chi \in \widehat{G}$ defines a maximal ideal. Conversely, After applying the Fourier transform we get the usual multiplication and ring structure, and then an approach similar to the Gelfand-Kolmogorov theorem gives the desired result.

For the infinite case, the maximal ideals of the Banach algebra $L^1(G, \mu)$ arise in this same way from homomorphisms into $\mathbb{C}$ (see the *Gelfand-Mazur Theorem*).

Thus, with the convolution algebra $\mathbb{C}[G]$ as our primary object, the natural geometric space that emerges is not the group G itself, but its Pontryagin dual, $\widehat{G}$. The Fourier transform is the mechanism that allows us to evaluate the functions of our algebra at the points of this "true" underlying space. This perspective, generalized by Gelfand, is the foundation of non-commutative geometry.

Concepts	**Classical Algebraic Geometry**
Algebra	Polynomial Ring $(k[x_1, \ldots, x_n], \cdot)$
Product	Polynomial Multiplication
Geometric Space	Affine Space k^n
A "Point" p	A tuple $(a_1, \ldots, a_n) \in k^n$
Evaluation Map	$\mathrm{ev}_p : f \mapsto f(p)$
Maximal Ideal for p	$\ker(\mathrm{ev}_p) = (x_i - a_i)$

Concepts	**Harmonic Analysis on G**
Algebra	Group Algebra $(\mathbb{C}[G], *)$
Product	Convolution
Geometric Space	The Dual Group $\widehat{G}$
A "Point" p	A character $\chi \in \widehat{G}$
Evaluation Map	Fourier "Evaluation" $\widehat{\mathrm{ev}}_\chi : f \mapsto \hat{f}(\chi)$
Maximal Ideal for p	$\ker(\widehat{\mathrm{ev}}_\chi) = I_\chi$

4

Quiver Representations and Path Algebras

(Currently, this chapter really focuses on Quiver theory. I will put my exploration of Lie Algebra Representations here)

(This motivation is if you know some differential geometry) Quiver's have their origin in the classification of Lie groups. A Lie group is a group $(G, \cdot)$ G is a *smooth manifold* and $\cdot : G \times G \to G$ is a *smooth map*. The classification of these types of groups (in particular, the continuous ones, since discrete lie groups is just the classification of all groups) is a problem completed by Killing. Out of his classification, there came out 5 exotic Lie groups lying outside of the other classifications. Why did these Lie groups come up? The answer of this question might come from a change in perspective: Any Lie group has a Lie Algebra – the tangent space at the identity of the group which also has an extra operation called the *lie braked* which allows to capture in some of the smooth properties of G into T_eG. These Lie brackets allow us to define a linear map $\text{ad}_x : T_eG \to \mathbb{C}$, $\text{ad}_x(y) = [x, y]$. The image of these operations are called the *ideals* of a lie algebra (not unlike rings, since an ideal $I \subseteq L$ is the set of all elements where for all $x \in L$ and $a \in I$, $[x, a] \in I$). Using this fact, we can ask for the classification of *simple lie algebras*. As an example of one:

$$\text{sl}(3, \mathbb{C}) = \left\{ \begin{matrix} a & b & c \\ d & e & f \\ g & h & i \end{matrix} \;\middle|\; a + e + i = 0 \right\}$$

which is closed under Lie braked $[A, B] = AB - BA$ (but not closed under multiplication). Now, since these are linear maps, they can be diagonalized. If we take $h_1, ..., h_k \in \text{sl}(3, \mathbb{C})$ to be diagonal elements (elements that are zero everywhere except possibly the diagonal), then $\text{ad}_{h_1}, ..., \text{ad}_{h_k}$ are all simultaneously diagonalizable. This leads us to be able to define *root systems*, which allows for *root decomposition* and *root systems*. This leads to two important theorems for classification of *complex lie algebras*:

(Cartan's Thorems)

1. Up to isomorphism, there is just one complex semisimple Lie algebra for each root system
2. With 5 exceptions, every finite dimensional semisimple Lie algebra over $\mathbb{C}$ is isomorphic to one of the classical Lie Algebras:

$$\mathrm{sl}(n, \mathbb{C}) \quad \mathrm{so}(n, \mathbb{C}) \quad \mathrm{sp}(2n, \mathbb{C})$$

   and 5 excpetional Lie algebras, labeled: e_6, e_7, e_8, f_4, g_2

4.1 Quiver's and their Decomposition

The following is just an accumulation of common terms in graph theory that we will need. The one notable exception is that we will rename oriented graphs to *quivers*

Definition 4.1.1: Quiver

A *quiver* Q is an oriented graph (V, E, s, t) where V represents the vertices (the vertex set), E is the set of edges (the "arrows") and $s : E \to V$ with $t : E \to V$ represents the orientation (s for source, t for terget), that is for any $e \in E$, if we have:

$$\overset{i}{\bullet} \xrightarrow{e} \overset{j}{\bullet}$$

then $s(e) = i$ and $t(e) = j$.

Usually, we will abuse notation say $Q = (V, E)$ is a quiver and ommit s and t. Most of the time, we will draw quivers instead of explicitely define them through a 4-tuple or 2-tuple

Example 4.1.2: Quivers

1. If $V = \{1\}$, $E = \{e\}$, and $s(e) = t(e) = 1$, then we can draw this quiver as

$$\begin{tikzcd} 1 \arrow[loop above] \end{tikzcd}$$

2. If $V = \{1, 2\}$, $E = \{e\}$, $s(e) = 1$, and $t(e) = 2$, then we can draw this quiver as

$$1 \xrightarrow{e} 2$$

3. If $V = \{1, 2\}, E = \{e, f\}$, with $s(e) = 1$, $t(e) = 2$ and $s(f) = 2$, $t(f) = 1$, then we can draw this quiver as

$$\begin{tikzcd} 1 \arrow[r, bend left, "e"] & 2 \arrow[l, bend left, "f"] \end{tikzcd}$$

If you are still uncomfortable, with quivers, try to come up with some as a challenge.

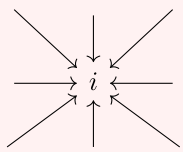
Definition 4.1.3: Path and Cycle

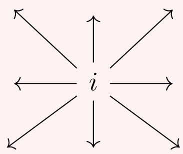
Let Q be a quiver.

1. A path γ is a sequence of edges
$$\gamma = e_k e_{k-1} \cdots e_1$$
where $t(e_i) = s(e_{i+1})$. for $1 \leq i \leq k-1$.
2. If γ is a path where $s(e_1) = a$ and $t(e_k) = b$, then we say that γ is a path from a to b, and we write $s(\gamma) = a$, $t(\gamma) = b$.
3. The number of number of edges in a path is called the *length* of he path.
4. If γ is a path and $s(\gamma) = t(\gamma)$, we call γ a *cycle*.
5. A cycle of length 1 is called a *loop*

Definition 4.1.4: Sink And Source

Let Q be a quiver

1. Let $e \in E$ and $i \in V$. We say that e *contains* i if $s(e) = i$ or $t(e) = i$. We say that e is *incoming at* i if $t(e) = i$ and that e is *outgoing at* i if $s(e) = i$
2. We say a vertex $i \in V$ is a *sink* if all edges containing i are incoming. Intuitvely, this can be visualized as:


3. We say a vertex $i \in V$ is a *source* if all edges containing i are outgoing. Intuitvely, this can be visualized as:


4. Let s be a sink or a source. Then define sQ to be a new quiver where all the arrows at s are reversed.

4.1.1 Quiver Representation and Varieties

Definition 4.1.5: Quiver Representation

Let Q be a quiver. Then a *quiver representation* V *of* Q consists of

1. a vector space $V(i)$ over a fixed field $\mathbb{F}$ for each vertex of Q
2. a linear transofmration $T_e : V(i) \to V(j)$ for each $e \in E$

Let $\mathrm{Rep}_{\mathbb{F}}(Q)$, or $\mathrm{Rep}(Q)$ be the collection of all quiver representations of Q.

Example 4.1.6: Quiver Representations

1. Let
$$1 \xrightarrow{e} 2$$
be a quiver. Then any linear transformation $T : V \to W$ is a representation of this quiver
2. Let
$$\begin{tikzcd} 1 \arrow[r, "e"] & 2 \arrow[loop above, "f"] \end{tikzcd}$$
be a quiver. Then we can take any linear transformation $T : V \to W$ and an identity map $\mathrm{id} : W \to W$. Note that f need not be represented by the identity: any linear transformation from W to itself is valid.
3. Take
$$1 \xrightarrow{e} 2 \xrightarrow{f} 3$$
Then one class of possible representations of this quiver consits of vector-spaces $V(1)$, $V(2)$, $V(3)$ where the sequence is exact at $V(2)$. Another representation can have T_e and T_f be scalar multiplications. Yet another can be the set of vector spaces and linear transformations T_e and T_f where $T_e, T_f \neq \mathrm{id}$ but $T_f \circ T_e = \mathrm{id}$. There are many more possibilities

Since each vertex has a representation, we can capture a notion of dimension of a quiver representation

Definition 4.1.7: Quiver Dimension

Let Q be a quiver and V a representation of Q. Then the *dimension* or *quiver dimension* of V is the tuple that is denoted:

$$\dim_Q V = (\dim V(i))_{i \in I}$$

where each i is a vertex. If there are finitely many vertices, then we can biject them to natural numbers and have:
$$\dim_Q V = (\dim V(1), \dim V(2), ..., \dim V(n))$$

Definition 4.1.8: Quiver Subrepresentation

Let Q be a quiver and V be a quiver representation. Then a *subrepresentation* W of V consists of a subspace $W(i) \subseteq V(i)$ for each $i \in V$ and a linear transformations restricted to the new domain, $T_e|_{W(i)} = S_e$ for each $e \in E$, such that each S_e, $S_e(W_i) \subseteq W_j$

the last condition is so that each T_e is still well-defined. As an exercise, see if you can come up with quivers and subrepresentations (you can keep it simple and consider a quiver $\bullet \to \bullet$).

Definition 4.1.9: Quiver Morphisms

Let Q be a quiver and let V and V be representations of Q. Then we say that $\varphi : V \to W$ is a *quiver morphism* if for every edge e with representation $T_e : V(i) \to V(j)$ in V and representation $S_e : W(i) \to W(j)$ in W, there exists linear transformations $\varphi(i)$ and $\varphi(j)$ such that the following diagram commutes:

$$\begin{tikzcd} V(i) \arrow[r, "T_e"] \arrow[d, "\varphi(i)"] & V(j) \arrow[d, "\varphi(j)"] \\ W(i) \arrow[r, "S_e"] & W(j) \end{tikzcd}$$

collection of all quiver morphisms between V and W is denoted $\mathrm{Hom}_{\mathbf{Q}}(V, W)$ or $\mathrm{Hom}(V, W)$ if it is unambiguous in context.

Proposition 4.1.10: Structure Of $\mathrm{Hom}(V, W)$

Let $V, W \in \mathrm{Rep}(Q)$. Then $\mathrm{Hom}(V, W)$ forms a vector-space structure over the base field $\mathbb{F}$.

***Proof*:**
left as an exercise

Example 4.1.11: Quiver Morphism

1. Let $V = \overset{\mathbb{F}^1}{\bullet} \xrightarrow{1} \overset{\mathbb{F}^1}{\bullet}$ and $W = \overset{\mathbb{F}^1}{\bullet} \xrightarrow{0} \overset{\{0\}}{\bullet}$ be two quiver representations. Then we can define a morphism:
$$\begin{tikzcd} V \arrow[d, "\varphi"] & \mathbb{F} \arrow[r, "1"] \arrow[d, "1"] & \mathbb{F} \arrow[d, "0"] \\ W & \mathbb{F} \arrow[r, "0"] & \{0\} \end{tikzcd}$$
Does there exist a nonzero morphism $\varphi : W \to V$?

2. Let $V = \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet} \xrightarrow{1} \overset{\mathbb{R}}{\bullet}$ and $W = \overset{\mathbb{R}^2}{\bullet} \xrightarrow{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} \overset{\mathbb{R}^2}{\bullet} \xrightarrow{\begin{pmatrix} 1 & 1 \end{pmatrix}} \overset{\mathbb{R}}{\bullet}$. Then: (commented to compile) is a quiver morphism. Does there exist a nonzero morphism $\varphi : W \to V$?

What we have technically defined is a *quiver representation morphism*. A quiver morphisms would usually be between two direct graphs where if $Q = (V, E, s, t)$ and $R = (V', E', s', t')$ are two quivers, then a quiver morphism would consit two functions $m_v : V \to V'$ and $m_e : E \to E'$ such that $m_v \circ s = s' \circ m_e$ and $m_v \circ t = t' \circ m_e$. This is ostensibly the same concept with different objects. All mosrphisms between Q and R would be represented as $\mathrm{Hom}_{\mathbf{Quiv}}(Q, R)$. Most of the time, we will care about the representation of quivers, and hence leaving the name of "quiver morphism" to representations. In either case, the collection of all quivers along with the just-defined quiver morphism defines a category **Quiv**, and the collection of all representations of a quiver $\mathrm{Rep}(Q)$ along with the previously-defined quiver morphism defines a category **Quiv(Q)** (you can check these as exercises)

The category **Quiv(Q)** has a a 0, namely the representation where all vertices are assigned the zero vector-sapces and the linear transformations are the identity. Since 0 exists, we can define kernels. Images also exist in this category:

Definition 4.1.12: Kernels And Images

Let $\varphi : V \to W$ be a quiver morphism. Then $\ker(\varphi)$ is the subrepresetnation of V consisting of $\ker(\varphi(i)) \subseteq V(i)$, and $\mathrm{im}(\varphi)$ is the subrepresentation of W consisting of $\mathrm{im}(V(i)) \subseteq W_i$

Note that $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are indeed subrepresentations: the fact that the vertices exist is a direct consequence of linear algebra. It remains to show that the diagram still commutes.

Definition 4.1.13: Quiver Isomorphisms

Let $V, W \in \mathrm{Rep}(Q)$. Then V is isomorphic to W if there exists a quiver morphism $\varphi : V \to W$ and $\psi : W \to V$. We denote two isomorphic quivers as $V \cong W$

Example 4.1.14: Quiver Isomorphism

In Jamail's notes. Some pretty cool ones. In pariutclar, the classification of quiver representations of the form $\bullet \xrightarrow{e} \bullet$ with each vector-space being finite dimensional.

Next, we define the dual of sub-representations. Since for every subspace of a vector space $W \subseteq V$ we can define V/W, the notion of a quotient quiver is defined similarly

Definition 4.1.15: Quotient Quiver

Let $V \in \mathrm{Rep}(Q)$ and $W \subseteq V$ a subrepresentation. Then V/W is defined as a new quiver where every vertex of V is replaced by the quotient of $V(i)/W(i)$.

Proposition 4.1.16: 1st Isomorphism for Quivers

Let $V, W \in \mathrm{Rep}(Q)$ and $\varphi : V \to W$ be a quiver morphism. Then:

$$\mathrm{im}(\varphi) = V/\ker(\varphi)$$

***Proof*:**
left as an exercise

Definition 4.1.17: Product Quiver

Let $V, W \in \text{Rep}(Q)$. Define $V \oplus W \in \text{Rep}(Q)$ as

1. $(V \oplus W)(i) = V(i) \oplus W(i)$
2. $T_e : (V \oplus W)(i) \to (V \oplus W)(j)$ is given by $T_e = T_{e,v} \oplus T_{e,w}$ (recall that $T \oplus S(a,b) = (T(a), S(b))$)

Definition 4.1.18: Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver. and $\underline{d} \in \mathbb{N}^n$. Then a *quiver variety* is:

$$\text{Rep}(\underline{d}, Q) := \prod_{e \in E} M_{d_j, d_i}(\mathbb{F})$$

where $i = s(e)$ and $j = t(e)$. Every point in $\text{Rep}(\underline{d}, Q)$ is a representation of Q with dimension vector $\underline{d}$.

A quiver variety let's us reduce a quiver where each vertex can have a different dimension (i.e. any possible total dimension) to only thinking about quivers with the same total dimension

Proposition 4.1.19: Action On Quiver Variety

Let $Q = (I, E, s, t)$ be a finite quiver with $I = \{1, ..., n\}$ and $E = \{e_1, ..., e_n\}$. Let $\underline{d} = (d_1, ..., d_n) \in \mathbb{N}^n$, and define the group $G(\underline{d}) = \prod_i GL_{d_i}(\mathbb{F})$. Then define the $G(\underline{d})$ action on $\text{Rep}(Q, \underline{d})$ as follows:

$$g = (\varphi_1, ..., \varphi_n)$$
$$(g \cdot V)_e = (\varphi_j A_e \varphi_i^{-1})_e$$

where $s(e) = i$ and $t(e) = j$.

***Proof*:**
If $g = (I)_e$ for each edge, then

$$(g \cdot V)_e = (I_j A_e I_i)_e = (A_e)_e$$

showing that $I \cdot V = V$. Associativity comes for free from assocaitivty of composition of linear maps.

I think the group in the theorme is called a *Weyl Group*.

Theorem 4.1.20: Isomorphism Equivalence

Let $\mathrm{Rep}(Q, \underline{d})$ be a quiver variety and $G(\underline{d})$ be the group as defined in proposition 4.1.19. Then two representations $V, W \in \mathrm{Rep}(Q, \underline{d})$ are isomorphic if and only if there exists a $g \in G(\underline{d})$ such that $g \cdot V$. That is, two quivers are isomorphic if they lie in the same orbit:

$$\{G(\underline{d})\text{-orbit of } \mathrm{Rep}(Q, \underline{d})\} \rightleftarrows \{\text{iso classes of rep with dim } \underline{d}\}$$

***Proof*:**

As we've seen in example 4.1.14, two quiver represenations (of the same dimension) of $\bullet \xrightarrow{e} \bullet$ are equivalent if T_e and S_e are equivalent, i.e., their range is of the same dimension, i.e. $\varphi_j A_e \varphi_i^{-1} = B_e$ for two invertible matrices φ_i and φ_j. However, this is exactly the condition for the orbits of this group-actions.

4.1.2 Decomposing Quivers

Similarly to all other algebraic structures we've encountered, we have a notion of simple objects:

Definition 4.1.21: Simple Representation

Let Q be a quiver. Then $V \in \mathrm{Rep}(Q)$ is called a *simple* or *irreducible* representation if the only subrepresentations of V are $\{0\}$ and V

The notion of simple objects is essentially trivial for vectorspaces since $V \cong \mathbb{F}^n$. In other words, every vector-space is decomposable into simple vector-spaces. The analogous notion of decomposability exists for quivers, however, they can be more complicated for quivers

Definition 4.1.22: Indecomposable Quiver

Let Q be a quiver. Then $V \in \mathrm{Rep}(V)$ said to be *decomposable* if there exists non-trival subrepresentation $V_1, V_2 \subseteq V$ such that:

$$V \cong V_1 \oplus V_2$$

and is said to be *indecomposable* otherwise

Example 4.1.23: Simple Quivers and decomposability

Let $Q = \bullet \xrightarrow{e} \bullet$. Then:

$$0 \xrightarrow{\quad 0 \quad} \mathbb{F} \qquad \mathbb{F} \xrightarrow{\quad 0 \quad} 0$$

are both simple representations. Note that the left quivers is a subrepresentations of $V = \mathbb{F} \xrightarrow{\mathrm{id}} \mathbb{F}$, and along with $\mathbb{F} \xrightarrow{\mathrm{id}} \mathbb{F}$ and $0 \xrightarrow{0} 0$, $\mathbb{F} \xrightarrow{\mathrm{id}} \mathbb{F}$ has 3 sub-quivers. Thus, V is not a simple quiver. However, V is *indecomposable*. To see this, notice that if V were decomposable, then we get:

$$\dim_Q(V_1 \oplus V_2) = \dim_Q(V_1) + \dim_Q(V_2)$$

and for the representation to be nontrival, we would have that one representation would have dimension $(1, 0)$, while the other has $(0, 1)$. However,V has no representation of dimension $(1, 0)$.

This means that the extension of quivers will be more elaborate than split exact sequences.

Theorem 4.1.24: Quiver Representation Decomposition

Let Q be a quiver and V be a representaiton where $\sum_i \dim V(i) < \infty$. Then there exists a finite set of indecomposable subrepesentations I_1, ..., I_n such that

$$V = I_1 \oplus I_2 \oplus \cdots \oplus I_n$$

***Proof*:**
Define the *total dimension* $\mathrm{Tdim}(V) = \sum_i \dim V(i)$. If $\mathrm{Tdim}(V) = 1$, then V must be only contain one vector-space $\mathbb{F}$ with all other vecor-spcaes being the trivial vector spaces. Assume for a moment that V has no oriented cycles. Then all maps from edges that are being the trivial map. Clearly, V is indecomposable, and so it is it's own decomposition.

On the other hand, if V has an oriented cycle, then without loss of generality we can consider

$$T = T_n \circ T_{n-1} \circ \cdots \circ T_1$$

so that we have $T : \mathbb{F} \to \mathbb{F}$. Since $\mathbb{F}$ has no non-trival subspace (being an indecomposable vector-space over $\mathbb{F}$), V is still indecomposable, and so is it's own decomposition.

Using strong induction, assume that all vector-spaces of total dimension $\mathrm{Tdim}(V) = n$ can be decomposed into a direct sum of indecomposable representations, and consider a representation of total dimension $n + 1$. If V is indecomposable, then we're done, so let's assume V is decomposableso that there exists two non-trival subrepresentations V_1, V_2 such that

$$V = V_1 \oplus V_2$$

Then since V_1, V_2 are both non-trivial, their total dimension must be strictly greater than and strictly less than that of V and nonzero. But then, by the inductive hypothesis, V_1 and V_2 can de decomposed into a finite direct sum of indecomposable modules. Hence:

$$V = V_1 \oplus V_2 = (I_{1,1} \oplus \cdots \oplus I_{1,n_1}) \oplus (I_{2,1} \oplus \cdots \oplus I_{2,n_2})$$

completing the proof.

Example 4.1.25: Decomposing Representations

1. Let $Q = \bullet \xrightarrow{e} \bullet$ be a quiver. Find all simple and indecomposable representations of Q. Using this knowledge, decompose any representation $V \xrightarrow{T_e} W$.

2. Let $Q = \bullet \xrightarrow{e} \bullet \xrightarrow{f} \bullet$. Find all simple and indecomposable representation of Q. Using this knoweldge, decompose any representation $V \xrightarrow{T_e} W \xrightarrow{T_f} L$.

3. Let $Q = \overset{\circlearrowleft}{\bullet}$ be a quiver. Given $\mathbb{F}$ is algebraically closed, find all simple and indecomposable representations of Q (this is heavily reliant on Jordan canonical forms from chapter **??**). Using this knowledge, decompose any representation $T : V \to V$.

4.1.3 Root Systems

Recall that if $(V, \langle \cdot, \cdot \rangle)$ ia an inner product space and $v \in V$, then

$$\frac{\langle w, v \rangle}{\langle v, v \rangle} v$$

is the projection of w onto v. Furthermore, define:

$$s_v(w) = w - 2\frac{\langle w, v \rangle}{\langle v, v \rangle} v$$

which reflects w across the hyperplane orthogonal to v.

Definition 4.1.26: Root System

Let V be an inner product space over $\mathbb{R}$. Then a *(reduced) root system* $R \subseteq V$ is a finite collection of non-zero vectors such that

1. $\operatorname{span}(R) = V$
2. For all $\alpha, \beta \in R$, $\eta_{\alpha,\beta} = \frac{2\langle \alpha, \beta \rangle}{\langle \beta, \beta \rangle} \in \mathbb{Z}$. In terms of angles, recall that $\theta = \cos^{-1}\left(\frac{\langle \alpha, \beta \rangle}{\|\alpha\|\|\beta\|}\right)$, so $n_{\alpha\beta} = \|\alpha\| \cos(\theta)$
3. For each $\alpha \in R$, the corresponding reflection $s_\alpha : V \to V$ has the property that for all $\beta \in R$, $s_\alpha(\beta) \in R$, i.e., R is symmetric under all α reflections.
4. If both $\alpha, c\alpha \in R$ then $c = \pm 1$ (this is true for reduced systems only)

Example 4.1.27: Root Systems

Let $V = \{(x, y, z) \in V \mid x + y + z = 0\}$, where V inherits the usual inner product from $\mathbb{R}^3$. Let

$$R = \{\pm(e_1 - e_2), \pm(e_2 - e_3), \pm(e_1, e_3)\} = \{\alpha, \beta, \gamma\}$$

Then R is a reduced root system. This set spans V because HERE.

Furthermore:

$$\eta_{\pm\alpha,\pm\beta} = \frac{2\langle \pm(e_1 - e_2), \pm(e_2 - e_3) \rangle}{\pm\langle (e_2 - e_3), \pm(e_2 - e_3) \rangle}$$

(finish rest later)

Include these

Notice that any root system can be simplified:

Definition 4.1.28: Polarization And Simple

Let R be a root sytem. Then if there exists a vector v such that $\langle v, \alpha \rangle \neq 0$ for all $v \in R$ then $R_+ = \{\alpha \in R \mid \langle v, \alpha \rangle > 0\}$ along with $R_+ = \{\alpha \in R \mid \langle v, \alpha \rangle < 0\}$ is called the *polarization of* R, and $R = R_+ \sqcup R_-$. The elements of R_+ are called *positive roots*. A positive root is called *simple* if it cannot be expressed as the sume of other positive roots.

Example 4.1.29: Simple Roots
homework, exercise 8 in Jimails notes

Theorem 4.1.30: Simple Root Decomposition

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$. Let Π denote the set of simple roots for this polarization. Then:

1. Every $\alpha \in R_+$ is a sum of simple roots
2. If $\alpha, \beta \in R_+$, then $\langle \alpha, \beta \rangle \leq 0$
3. The set Π of simple roots froms a basis for V

***Proof*:**
exercise 9

(words about root systems being isomorphic, which I'm guessing is a map from one vec sp from another s.t. all the properties of root systems are preserved)

Definition 4.1.31: Cartan Matrix

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and let $r = |\Pi|$. Then define *Cartan matrix* A to be the $r \times r$ matrix defined by:

$$A_{ij} = n_{\alpha_j \alpha_i} = \frac{2\langle \alpha_i, \alpha_j}{\alpha_i, \alpha_i}$$

Example 4.1.32: Cartan Matrix
exercise 10

(word to motivate the following visualization)

Definition 4.1.33: Dynkin Diagrams

Let R, V be a (reduced) root system with polarization $R = R_+ \sqcup R_-$, let Π denote the set of simple roots for this polarization, and identify Π with the set $\{1, ..., k\}$. Then we can construct a graph following these steps:

1. The vertices are given by simple roots Π (or equivalently, by indices $i \in \{1, ..., k\}$)
2. We join a vertex i and j by n_{ij} edges, which depends on the angle θ where
$$\theta = \frac{\pi}{2} \Rightarrow n_{ij} = 0 \qquad \theta = \frac{2\pi}{3} \Rightarrow n_{ij} = 1 \qquad \theta = \frac{3\pi}{4} \Rightarrow n_{ij} = 2 \qquad \theta = \frac{5\pi}{6} \Rightarrow n_{ij} = 3$$
3. Roots joined by a single edge have the same length. Roots joined by muliple edges have an arrowhead ">" on the edge pointing towards the root with sorther length (imagine this as an inequality between the length of the roots)

Example 4.1.34: Dynkin Diagrams

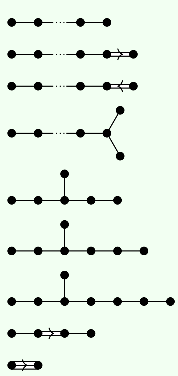
1. finding dynkin diagram for many root systems

We know proceed to classify root systems. As with other definition, if R is a root system and $R = R_1 \sqcup R_2$ where R_1, R_2 are proper subsets of R that are root systems, we say that R is reducible, and irreducible otherwise.

Theorem 4.1.35: Classification of Root Systems

Let R be a (reduced) irreducible root system. Then it's dynkin Diagram is one of the following:

A_n B_n C_n D_n E_6 E_7 E_8 F_4 G_2



***Proof*:**
here

(representing root systems via it's symmetries)

Definition 4.1.36: Weyl Group

Let $R \subseteq V$ be a root system. Let $W \subseteq \mathrm{GL}(V)$ be the group generated by the reflections s_α, $\alpha \in R$. Then W is called the *Weyl group* of R.

Since it is a group generated by reflections, every element acts linearly on V. Furthermore, it perserves lengths and angles, and so it is an element of $O(V)$ (**??**)

Example 4.1.37: Weyl Group
p. 11 Jimail

Definition 4.1.38: Simple Reflection

Let $R \subseteq V$ be a root system with Weyl group W, and $R = R_+ \sqcup R_-$ is a polarizatoin with simple roots Π.
Then for each $\alpha_i \in \Pi$, the corresponding reflection $s_i = s_{\alpha_i} \in W$ is called a *simple reflection.*

Theorem 4.1.39: Weyl Groups Representing Root Systems

Let $R \subseteq V$ be a root system with Weyl group W, and $R = R_+ \sqcup R_-$ a polarization with simple roots Π. Then:

1. The simple roots s_i (for $i \in \Pi$) generate W
2. $W(\Pi) = R$. Hence, the root system can be recovered from the simple roots; we apply all Weyl group elements to Π to produce the full list of possible roots

***Proof*:**
exercise?

(reduced expressions and longest elements)

Definition 4.1.40: Length Of Weyl Group Elements

Let W be the Weyl group of a root system R, and Π the simple roots. Then the *length* of $w \in W$ is the shortest length of word for w as a product of simple reflections. We denote the length of w by $l(w)$.

In other words, if $w = s_{i_\ell} \cdots s_{i_1}$ and no shorts sequence exists, then $l(w) = \ell$.

Theorem 4.1.41: Generating Roots Using Weyl Group

Let W be the Weyl group of a root system R with polarization $R = R_+ \sqcup R_-$ and Π the corresponding simple roots.

1. There exists a unique element $w_0 \in W$ such that $\ell(w_0) = |R_+|$
2. The element w_0 has the property that $w_0(R_+) = R_-$
3. If $w = s_{i_k} \cdots s_{i_1}$ is an expression for w_0 of length $l(w_0)$, then

$$\begin{aligned} \gamma_1 &= \alpha_{i_1} \\ \gamma_2 &= s_{i_1}(\alpha_{i_2}) \\ &\vdots \\ \gamma_k &= s_{i_k} \cdots s_{i_1}(\alpha_{i_k}) \end{aligned}$$

   is an enumeration of the positive roots
4. The elements γ_j defined above satisfy the propety that the roots γ_j, $s_{i_1}(\gamma_j)$, ..., $s_{i_1}(\gamma_j) \in R_+$ are positive, but the roots $s_{i_j} s_{i_{j-1}} \cdots s_{i_1}(\gamma_j) \in R_-$ are negative.

***Proof*:**
Jimail didn't prove – we don't have enough tools yet

Now that we have a better undersatnding of Weyl groups, we seek to link some of their structure to structure of simple roots. As we saw, the relations between simple roots can be captured via Dynkin Diagrams. We will seek for elements of the Weyl group which are "compatible" with the Dynkin diagrams. (for some reaosn), let Γ be a dynkin diagram of type A, D, or E (i.e. a Dynkin diagram with no multiple edges). Let R denote the corresponding root system. For any orientation Ω of Γ, Let (γ, Ω) represent the correspondign quiver. Recall that if i is a sink or source in Ω, we have an operator $s_i^{\pm}$ that reverses the arrows at i. Denote thie operation by

$$(\Gamma, \Omega) \mapsto (\Gamma, s_i^{\pm}\Omega)$$

where we would put a $+$ if it is a sink-to-source operation, and a $-$ if it is a source-to-sink operation

Definition 4.1.42: Adapted Word

Let Γ be a Dynkin diagram of type A, D, or E (i.e. a diagram with no multiple edges, or "simply laced"), W the corresponding Weyl group, and Ω an orientation of Γ. Let $w \in W$. Then the reduced expression $w = s_{i_k} \cdots s_{i_1}$ for w is *adapted for* Ω if

1. The verex i_1 is a sink for (Γ, Ω)
2. The verex i_2 is a sink for $(\Gamma, s_{i_1}^+\Omega)$

   $\vdots$

3. The verex i_k is a sink for $(\Gamma, s_{i_{k-1}}^+ \cdots s_{i_1}^+\Omega)$

i.e. at each step, we can apply the operator $s_{i_j}^+$ to the quiver $(\Gamma, s_{i_{j-1}}^+ \cdots s_{i_1}^+\Omega)$

Example 4.1.43: Adapted Dynkin Diagrams
p. 15 Jaimal

Theorem 4.1.44: Lusztig's Theorem

Let R be an A, D, or E root system with dynkin diagram Γ and Weyl group W. For any orientation Ω of Γ, there exists an expession for the longest element $w_0 \in W$ which is adapted for Ω

***Proof*:**
exercise 19

4.1.4 Gabriel's Theorem

In this section, we come back to Quiver's. Recall in example 4.1.25 it was asked to show how find the indecomposable quivers with underlying graphs of type A_1, A_2, and a cyclic quiver. The result was that A_2 and A_3 (with the given orientation) reduced to finitely many indecomposable representations. However, if there is a cycle, the quiver did not afford such a representation. A quiver which has finite decomposition is naturally much easier to study (TBD). The question then becomes:

> Which quivers Q can be decomposed into finitely many indecomposable representations?
> Can we classify all such quivers?

The answer of this question is *Gabriel's Theorem*; the goal of this section is to prove this result. Gabriel's Theorem is the key result linking the representation of quivers to Lie algebras, a pivotal tool in studying manifolds with group structures.

(a,b) To make the connection more precise, some terms need to be defined:

Definition 4.1.45: Collection Of Indecomposables

Let Q be a quiver, and let $\text{Rep}(Q)$ be it's representations. Then the set $\text{Ind}(Q) = \{[X] \mid X \in \text{Rep}(Q) \text{ is indecomposable }\}$ is the set of isomrophism classes of indecomposable quivers of $\text{Rep}(Q)$.
We say that a quiver is of *finite type* if $\text{Ind}(Q)$ is finite.

Before continuing with proving he result, it is useful to at least see how such a connection might even be hypothesized:

Example 4.1.46: Motivating Connection

Consider the quiver $\overset{1}{\bullet} \longrightarrow \overset{2}{\bullet} \longrightarrow \overset{3}{\bullet}$. The underlying graph is A_3, so let α_1, α_2, and α_3 denote the simple roots for the root system A_3. Using these, we can deduce the positive roots are:

$$\{\alpha_1, \alpha_2, \alpha_3, \alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3\}$$

We furthermore saw in example 4.1.25 that the indecomposable representations are:

$$\mathbb{F} \xrightarrow{0} 0 \xrightarrow{0} 0 \qquad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{0} 0$$

$$0 \xrightarrow{0} \mathbb{F} \xrightarrow{0} 0 \qquad 0 \xrightarrow{0} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

$$0 \xrightarrow{0} 0 \xrightarrow{0} \mathbb{F} \qquad \mathbb{F} \xrightarrow{1} \mathbb{F} \xrightarrow{1} \mathbb{F}$$

Here is the key insight: notice we can correspond the *dimension vector* of these repesenations to the positive roots!

Theorem 4.1.47: Gabriel's Theorem

Let Q be a quiver. then Q is of finite type if and only if the underlying graph of Q is a Dynkin Diagram of type A, D, or E. In this case, the dimension vector:

$$\dim_Q : \text{Ind}(Q) \to R_+ \qquad [I] \mapsto \dim_Q(I)$$

bijects the indecomposable representations (up to isomorphism) and positive roots in the corresponding root system.

The proof will be relegated to page pageref:HERE

The theorem hints to us that indecomposable quivers somehow relate to the simple reflections of a Weyl group. However, a Dynkin Diagram is inherently unoriented, while Quivers are by definition an oriented graph. The first thing we need to do to fix is find the equivalent of a "reflection" for quivers.

The idea for the following definition is as follows: Take the quiver and the reflection

$$\overset{1}{\bullet} \xrightarrow{e} \overset{2}{\bullet} \qquad \overset{1}{\bullet} \xleftarrow{f} \overset{2}{\bullet}$$

Now, given a representation assigning $V(1)$, $V(2)$, and T_e, how should we assign values to the reflection? One way of doing it is by taking $T_e : V(1) \to V(2)$, and mapping it to $S_f : \ker(T_e) \to V(1)$ where $S_f(x) = 0$. Visually, we see that:

$$\left(\underset{\bullet}{V(1)} \xrightarrow{T_e} \underset{\bullet}{V(2)} \right) \mapsto \left(\underset{\bullet}{V(1)} \xleftarrow{S_f} \underset{\bullet}{\ker(T_e)} \right)$$

this map considers what happens at a sink. Another approach maps by considering the source:

$$\left(\underset{\bullet}{V(1)} \xrightarrow{T_e} \underset{\bullet}{V(2)} \right) \mapsto \left(\underset{\bullet}{\mathrm{coker}(T_e)} \xleftarrow{S_f} \underset{\bullet}{V(2)} \right)$$

hwere $S_f(x) = \overline{T_e(x)}$. Both of these asignements are in fact functorial, If we have a quiver morphism $\varphi : V \to W$ represented by:

$$\begin{tikzcd} V(1) \arrow[r, "T_e"] \arrow[d, "\varphi(1)"] & V(2) \arrow[d, "\varphi(2)"] \\ W(1) \arrow[r, "S_e"] & W(2) \end{tikzcd}$$

Then the appropriate morphim φ maps to is represented in these diagrams:

$$\begin{tikzcd} V(1) \arrow[d, "\varphi(1)"] & \ker(T_e) \arrow[l, "T_f|_{\ker(T_e)}"'] \arrow[d, "\varphi(1)|_{\ker(T_e)}"] \\ W(1) & \ker(S_e) \arrow[l, "S_f|_{\ker(S_e)}"'] \end{tikzcd} \qquad \begin{tikzcd} \mathrm{coker}(T_e) \arrow[d, "\overline{\varphi(2)}"] & V(2) \arrow[l, "\overline{T_f}"'] \arrow[d, "\varphi(2)"] \\ \mathrm{coker}(S_e) & W(2) \arrow[l, "\overline{S_f}"'] \end{tikzcd}$$

which both commute (where it was shown that $\varphi|_{\ker(T_e)}(\ker(T_e)) \subseteq \ker(S_e)$ when we showed the cateogry of quivers contains kernels). This will essentially represent how a reflection functor between two quivers will be defined – locally.

The different between these two functors is where there is a sink or a source. If i is a sink, then for any given quiver representatin $V \in \mathrm{Rep}(Q)$, let

$$\Psi_i : \bigoplus_{e:i\to j} V(j) \to V(i)$$

represent the collection of all linear transformation, where for each summand we have $T_e : V(i) \to V(j)$. If i is a source, then let:

$$\Phi_i : V(i) \to \bigoplus_{e:i\to j} V(j)$$

where each summand is mapped to T_e.

Definition 4.1.48: Reflection Functor

Let Q be a quiver. Define a functor from $\text{Rep}(Q)$ to $\text{Rep}(s_i^{\pm}Q)$ as follows:

1. For any $V \in \text{Rep}(Q)$, define $S(V)_i^+ \in \text{Rep}(s_i^+(Q))$ for a sink i, where at each vertex k,
$$S_i^+(V)(k) = \begin{cases} V(k) & i \neq k \\ \ker(\Psi_i) & i = k \end{cases}$$
and for each linear transformation T_e:
$$S_i^+(T_e) = \begin{cases} T_e & s(k) \neq i \\ \pi_k \circ \iota_i & \text{if } f : i \to k \text{ is in } s_i^+(Q) \end{cases}$$
where $\iota_i : \ker(\Psi) \to \bigoplus_{e:j\to i} V(j)$ is the inclusion map and $\pi_k : \bigoplus_{e:j\to i} V(j) \to V(k)$ is the projection map onto the kth component.
2. For any $V \in \text{Rep}(Q)$, define $S(V)_i^- \in \text{Rep}(s_i^n(Q))$ for a source i, where at each vertex k,
$$S_i^+(V)(k) = \begin{cases} V(k) & i \neq k \\ \text{coker}(\Phi_i) & i = k \end{cases}$$
and for each linear transformation T_e:
$$S_i^+(T_e) = \begin{cases} T_e & s(k) \neq i \\ q_k \circ \iota_i & \text{if } f : i \to k \text{ is in } s_i^+(Q) \end{cases}$$
where $\iota_i : V(k) \to \bigoplus_{e:j\to i} V(j)$ is the inclusion map and $q : \bigoplus_{e:j\to i} V(j) \to \text{coker}(\Phi_i)$ is the quotient map

Then these two functors, $S_i^{\pm} : \text{Rep}(Q) \to \text{Rep}(s_i^{\pm}Q)$ are called the *Bernstein-Gelfand-Ponomarev reflection functor*, or *reflection functor* for short.

Example 4.1.49: Reflection Functor

p. 70 Jaimal. The key idea is that these functors give different quiver representations, even if it on A_2 with the only orientation and doing S_1^+ then S_2^-.

A couple of properties of the functors needs to be established. For notational simplicity, let $\text{Rep}^{i,-}(Q)$ denote the full subcategory consisting of representations of Q for which the map
$$\Psi_i : \bigoplus_{e:j\to i} V(j) \to V(i)$$
is surjective, while $\text{Rep}^{i,+}(Q)$ denotes the full subcategory consisting of representations of Q for which the map
$$\Phi_i : V(i) \to \bigoplus_{e:i\to j} V(j)$$
is injective.

Proposition 4.1.50: Properties Of Reflection Functor

Let Q be a quiver.

1. If i is a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_i^+ \in \mathrm{Rep}^{i,+}(s_i^+(Q))$ (similarly for source)
2. If i a sink, then for any $V \in \mathrm{Rep}(Q)$, $S_j^- S_i^+ \in \mathrm{Rep}^{i,-}(Q)$ (similarly for source)
3. The restriction of the reflection functors are inverse equivalences (adjoints?)
$$S_i^+ : \mathrm{Rep}^{i,-}(Q) \to \mathrm{Rep}^{i,+}(s_i^+ Q)$$
$$S_i^- : \mathrm{Rep}^{i,+}(Q) \to \mathrm{Rep}^{i,-}(s_i^- Q)$$
4. If V is simple at vertex i, then $S_i^{\pm}(V) = 0$.
5. If V is a non-zero indecomposable representation of Q that is not simple at i, then $S_i^{\pm}(V)$ is a nonzero indecomposable representation of $s_i^{\pm}(Q)$, and $V \cong S_j^{\mp} S_i^{\pm}(V)$.
6. $S_i^{\pm}(V)$ is a nonzero indecomposable representation of $s_i^{\pm}(V)$ if and only if $s_i(\dim_Q(V)) \geq 0$
7. If $V \in \mathrm{Rep}^{i,\pm}(Q)$ is a nonzero indecomposable representation, then $\dim_Q(S_i^{\pm}(V)) = s_i \dim_Q(V)$

***Proof*:**
exercise

Finally, we define a new object with purpose HERE (Apparently, these definitions can be given if we use some homological algebra. Perhaps revamp this after you do the homological algebra section, or keep it at this level with the mention of the version of greater generality for those who want to skip Hom Alg)

Definition 4.1.51: Root Lattice

Let Q be a quiver with vertex set I. Then if the underlying graph of Q is a dynkin diagram, $\mathbb{Z}^I$ is called the *root lattice*. For $(v_i)_{i\in I} \in \mathbb{Z}^I$, we say that $v \geq 0$ if $v_i \geq 0$ for all $i \in I$. Furthermore, we let $\mathbb{Z})+^I := \left\{v \in \mathbb{Z}^I \;\middle|\; v \geq 0\right\}$

(The following theorem seems to make more sense with some homological algebra, where the domain of the function should actually be the Grothendieck group of a $\mathrm{Rep}(Q)$) (Are objects in the following theorem representations?)

Theorem 4.1.52: Root Lattice Map

here exists a map $\dim_Q : \{\text{iso classes of Rep's}\} \to \mathbb{Z}_+^I$

***Proof*:**
not proved

Definition 4.1.53: Euler And Tits Form

Let $Q = (I, E)$ be a quiver. Define a bilinar form on $\mathbb{Z}^I$ as follows: for any $v, w \in \mathbb{Z}^I$:

$$\langle v, w \rangle_Q = \sum_i v_i w_i - \sum_e v_{s(e)} w_{t(e)}$$

This is called the *Euler form* of Q. The *symmetrized Euler form* of Q is given by

$$(v, w)_Q = \langle v, w \rangle_Q + \langle w, v \rangle_Q$$

Finally, the *tits form* of Q is the quadratic form $q_Q(v) = \langle v, v \rangle = \frac{1}{2}(v, v)$

Example 4.1.54: Euler And Tits Forms

Let $Q = \overset{1}{\bullet} \xrightarrow{e} \overset{2}{\bullet} \xrightarrow{f} \overset{3}{\bullet}$. Then the Euler's form is:

$$\langle v, w \rangle_Q = \sum_{i=1}^{3} v_i w_i - v_1 w_2 - v_2 w_3$$

$$q_Q(v) = \sum_{i=1}^{3} v_i v_i - v_1 v_2 - v_2 v_3$$

The tits form is also positive definite, since if it weren't, we would be able to pick 3 positive integer's satisfying:

$$v_2(v_1 + v_3) > v_1^2 + v_2^2 + v_3^2$$

However, this is no possible (we can show this with elementary calculus and optimization, or take a couple of space cases and induction).

Now let $Q' = \overset{1}{\bullet} \xleftarrow{e} \overset{2}{\bullet} \xrightarrow{f} \overset{3}{\bullet}$. Then:

$$\langle v, w \rangle_Q = \sum_{i}^{3} v_i w_i - v_2 w_1 - v_2 w_3$$

and:

$$q_Q(v) = \sum_{i}^{3} v_i v_i - v_2 v_1 - v_2 v_3$$

then we see that the tits form of Q and Q' agree.

Theorem 4.1.55: Positive Definite Forms And Dynkin Diagrams

Let Γ be a connected graph. The Tits form q_Γ is positive definite if and only if Γ is a Dynkin diagram of type A, D, or E

***Proof*:**
ommited by Jimail

(exercise showing not true for not A, D, E)

Finally, we will state one more theorem we need for the proof

Theorem 4.1.56: Coxeter Element Lemma

Let R be a root system, $\Pi = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ a set of simple roots and W its Weyl group. Let $s_i = s_{\alpha_i}$ denote the simple reflections corresponding to the simple root α_i. let $C = s_{i_n} \cdots s_{i_1}$ where $i_1, i_2, ..., i_n$ is some ordering of the simple roots (such an element is called a *Coxeter element*).

Then for any $v \in \mathbb{Z}^I$, there exists a $m \geq 0$ such that $C^m(v) \ngeq 0$

***Proof*:**
ommited

***Proof*:**
(of Gabriels theorem) here

4.1.5 Path Algebra of Quivers

(there are some really important insights that feel like they should actually be the priority of this section!! Essentially, we build-up more on how to think about algebras in terms of quotients of free modules, build-up how "path algebras" naturally come into play in representing quivers, and finish off with the Morita equivalence between R-Mod and R/I-Mod where $I = \mathrm{Ann}_R(M)$).

Let $Q = (V, E)$ be a quiver. For vertices i, j of Q, Define the set $\mathrm{Path}(i, j) = \mathrm{span}\{\text{paths } \gamma \text{ from } i \text{ to } j\}$ (i.e. consider the path as a basis to generate a vector-space over k, where k is usually understood), which we will call the *path space*. the length of a particular path is the number of edges between each point. The length of the path from i to i is zero. If γ is a path, and if we let i, j represent a path from it to itself, then if γ is a path from i to j, we will say that γi and $j\gamma$ are the same path.

Lemma 4.1.57: Properties Of Path

here

***Proof*:**

1. We want to show that P_i has the structure of a representation Rep(Q), which consists of a collection of vector-spaces and linear transformation's between them. First, each $P_i(j)$ is a vector-space, as pointed out earlier. Next, we may define a "linear" map $T_e : P_i(j) \to P_i(k)$ to be $\gamma \mapsto e\gamma$ (i.e. append another edge to the path). Notice that since we are treating paths as basis elements to some freely generated vector-space $P_i(j)$, then we get by construciton that $T_e(c \cdot \gamma) = ec \cdot \gamma = ce\gamma = cT_e(\gamma)$. Next, taking $\gamma + \epsilon$ to be a formal sum of two paths, we can naturally extend the action of concatinating to the formal sum:

$$e(\gamma + \epsilon) = e\gamma + e\epsilon$$

   which will mirror the linear maps within a represetnation Rep(Q). Furthermore, if e, f are neigbhouring edges, then $T_{ef} = T_e T_f$. This now mirrors how Rep(Q) is both a collection of vector-spaces and a collection of maps between them.

2. We are now going to treat P_i as a representation, since we considered them to be related to them in the above example. We want to prove that the collection of vector-spces P_i along with the linear transformations T_e are indecomposable, that is we cannot find subrepresentations. This comes from the construction of our map, in particular that we have that for any collection of paths that form a basis for $P_i(j)$, we have that they are eached mapped to by some paths through the appropriate T_e for an appropriate edge e. To make it more intuitive to see, consider for example P_i of the bellow quiver with a "cycle":

$$\begin{tikzcd} & j \arrow[dr, "b"] & \\ i \arrow[ur, "a"] \arrow[rr, "c"'] & & k \end{tikzcd}$$

   Then $\dim(V(i)) = 1$, $\dim(V(j))) = 1$, and $\dim(V(k)) = 2$, with the path i mapping to ci in $V(k)$ and mapping to bi in $V(j)$, while the path bi of $V(j)$ maps to ibc, in $V(k)$. Then notice that $V(k)$ is "surjected" upon by both $V(i)$ and $V(j)$ combined, and all the maps are injective. If the above quiver were to split, then either the dimension of $V(i)$ or $V(j)$ would have to get smaller (since $V(k)$ would have to diminish a dimension): if $\dim(V(j)) = 0$, then that would break the surejctivity of T_a, while if $\dim(V(i)) = 0$, then the map $T_c \neq T_c|_A \oplus T_c|_B$ for the two subquivers A and B.

   If the quiver has a proper cycle (i.e. $P_i(i)$ has more than $\{i\}$ as a basis), then since every map must be injective (mapping onto the appropriate basis) we have that each vector-space is 1-dimensional and an isomorphism, which we've shown in chapter 1 with a dimension argument that this representation is indecomposable.

   If The quiver does not have a cycle:

$$\begin{tikzcd} j & & k \\ & i \arrow[ul, "a"] \arrow[ur, "b"'] & \end{tikzcd}$$

   Then if for the sake of contradiction P_i were decomposable, then if $\dim(V(i)) = 0$ in one of the decomposibition, so would $V(j)$ and $V(k)$, making it a trivial decomposition, while if $\dim(V(i)) = 1$, both $\dim(V(j))$ and $\dim(V(k))$ would have to be 1 or else the map wouldn't be well-defined, and hence this quiver is indecomposable.

More generally, any quiver Q can be broken down into one of these two cases: If the quiver has a "cycle", then since every map is injective and must surject onto the vertex where two edges meet, we can repeat the above argument to show that the two scenarios which we may have while trying to decompose a quiver lead to a contradiction, and if the quiver has no cycles, we repeat the argument we've done for the above quiver.

3. If P_i were not a sink, then if e is an edge where that makes i not a sink, say $s(e) = i$ and $t(e) = j$, then we can take a subquiver where the vector-space at $P_i(i) = 0$, and we have a zero-map from $P_i(i)$ to $P_i(j)$, giving us a subquiver.

   Conversely, if P_i were a sink, the only path from i would be to i, and so all other vertices would have a 0 dimensional vector-space, while $P_i(i)$ would be 1-dimensional, which makes P_i into a simple representation

4. For notational simplicity, if $\gamma = k_n k_{n-1} \cdots k_1$ is some path, let

$$f_\gamma(v) = f_{k_n}(f_{n-1}(\cdots f_1(v)))$$

   so that $f_{ab}(v) = f_a(f_b(v))$.

   For the proof, pick any $v \in V(i)$, and consider the following commutative diagram:

$$\begin{tikzcd} P_i(a) \arrow[r, "T_e"] \arrow[d, "\varphi_a"] & P_i(b) \arrow[d, "\varphi_b"] \\ V(a) \arrow[r, "S_e"] & V(b) \end{tikzcd}$$

   Then define $\varphi^v : P_i \to V$, $\varphi^v(\gamma) = S_\gamma(v)$. Then

$$S_e(\varphi_a^v(\gamma)) = S_e S_\gamma(v) = S_{e\gamma}(v) = \varphi_b^v(e\gamma) = \varphi_b^v(T_e(\gamma))$$

   proving commutativity of the diagram. If $v \neq w$, then since $\varphi_i^v(i) = v \neq w = \varphi_i^w(i)$, $\varphi^v \neq \varphi^w$. Next, if $\varphi \in \mathrm{Hom}_Q(P_i, V)$, then if $(\gamma_1, \gamma_2, ..., \gamma_n)$ is a basis for $V(i)$, consider

$$(\varphi_i(\gamma_1), ..., \varphi_i(\gamma_n)) = (v_1, ..., v_n)$$

   and define $v = \sum_i v_i$. Then extending linearly we get that $\varphi_i\left(\sum_i c_i\gamma_i\right) = \sum_i c_i v_i$, and then extending φ_i to any other φ_j through a chain of commutative diagrams, we get that all of φ is determined by where v maps to in any path in V. Thus, $\varphi = \varphi^v$, and so $\mathrm{Hom}(P_i, V)$ and $V(i)$ are in bijective correspondence. Finally, By linearly of each map in φ, and each map assigned to an edge in P_i and $V(i)$, it we see that:

$$\varphi^{v+cw} = \varphi^v + c\varphi^w \in \mathrm{Hom}(P_i, V)$$

   giving us that the map is in fact a linear map, completing the proof.

Definition 4.1.58: Path Algebra

Let $Q = (V, E)$ be a quiver. Then the *path algebra* $P(Q)$ is defined as follows:

1. As a vector-space $P(Q) = \text{span}\{\text{paths in } Q\}$, where each vertex $i \in V$ we include an element e_i representing the constant path $i \to i$
2. multiplication is defined via concatenation when legal, that is:

$$p \cdot q = \begin{cases} pq & \text{if } t(q) = s(p) \\ 0 & \text{otherwise} \end{cases}$$

4.2 Bound Quiver Algebras

The path algebra of definition 4.1.58 takes a combinatorial object, a quiver, and turns it into an associative algebra whose modules are exactly the quiver representations of definition 4.1.5. This is already a bridge between two worlds, but it runs in only one direction: every quiver produces an algebra. The converse question is the one that completes the picture. Given an abstract finite-dimensional algebra A over a field k, is there a quiver Q whose path algebra is A, or close enough to A that the two have the same representation theory? A path algebra is rarely finite-dimensional (any quiver with a cycle has paths of every length), so one cannot hope for equality on the nose. The fix is to impose relations, cutting the path algebra down by an ideal, and the content of this section is that this always suffices: over an algebraically closed field, every finite-dimensional algebra is, up to the coarsening supplied by Morita equivalence, a path algebra modulo relations. This is the structure theorem of Gabriel, and it makes the theory of quivers the universal combinatorial model for finite-dimensional representation theory.

The route to this statement passes through three reductions. First we cut a path algebra by an *admissible* ideal, which is what makes it finite-dimensional while keeping its arrows intact. Then we isolate the class of algebras for which the reconstruction is cleanest, the *basic* algebras, and show via Morita theory that restricting to them loses nothing. Finally we read a quiver directly off an abstract algebra A (its *Gabriel quiver*) using the Jacobson radical of chapter 2, and prove that A is the path algebra of that quiver modulo an admissible ideal. Throughout, A denotes a finite-dimensional algebra over k, modules are left modules, and $\text{Jac}(A)$ is the Jacobson radical. We write kQ for the path algebra $P(Q)$ of definition 4.1.58, following the standard notation, and recall its multiplication convention: for paths p and q the product $p \cdot q$ is the concatenation "q then p", nonzero exactly when $t(q) = s(p)$. In particular a path γ from i to j satisfies $e_j \gamma e_i = \gamma$, so it lives in $e_j(kQ)e_i$; this ordering of the idempotents matters when counting arrows.

The first task is to identify, inside kQ, the piece built from arrows of positive length. The constant paths e_i span a semisimple part, a product of copies of k, and everything else is generated by the arrows. Collecting the arrow-generated part into an ideal gives the object that measures "length at least one", and its powers measure length.

Definition 4.2.1: Arrow Ideal and Admissible Ideal

Let Q be a finite quiver and kQ its path algebra over k.

1. The *arrow ideal* $R_Q \subseteq kQ$ is the two-sided ideal generated by the arrows (the paths of length 1).
2. A two-sided ideal $I \subseteq kQ$ is *admissible* if there exists an integer $m \geq 2$ with

$$R_Q^m \subseteq I \subseteq R_Q^2$$

Two facts make this definition run, both immediate from the way paths concatenate. First, R_Q^ℓ is spanned by the paths of length at least ℓ: a product of ℓ arrows is a path of length ℓ (when the concatenation is legal) and zero otherwise, and every path of length ℓ factors as such a product. Consequently $kQ = kQ_0 \oplus R_Q$ as vector spaces, where $kQ_0 = \bigoplus_i ke_i$ is the span of the constant paths, and more generally the paths of a fixed length span a complement to $R_Q^{\ell+1}$ inside R_Q^ℓ. Second, the sandwiching $R_Q^m \subseteq I \subseteq R_Q^2$ has a concrete meaning at each end. The lower bound $R_Q^m \subseteq I$ forces I to contain all sufficiently long paths, which is exactly what truncates kQ/I to finite dimension. The upper bound $I \subseteq R_Q^2$ forbids relations of length 0 or 1: no relation may involve a constant path or set a single arrow equal to a combination of other arrows. This is the condition that keeps the arrows of Q visible in the quotient, so that Q can be recovered from the algebra kQ/I rather than being partly erased by the relations.

Definition 4.2.2: Bound Quiver Algebra

Let Q be a finite quiver and $I \subseteq kQ$ an admissible ideal. The quotient

$$kQ/I$$

is called the *bound quiver algebra* (or the path algebra of Q *bound by* the relations I). The pair (Q, I) is called a *bound quiver*.

Because $R_Q^m \subseteq I$, the quotient kQ/I is spanned by the images of the paths of length less than m, of which there are finitely many when Q is finite; so a bound quiver algebra is always finite-dimensional. The constant paths e_i descend to a complete set of orthogonal idempotents summing to 1, and R_Q/I is a nilpotent ideal with $(R_Q/I)^m = 0$. In fact R_Q/I is precisely the Jacobson radical of kQ/I: the quotient by it is $kQ_0 = \prod_i k$, a product of copies of k, which is semisimple, and a nilpotent ideal is always contained in the radical, so the two coincide. This last observation is the seed of the reconstruction theorem: the arrows of Q will be recovered from $\mathrm{Jac}(kQ/I)/\mathrm{Jac}(kQ/I)^2$.

Example 4.2.3: Loops, Truncation, and the A_2 Quiver

We compute three bound quiver algebras from the ground up.

1. *The dual numbers as a bound one-loop quiver.* Let Q be the quiver with one vertex 1 and one loop $\alpha\colon 1 \to 1$:

$$\begin{tikzcd} 1 \arrow[loop above, "\alpha"] \end{tikzcd}$$

The paths in Q are $e_1, \alpha, \alpha^2, \alpha^3, \ldots$, so as a vector space $kQ = k[\alpha]$ is the polynomial ring in one variable (identifying e_1 with 1 and concatenation of the loop with multiplication). The arrow ideal is $R_Q = (\alpha)$, and $R_Q^m = (\alpha^m)$. Take $I = (\alpha^2) = R_Q^2$, which is admissible with $m = 2$ (indeed $R_Q^2 \subseteq I \subseteq R_Q^2$). Then

$$kQ/I = k[\alpha]/(\alpha^2)$$

is the algebra of *dual numbers*, two-dimensional over k with basis $\{e_1, \alpha\}$ and the single relation $\alpha^2 = 0$. Its radical is $R_Q/I = (\alpha)$, one-dimensional, with square zero, and kQ/I modulo its radical is k: there is exactly one simple module, and it is one-dimensional. This is the smallest example of a bound quiver algebra that is not itself a path algebra, the relation $\alpha^2 = 0$ being unavoidable if one wants a finite-dimensional quotient of $k[\alpha]$.

2. *Truncating a longer loop.* Keeping the same one-loop quiver but taking instead $I = (\alpha^3) = R_Q^3$ gives $kQ/I = k[\alpha]/(\alpha^3)$, three-dimensional with basis $\{e_1, \alpha, \alpha^2\}$. Here $m = 3$; the admissibility bound $I \subseteq R_Q^2$ holds because $\alpha^3 \in R_Q^3 \subseteq R_Q^2$, and the relation still involves no constant path and no bare arrow. More generally $k[\alpha]/(\alpha^n)$ is the one-loop quiver bound by (α^n) for each $n \geq 2$, and these exhaust the local algebras arising from a single loop.

3. *The A_2 quiver, hereditary case.* Let Q be the quiver

$$1 \xrightarrow{\alpha} 2$$

with two vertices and a single arrow $\alpha\colon 1 \to 2$. The only paths are e_1, e_2, and α, so kQ is already three-dimensional and $R_Q^2 = 0$ (there is no legal concatenation of two arrows). The only admissible ideal is therefore $I = 0$: admissibility requires $I \subseteq R_Q^2 = 0$, so nothing may be quotiented, and $R_Q^2 = 0 = I$ supplies the lower bound with $m = 2$. Thus $kQ/I = kQ$, a path algebra with no relations. An algebra whose only admissible bound is the zero ideal, equivalently one whose radical squares to zero and imposes no relations, is called *hereditary*; the A_2 path algebra is the prototype.

Concretely kQ is isomorphic to the lower-triangular matrix algebra

$$\begin{pmatrix} k & 0 \\ k & k \end{pmatrix}$$

via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, and $\alpha \mapsto E_{21}$, where E_{ij} is the elementary matrix with a 1 in position (i, j). Lower- rather than upper-triangular is forced by the multiplication convention: the arrow $\alpha\colon 1 \to 2$ satisfies $e_2 \alpha e_1 = \alpha$, so its image must land in position $(2, 1)$, the strictly lower-triangular slot. The check is direct on the multiplication table: $e_2 \cdot \alpha \cdot e_1 = \alpha$ matches $E_{22}E_{21}E_{11} = E_{21}$, and every other product of basis elements is forced to zero on both sides. (The upper-triangular algebra is the opposite algebra, the A_2 path algebra with the arrow reversed to $2 \to 1$; we recover it from the radical in example 4.2.9.) We return to this identification once we have located the indecomposable projectives.

The dual-numbers example already shows the shape of the reconstruction: a local algebra with radical square zero came out as a single loop bound by α^2, and the loop is exactly a basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$. Before making this into a theorem, one must decide which algebras to reconstruct. Not every finite-dimensional algebra is a bound quiver algebra on the nose. The matrix algebra $M_2(k)$, for instance, is simple, so its only simple module is two-dimensional, whereas a bound quiver algebra always has all its simples one-dimensional over k when k is algebraically closed (they are the vertex simples S_i). The class for which the reconstruction is exact is singled out by demanding that the semisimple quotient $A/\mathrm{Jac}(A)$ be as small as possible: a product of division rings, with no matrix blocks of size larger than one.

Definition 4.2.4: Basic Algebra

A finite-dimensional k-algebra A is *basic* if the quotient $A/\mathrm{Jac}(A)$ is a product of division rings,

$$A/\mathrm{Jac}(A) \cong D_1 \times \cdots \times D_n$$

with each D_i a division k-algebra. Equivalently, in the Artin–Wedderburn decomposition of $A/\mathrm{Jac}(A)$ (corollary 1.2.18) every matrix block $M_{n_i}(D_i)$ has $n_i = 1$.

When k is algebraically closed the finite-dimensional division k-algebras collapse to k itself (corollary 1.2.20), so a basic algebra over such a k has

$$A/\mathrm{Jac}(A) \cong k \times \cdots \times k$$

a product of n copies of k. Two consequences are worth stating plainly. First, the simple A-modules are the n simple modules of this product, each one-dimensional over k; a basic algebra over an algebraically closed field has *all simple modules one-dimensional*. Second, the number n of simple modules equals the number of factors, and it becomes the number of vertices of the reconstructed quiver. The condition "every matrix block has size one" is precisely the condition that the regular representation of $A/\mathrm{Jac}(A)$ contains each simple exactly once, so that A is, in the language of idempotents, a sum of pairwise non-isomorphic indecomposable projectives.

Example 4.2.5: Basic and Non-Basic Algebras

The bound quiver algebras of example 4.2.3 are all basic: for $k[\alpha]/(\alpha^2)$ the radical quotient is k, and for the A_2 path algebra it is $k \times k$, both products of copies of k. The upper-triangular algebra of 2×2 matrices is basic for the same reason. By contrast $M_2(k)$ is *not* basic: it is already semisimple, so $\mathrm{Jac}(M_2(k)) = 0$ and $M_2(k)/\mathrm{Jac}(M_2(k)) = M_2(k)$ is a single 2×2 block. The group algebra $k[S_3]$ over $k = \mathbb{C}$ is not basic either, since by Maschke's theorem it is semisimple and Artin–Wedderburn gives $\mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, a product with a 2×2 block coming from the two-dimensional irreducible representation of S_3. These non-basic examples are not pathological; they are the reason the theorem is stated up to Morita equivalence rather than isomorphism.

The point of Morita equivalence is that it does not see the difference between an algebra and its basic model. Two algebras are Morita equivalent when their module categories are equivalent, and Morita's theorem identifies exactly which algebras share a module category with a given one: they are the endomorphism algebras of progenerators. That theory, including generators, progenerators, and the classification of Morita equivalences by bimodules, is developed in *Core Algebra* [5, Morita theory]; what is used here is the criterion for a corner ring, that Ae is a progenerator exactly when the idempotent e is full, in which case A and eAe are Morita equivalent. The block structure of the radical quotient is this book's own and is settled below.

Proposition 4.2.6: Reduction to the Basic Case

Let A be a finite-dimensional k-algebra. Then there is a basic finite-dimensional k-algebra A^b, the *basic algebra of* A, that is Morita equivalent to A: the categories of left modules A**-Mod** and A^b**-Mod** are equivalent. The algebra A^b is unique up to isomorphism, and A and A^b have the same number of simple modules and the same radical quotient block *data* in the sense that $A^b/\mathrm{Jac}(A^b) \cong D_1 \times \cdots \times D_n$ where $M_{n_i}(D_i)$ are the Artin–Wedderburn blocks of $A/\mathrm{Jac}(A)$.

***Proof*:**
Step 1: the idempotent. Write $1 = e_1 + \cdots + e_r$ as a sum of primitive orthogonal idempotents, following theorem 2.3.2, so that $A = \bigoplus_j Ae_j$ expresses the regular module as a direct sum of indecomposable projectives $P_j = Ae_j$. Two of these are isomorphic exactly when the corresponding simples $P_j/\mathrm{rad}(P_j)$ and $P_{j'}/\mathrm{rad}(P_{j'})$ agree, so grouping the e_j by the isomorphism class of P_j and choosing one representative from each class produces an idempotent

$$e = e_{i_1} + \cdots + e_{i_n}$$

with n the number of simple modules. It is idempotent because the e_{i_j} are orthogonal. Set $A^b := eAe$.

Step 2: e is full. Each e_{i_j} satisfies $e_{i_j}e = e_{i_j}$, again by orthogonality, so $Ae_{i_j}A = Ae_{i_j}eA \subseteq AeA$. For an arbitrary l the projective Ae_l is isomorphic to Ae_{i_j} for some j by the choice of representatives, and isomorphic projectives generate the same two-sided ideal, so $Ae_lA = Ae_{i_j}A \subseteq AeA$. Summing,

$$1 = \sum_{l=1}^{r} e_l \in \sum_{l=1}^{r} Ae_lA \subseteq AeA$$

so $AeA = A$ and e is full.

Step 3: the equivalence. By [5, the corner ring of a full idempotent], a full idempotent makes Ae a progenerator with $\mathrm{End}_A(Ae)^{\mathrm{op}} \cong eAe$, so A and $A^b = eAe$ are Morita equivalent. The equivalence is $\mathrm{Hom}_A(Ae, -)$, which sends an A-module M to eM.

Step 4: A^b is basic, with the stated radical quotient. The equivalence carries the indecomposable projectives of A to those of A^b and preserves isomorphism, and $Ae = \bigoplus_j Ae_{i_j}$ retains exactly one indecomposable projective from each isomorphism class. So the regular representation of A^b contains each simple exactly once, which is what it means for A^b to be basic, and its radical quotient has all blocks of size one. The division rings are unchanged: the equivalence induces a bijection on simple modules and a ring isomorphism on their endomorphism rings, so the blocks are over the same $D_i = \mathrm{End}_A(S_i)^{\mathrm{op}}$ furnished by Schur's lemma (lemma 1.1.5). Uniqueness of A^b up to isomorphism follows because any two choices of representatives give isomorphic modules Ae, hence isomorphic endomorphism rings, completing the proof.

The reduction lets us prove the structure theorem only for basic algebras and inherit it for all finite-dimensional algebras through the equivalence of module categories: whatever we say about the representations of $A^b = kQ_A/I$ transfers verbatim to the representations of A. From now on A is basic, and to make the reconstruction concrete assume k is algebraically closed, so that all simple modules are one-dimensional and the radical quotient is a product of copies of k. This is the setting of Gabriel's theorem, and it is where the assumption $k = \overline{k}$ earns its place: over a non-closed field the division algebras D_i reappear and the vertices of the quiver would have to be weighted by them.

The quiver attached to an abstract algebra can now be defined. Its vertices are the simple modules; its arrows count how simples fail to split off one another, measured either homologically by Ext^1 or internally by the radical filtration. That these two counts agree is the bridge between the abstract and combinatorial descriptions.

Definition 4.2.7: Ext-Quiver (Gabriel Quiver)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with simple modules $S_1, \dots, S_n$ up to isomorphism and a complete set of primitive orthogonal idempotents $e_1, \dots, e_n$ with $S_i = Ae_i/\mathrm{rad}(Ae_i)$. The *Ext-quiver* (or *Gabriel quiver*) Q_A is the quiver with

1. vertex set $\{1, \dots, n\}$, one vertex for each simple module S_i, and
2. exactly
$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
arrows from i to j.

The order of the idempotents in the formula is fixed by the multiplication convention of definition 4.1.58 and is worth pausing on, since it is easy to get backwards. In the path algebra a would-be arrow from i to j satisfies $e_j \alpha e_i = \alpha$, so it lives in $e_j(kQ)e_i$; matching this, the arrows from i to j in Q_A are counted by $e_j(\mathrm{Jac}/\mathrm{Jac}^2)e_i$, with e_j on the left and e_i on the right. A source using the opposite (left-to-right) convention for path multiplication would write $e_i(\mathrm{Jac}/\mathrm{Jac}^2)e_j$ instead; the two conventions transpose the quiver, so one must commit to one. Here the commitment is to the book's convention throughout. The equality of the two counts, homological and idempotent-theoretic, is the following lemma, proved here because it is what makes the definition well-posed: the arrow count must not depend on the choice of description.

Lemma 4.2.8: Arrows Count Radical Extensions

Let A be a finite-dimensional k-algebra with primitive orthogonal idempotents $e_1, \dots, e_n$ and simple modules $S_i = Ae_i/\mathrm{rad}(Ae_i)$, and assume k is algebraically closed. Then for all i, j,
$$\mathrm{Ext}^1_A(S_i, S_j) \cong e_j\big(\mathrm{Jac}(A)/\mathrm{Jac}(A)^2\big)e_i$$
as k-vector spaces. In particular the two are equidimensional.

***Proof*:**
Write $J = \mathrm{Jac}(A)$ and $P_i = Ae_i$, the indecomposable projective with top $S_i = P_i/\mathrm{rad}(P_i)$; recall from definition 2.3.3 that $\mathrm{rad}(P_i) = Je_i$, so that $S_i = Ae_i/Je_i$.

Step 1: A projective presentation of S_i. The surjection $P_i \twoheadrightarrow S_i$ has kernel $\mathrm{rad}(P_i) = Je_i$, giving the short exact sequence
$$0 \to Je_i \to P_i \to S_i \to 0$$
Applying $\mathrm{Hom}_A(-, S_j)$ produces the exact sequence
$$0 \to \mathrm{Hom}_A(S_i, S_j) \to \mathrm{Hom}_A(P_i, S_j) \to \mathrm{Hom}_A(Je_i, S_j) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

where the last term is $\mathrm{Ext}^1_A(S_i, S_j)$ and not $\mathrm{Ext}^1(P_i, S_j)$ because P_i is projective, so $\mathrm{Ext}^1_A(P_i, S_j) = 0$ and the connecting map surjects onto $\mathrm{Ext}^1(S_i, S_j)$. The definition and basic properties of Ext^1 as the obstruction to lifting homomorphisms along a projective presentation are developed in the companion volume on core algebra, `NateCoreAlgebra` [5]; we use only that a projective resolution computes it and that projectives have no higher self-extensions.

Step 2: Evaluating the outer terms. For any simple S_j and any module M, a homomorphism $M \to S_j$ factors through $M/\mathrm{rad}(M)$ because $\mathrm{rad}(M)$ maps into $\mathrm{rad}(S_j) = 0$. Applying this to P_i gives $\mathrm{Hom}_A(P_i, S_j) = \mathrm{Hom}_A(S_i, S_j)$, so the first two terms of the sequence in Step 1 cancel and the map between them is an isomorphism. The sequence therefore collapses to

$$0 \to \mathrm{Hom}_A(Je_i, S_j)/\big(\text{image of } \mathrm{Hom}_A(P_i, S_j)\big) \to \mathrm{Ext}^1_A(S_i, S_j) \to 0$$

but the image of $\mathrm{Hom}_A(P_i, S_j)$ in $\mathrm{Hom}_A(Je_i, S_j)$ is zero: a homomorphism $P_i \to S_j$ vanishes on $\mathrm{rad}(P_i) = Je_i$ by the factoring just noted. Hence the connecting map is an isomorphism

$$\mathrm{Ext}^1_A(S_i, S_j) \cong \mathrm{Hom}_A(Je_i, S_j)$$

Step 3: The Hom into a simple is a radical layer. Again by the factoring through the top, any homomorphism $Je_i \to S_j$ kills $\mathrm{rad}(Je_i) = J^2e_i$, so

$$\mathrm{Hom}_A(Je_i, S_j) = \mathrm{Hom}_A\big(Je_i/J^2e_i,\ S_j\big)$$

The module Je_i/J^2e_i is semisimple, being a module over the semisimple algebra A/J, and S_j is simple, so by Schur's lemma (lemma 1.1.5) and $\mathrm{End}_A(S_j) = k$ (from $k = \overline{k}$) the space $\mathrm{Hom}_A(Je_i/J^2e_i, S_j)$ has dimension equal to the multiplicity of S_j in Je_i/J^2e_i. That multiplicity is $\dim_k e_j(Je_i/J^2e_i)$, since for a semisimple module N over $A/J = \prod_\ell k$ the multiplicity of S_j in N is $\dim_k e_jN$: the idempotent e_j projects N onto its S_j-isotypic part, which over an algebraically closed field is a sum of copies of S_j each one-dimensional. Therefore

$$\dim_k \mathrm{Ext}^1_A(S_i, S_j) = \dim_k e_j(Je_i/J^2e_i) = \dim_k e_j(J/J^2)e_i$$

the last equality because $e_j(J/J^2)e_i = e_j(Je_i)/e_j(J^2e_i)$ and right-multiplication by e_i identifies Je_i/J^2e_i with the e_i-column of J/J^2. Assembling the three isomorphisms gives $\mathrm{Ext}^1_A(S_i, S_j) \cong e_j(J/J^2)e_i$, as we sought to show.

The lemma says the arrows of Q_A have two readings. Homologically, an arrow $i \to j$ records a one-dimensional worth of non-split extension of S_i by S_j: a module of length two with S_j as a submodule and S_i on top that does not break as $S_i \oplus S_j$. Internally, the same arrow is a basis vector of the (j, i) block of the first radical layer $\mathrm{Jac}/\mathrm{Jac}^2$. The first reading is the reason the quiver is called the Ext-quiver and explains why it is an invariant of the module category, hence of the Morita equivalence class; the second reading is the one to compute with, because $\mathrm{Jac}/\mathrm{Jac}^2$ is a finite matrix of idempotent-sandwiched subspaces. The running examples compute it.

Example 4.2.9: Gabriel Quivers of the Running Algebras

Reading off Q_A for two algebras confirms it recovers the starting quiver.

1. *The dual numbers.* Let $A = k[\alpha]/(\alpha^2)$. This is local: its only maximal ideal is (α), so $\mathrm{Jac}(A) = (\alpha)$ and there is a single simple module $S_1 = A/(\alpha) = k$, giving one vertex. The first radical layer is $\mathrm{Jac}/\mathrm{Jac}^2 = (\alpha)/(\alpha^2)$, one-dimensional with basis the image of α, and

with $e_1 = 1$ the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_1$ is all of it. So Q_A has one vertex and one arrow, a loop:

$$\begin{tikzcd} 1 \arrow[loop above, "\alpha"] \end{tikzcd}$$

exactly the quiver we bound in example 4.2.3. The relation to be imposed will be $\alpha^2 = 0$, recovering A.

2. *The upper-triangular algebra.* Let $A = \begin{pmatrix} k & k \\ 0 & k \end{pmatrix}$, the 2×2 upper-triangular matrices. Its radical is the strictly upper-triangular part, $\mathrm{Jac}(A) = kE_{12}$, one-dimensional with square zero. The idempotents $e_1 = E_{11}$ and $e_2 = E_{22}$ are primitive orthogonal with $e_1 + e_2 = 1$, and $A/\mathrm{Jac}(A) \cong k \times k$, so there are two vertices with simples S_1, S_2. Since $\mathrm{Jac}^2 = 0$ we have $\mathrm{Jac}/\mathrm{Jac}^2 = kE_{12}$, and we locate E_{12} by its idempotent sandwich:

$$e_1E_{12}e_2 = E_{11}E_{12}E_{22} = E_{12}, \qquad e_2E_{12}e_1 = E_{22}E_{12}E_{11} = 0$$

so the one basis vector of $\mathrm{Jac}/\mathrm{Jac}^2$ sits in the block $e_1(\mathrm{Jac}/\mathrm{Jac}^2)e_2$. By definition 4.2.7 the block $e_j(\cdots)e_i$ counts arrows $i \to j$, so with $j = 1$, $i = 2$ this is one arrow $2 \to 1$:

$$2 \xrightarrow{\alpha} 1$$

Under the identification $\alpha \mapsto E_{12}$ this is the A_2 quiver of example 4.2.3, up to relabelling the vertices, and $\mathrm{Jac}^2 = 0$ forces the admissible ideal to be $I = 0$. The algebra is hereditary, and Q_A carries no relations: the upper-triangular algebra *is* its own path algebra kQ_A.

The two examples describe the same three-dimensional hereditary algebra, once as a bound quiver algebra and once reconstructed from its radical, and the indecomposable projectives match under the identification. Take the bound-quiver presentation of example 4.2.3, the A_2 path algebra with arrow $\alpha\colon 1 \to 2$ realized as the lower-triangular matrices via $e_1 \mapsto E_{11}$, $e_2 \mapsto E_{22}$, $\alpha \mapsto E_{21}$. The indecomposable projective $P_i = kQ \cdot e_i$ is spanned by the paths starting at i: P_1 has basis $\{e_1, \alpha\}$, the trivial path at 1 together with the arrow out of 1, and is two-dimensional with top S_1 and socle S_2, while P_2 has basis $\{e_2\}$ and is the simple S_2 itself. On the matrix side these are the column modules AE_{11} and AE_{22}: the first column of the lower-triangular algebra is two-dimensional, spanned by E_{11} and E_{21} (the entries in rows 1 and 2 of column 1), and the second column is one-dimensional, spanned by E_{22}. So the indecomposable projectives read off from the primitive idempotents of theorem 2.3.2 are exactly the projective quiver representations P_i of the path algebra. The Gabriel reconstruction of example 4.2.9 applied instead the upper-triangular representative, whose radical kE_{12} sits in the opposite corner and so produces the arrow $2 \to 1$; the two matrix algebras are opposite to one another (transpose exchanges upper and lower triangular), which is exactly the transposition of the quiver that the two multiplication conventions differ by. Either way the representation theory of A_2 is the same, and the two constructions coincide.

Everything is now in place for the structure theorem. It asserts that the passage from a basic algebra to its Gabriel quiver and back to a bound quiver algebra is a round trip: A is isomorphic to kQ_A modulo an admissible ideal. The proof builds a surjection $kQ_A \twoheadrightarrow A$ by sending vertices to the primitive idempotents and arrows to a lift of the radical layer, then shows the kernel is admissible using that $\mathrm{Jac}(A)$ is nilpotent.

Theorem 4.2.10: Structure of Basic Algebras (Gabriel)

Let A be a basic finite-dimensional algebra over an algebraically closed field k, with Gabriel quiver Q_A (definition 4.2.7). Then there is an admissible ideal $I \subseteq kQ_A$ and an isomorphism of k-algebras

$$A \cong kQ_A/I$$

Consequently, by the reduction to the basic case (proposition 4.2.6), every finite-dimensional k-algebra over an algebraically closed field is Morita equivalent to a bound quiver algebra kQ/I.

The quiver Q_A is determined by A, but the ideal I is not: different admissible ideals can present isomorphic algebras, so the theorem produces *some* admissible presentation rather than a canonical one. This is the one point where the correspondence between algebras and bound quivers fails to be a clean bijection, and it is the reason bound quiver algebras are studied through their quiver plus a choice of relations rather than through the quiver alone.

***Proof*:**
Write $J = \mathrm{Jac}(A)$, and recall J is nilpotent, say $J^m = 0$ for some $m \geq 2$, because A is finite-dimensional (a nilpotent radical is the Artinian case established in theorem 2.1.8). Let $e_1, \ldots, e_n$ be the complete set of primitive orthogonal idempotents underlying Q_A, with $1 = e_1 + \cdots + e_n$ and $S_i = Ae_i/Je_i$.

Step 1: Send vertices to idempotents and arrows to a radical lift. For each vertex i of Q_A set the constant path $\varepsilon_i \in kQ_A$ to map to $e_i \in A$. For the arrows, fix for each pair (i, j) a k-basis of the block $e_j(J/J^2)e_i$; by definition 4.2.7 and lemma 4.2.8 the arrows of Q_A from i to j are in bijection with this basis. For each arrow $\alpha\colon i \to j$ choose a lift $x_\alpha \in e_jJe_i \subseteq A$ of the corresponding basis vector of $e_j(J/J^2)e_i$; such a lift exists because $e_j(J/J^2)e_i$ is a quotient of e_jJe_i. Note $x_\alpha \in e_jJe_i$ means $e_jx_\alpha e_i = x_\alpha$, which is the algebra image of the path relation $e_j\alpha e_i = \alpha$, so the assignment respects the idempotent bookkeeping.

Step 2: Extend to an algebra homomorphism. Because kQ_A is free on paths, an algebra homomorphism out of it is determined by the images of the vertices and arrows, subject only to the relations $\varepsilon_i\varepsilon_{i'} = \delta_{ii'}\varepsilon_i$, $\sum_i \varepsilon_i = 1$, and the source-target matching for arrows. The chosen images e_i and x_α satisfy all of these: the e_i are orthogonal idempotents summing to 1, and $x_\alpha \in e_jAe_i$ matches the source i and target j of α. Extend multiplicatively, sending a path $\alpha_r \cdots \alpha_1$ to the product $x_{\alpha_r} \cdots x_{\alpha_1} \in A$ (in the algebra's own multiplication, so a path from i to j lands in e_jAe_i). This defines a k-algebra homomorphism

$$\varphi\colon kQ_A \to A$$

Step 3: φ is surjective. The image contains all e_i and, since each x_α together with the x_α-products spans, it contains a lift of a basis of J/J^2. We show $\mathrm{im}\varphi = A$ by showing it contains all of $A = A/J^0 \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^m = 0$ layer by layer. The images $\varphi(\varepsilon_i) = e_i$ generate $A/J = \prod_i ke_i$ over k, so $\mathrm{im}\varphi + J = A$. Next, the products of length one, $x_\alpha = \varphi(\alpha)$, form lifts of a basis of J/J^2, so $\mathrm{im}\varphi + J^2 \supseteq J$, hence $\mathrm{im}\varphi + J^2 = A$. Inductively suppose $\mathrm{im}\varphi + J^\ell = A$; then J^ℓ is spanned modulo $J^{\ell+1}$ by products of ℓ elements of J, each of which may be taken from the spanning set $\{x_\alpha\}$ of J modulo J^2, so products $\varphi(\alpha_\ell \cdots \alpha_1)$ of ℓ arrows span J^ℓ modulo $J^{\ell+1}$. Thus $\mathrm{im}\varphi + J^{\ell+1} = A$. Since $J^m = 0$, after m steps $\mathrm{im}\varphi = \mathrm{im}\varphi + J^m = A$, so φ is surjective.

Step 4: The kernel is admissible. Let $I = \ker\varphi$, so $A \cong kQ_A/I$. It remains to check $R_Q^{m'} \subseteq I \subseteq$

R_Q^2 for some m', where R_Q is the arrow ideal of kQ_A.

For the lower bound, $\varphi(R_Q^m) \subseteq J^m = 0$: a path of length at least m is a product of at least m arrows, and each arrow maps into J, so a length-$\geq m$ path maps into $J^m = 0$. Hence $R_Q^m \subseteq \ker \varphi = I$, giving the lower bound with $m' = m$.

For the upper bound $I \subseteq R_Q^2$, suppose $u \in I$ and write $u = u_0 + u_1 + u_2$ where $u_0 \in kQ_{A,0}$ is the constant-path part, u_1 is the arrow part (length exactly one), and $u_2 \in R_Q^2$ collects the length-≥ 2 paths. We must show $u_0 = 0$ and $u_1 = 0$. Applying φ and reducing modulo J: $\varphi(u_1) \in J$ and $\varphi(u_2) \in J^2 \subseteq J$, so $0 = \varphi(u) \equiv \varphi(u_0) \pmod J$. But φ restricted to constant paths is the isomorphism $kQ_{A,0} = \prod_i k\varepsilon_i \xrightarrow{\sim} \prod_i ke_i = A/J$ (the e_i are linearly independent modulo J because they are orthogonal idempotents with distinct simple tops), so $\varphi(u_0) \equiv 0 \pmod J$ forces $u_0 = 0$. Now reduce modulo J^2: with $u_0 = 0$ we have $0 = \varphi(u) = \varphi(u_1) + \varphi(u_2)$, and $\varphi(u_2) \in J^2$, so $\varphi(u_1) \equiv 0 \pmod{J^2}$. But $\varphi(u_1)$ is the image in J/J^2 of a linear combination of arrows, and the arrows were chosen to map to a *basis* of J/J^2; a linear combination of basis vectors that is zero in J/J^2 has all coefficients zero, so $u_1 = 0$. Therefore $u = u_2 \in R_Q^2$, proving $I \subseteq R_Q^2$.

Combining the two bounds, $R_Q^m \subseteq I \subseteq R_Q^2$, so I is admissible and $A \cong kQ_A/I$ is a bound quiver algebra. The final clause on general algebras follows by composing with the Morita reduction of proposition 4.2.6: an arbitrary finite-dimensional A is Morita equivalent to its basic algebra A^b, and $A^b \cong kQ_{A^b}/I$ by what was just proved, completing the proof.

Three features of the argument deserve a closing word. The surjectivity in Step 3 is a radical-filtration induction: because $\mathrm{Jac}(A)$ is nilpotent, generating $A/\mathrm{Jac}(A)$ by the idempotents and the first radical layer $\mathrm{Jac}/\mathrm{Jac}^2$ by the arrows suffices to generate all of A, since higher layers are spanned by products. The admissibility of the kernel in Step 4 is the exact counterpart of the two bounds in definition 4.2.1: the lower bound $R_Q^m \subseteq I$ is nilpotence of the radical, and the upper bound $I \subseteq R_Q^2$ is the choice of arrows as a basis of $\mathrm{Jac}/\mathrm{Jac}^2$, which is precisely why an admissible ideal is forbidden to contain constant paths or bare arrows. And the theorem is the structure theorem of Gabriel, to be kept distinct from Gabriel's other theorem in this chapter (theorem 4.1.47), which classifies the quivers of finite representation type as the simply-laced Dynkin diagrams A, D, E. The two results are complementary: the present one says every finite-dimensional algebra is a bound quiver algebra, and the Dynkin classification says which of the unbound path algebras have only finitely many indecomposable modules.